



Harman

5G NAD Module

Hardware Specification v3.4

Change History

Rev	Date	Change Description	Author
0.1	28-June-19	Initial Release (Preliminary)	NAD team
0.2	8-July-19	Updated ESD test requirement, clarified signals (MIPI, ADC, I2S, GND) in pinout diagram and pinout table. Updated ADC1 detail pin description.	
0.3	10-July-19	Updated Power supply specifications and descriptions, Boot Configuration Pin descriptions, additional information for PM_GPIO added to pinout table.	
0.4	25-Oct-19	Added routing recommendations, thermal section, pin descriptions, Edited Variants/bands in Table 1-1 Part Number Table updated Table 3-1 . Section 2.1 Added System Block Diagram Added Peak Current Consumption limits Table 6-3 Edited Section 11.1.1 NAD Antenna Breakout Added Max V parameter for 3v3_PMX Table 6-2 Support for External MIPI Antenna Switch Section 4.5.14	
0.5	4-Nov-19	Updated VREF_RGMII to only use 1.8V Updated pinout signal descriptions Updated Mech drawing side view Figure 7-3 Removed all references to modes other than 5G NR, LTE, WCDMA, GSM Section 12 Thermal Considerations, updated thermal via locations and heatsinking	
0.6	22-Nov-19	Table 4-4 Update descriptions to NO CONNECT for VREF_SDC, RF_GPIO_N19, RF_GPIO_N20, RF_GPIO_M19, RF_GPIO_M20; S18, T17 descriptions changed Figure 4-1 Pin Assignments updated Table 3-1 NAD Model Names updated Figure added for DSDA footprint	
0.7	11-Dec-19	Table 4-4 U18 is NO CONNECT Table 1-1 Supported bands changed for variants CN, EU, JP	
0.8	27-Dec-19	Change Tracking Record Table added Updated descriptions to VUC_BOOTLOADER_MODE, VUC2NAD_BANK_SELECT Removed references to VREF_2V7 MDM2VUC name changed to NAD2TCU Figure 7-1, Figure 7-2, Figure 7-3 Mechanical Dimensions updated Figure 7-2 PCB Footprint - thermal vias reduced to 0.6mm, same as signal vias	
0.9	14-Jan-20	Table 1-1 Supported bands updated Table 3-1 Model Names updated Table 4-6 Antenna Pin detail added Figure 11-3 Antenna Pad drawings updated to include GNSS2 ANT Table 11-7 Updated constraints for RGMII routing Figure 8-1, Figure 7-2 Updated for clarity, new pad dimensions Section 8.3 PCB CTE _{xy} Requirement, PCB Coplanarity Requirement	
0.9.3	16-Jan-20	Table 4-4 (note 3) recommend GPIO M18 Figure 8-1 pin #'s now indicated, including DSDA "Known Issues" Section added	
1.0	18-Feb-20	Section 1.2.1 CDD support added Figure 4-1 Pin Assignments Array updated Figure 8-1 All footprint information consolidated into single drawing Figure 11-6 Power routing pinout updated Figure 4-1, Figure 11-6 "CV2X_TR" ==> "CV2X_TR1", pad group P3 added Table 4-4 pin E15 and C22 functional group names changed; "4V_5GPA" renamed as "4V_DSDA" Table 6-2 removed 4V_5GPA supply & RMS current specs; 5V_CV2X current updated Table 6-3 5V_CV2X current updated Table 4-5 "4V_5GPA" renamed as "4V_DSDA" Table 1-1 Supported bands updated (JP => RoW) Section 4.5.18 updated value of Rdat to 10 kΩ Table 4-4 VUC2NAD_BANK_SELECT function defined	
1.1	28-Feb-20	Table 1-1 Supported bands updated RoW	
1.2	9-Apr-20	Table 4-4 "CV2X_TR" ==> "CV2X_TR1" (other locations updated in v1.0), polarity stated, P3 pad-type corrected DSDA footprint drawing updated Table 4-4 , Sect. 4.5.8 Changed name VUC2MDM_HOLDUP ==> VUC2NAD_HOLDUP Changed name VUC_BOOTLOADER_MODE ==> VUC_BOOT_MODE Section 4.5.9 Added section for VUC2NAD_BANK_SELECT Table 6-3 Peak Current values updated Figure 8-1 NAD footprint updated for B2	

Rev	Date	Change Description	Author
		Section 4.5.1 Supply pin location and net name corrected for large electrolytic capacitor Section 4.5.1 Note that 4V_QET and 4V_PMX are tied together externally	
1.3	24-Apr-20	Table 4-6 Antenna Pin usage updated	
1.4	24-Apr-20	Figure 8-1 NAD footprint updated for B2+	
1.5	4-May-20	Section 8.1 Corrected text description, changed thermal "vias" to "pads" Table 4-4 Update NAD_STATUS[1:0] signal names to NAD_READY and NAD_LPM Section 4.5.9 VUC2NAD_BANK_SELECT description updated	
1.6	3-June-20	Table 4-4 Included Y08 in GND pin list Figure 4-1 Pin assignment array updated (to match earlier name changes) Figure 11-6 Power routing pinout updated (to match earlier name changes) Section 6.8.1 Added TX power targets and tolerances Section 6.8.2 Added Rx sensitivity specs & references Section 8.2 Manufacturing recommendations updated (aperture/solder) Table 6-2 E-call operating temperature (Qualcomm SoC Junction) added with footnote	
1.7	27-July-20	Table 6-3 Corrected descriptions of C-V2X operation Table 1-1 Band support updated; NSA bands denoted Section 3.2 Module Marking/Label updated Figure 5-1, Figure 5-2 Power Up/Down Timing diagrams changed Section 12 Thermal topics updated Table 12-1 Thermal Mitigation States updated Section 12.2 Heat sinking approach verbiage changed. Table 4-4, Section 4.5.9 VUC2NAD_BANK_SELECT polarity changed. Table 4-4 Pin S18 changed to Free GPIO Removed signal and references to VUC_BUSY Removed signal and references to NAD2TCU_WAKEUP Removed signal and references to VUC_BOOT_MODE Removed signal and references to NAD_READY Removed signal and references to VBATT_LOW Removed signal and references to TEST_BOOT Section 4.5.5 NAD Status ==> NAD Low Power Mode, description updated Section 4.5.6 RESET pin description updated Section 5.1 Module ON Timing updated Section 5.2 Module OFF Timing updated Section 4.3 Pad Voltage Groupings updated Section 8.3.1 Flatness drawing added, limit specified Thermistor Locations added Table 3-1 Model Names table updated Table 4-3 I/O Voltage Levels defined Table 4-19 VUC2NAD_HOLDUP function description All instances of FORCE_USB_BOOT changed to FORCED_USB_BOOT	
1.8	12-Aug-20	Table 1-1 Removed LTE bands 6 and 9 (not supported), notes corrected Figure 3-1 NAD Label drawing updated Section 8.3.1 Flatness drawing fixed Section 3.2 NAD Label drawing corrected Section 4.5.6 RESET_N description updated; PMX_RESIN_N ==> RESET_N Section 1.4.3 EN-DC supported combinations added Section 6.5 ESD ratings Section 12.1 Thermistors description updated PCB Thermistor names updated	
1.9	03-Sept-20	Section 8.3.1 Solder Pre-tinning use clarified explicitly	

Rev	Date	Change Description	Author
2.0	04-Sept-20	Section 6.8.1 RF TX power window +/- Section 6.5 ESD CDM Ratings update	
2.01	08-Sept-20	Section 6.5 ESD CDM Ratings corrected	
2.1	07-Oct-20	"Known Issues" Section removed, info available in HW release notes Table 1-1 Add band NR n25 and LTE b71 support Section 11.1.1 Antenna Layout and Routing guidance updated Section 11.1.1.2 Removed image of online tool FORCED_USB_BOOT section removed as it is not used Section 7.1 Module Exploded view updated Section 7.2 Module Top view updated Section 7.4 Module Height/Tolerance detail added Section 7.3 Side view conflicting dimensions removed Section 4.5.39 Added C-V2X SA2150 EAP Interface detail Removed "b41 + n41" from EN-DC list Figure 4-1 Pin Functional Groups diagram updated DSDA detail items removed Sections 13, 14 Added placeholders for certification/compliance Table 8-1 Pad dimensions table added Section 4.5.6 RESET_N detail Title page now reads "Hardware Specification" Section 4.5.32 RGMII delay detail added Table 4-3 pad S20 changed to "reserved", removed from Table 4-50 Table 4-6 unused antennas to remain open Section 1.2.1 LTE UL CA Intraband support Section 7.4 Module Height pre-tinned and non-pre-tinned detail	
2.2	21-Oct-20	Section 4.5.8 VUC2NAD_HOLDUP description updated Section 4.5.9 VUC2NAD_BANK_SELECT description updated Figure 3-1 NAD label slight correction Table 6-3 SIM1 Peak Current case added, footnotes updated	
2.3	07-Feb-21	Section 6.8.1 WCDMA TX Max Power target set to 23.0 dbm Section 12.1 explicitly name IC components for clarity Section 5.8.3 HPUE supported bands Table 6-2 3V3_PMX max voltage is 4.2V Section 4.5.10 Antenna ADC section Section 1.3 Pin count corrected to 725 Section 1.4.1 Updated variant band table w/NAX Section 11.3 Change UIMX_DATA pull up value to 10Kohm Section 12.1 Thermal Mitigation updated Table 4-7 Recommended Antenna isolation Section 7 Mechanical Drawing reference added	
2.4	02-Apr-21	Acronym "SoC" definition updated ENDC support updated Table 4-6 LTE_ANT_2 function corrected Table 1-1 remove b34 & add bands 6 and 9 in FE5RW0020; added CN variants Table 6-26 Added GSM multi-slot power reduction Section 6.8.1 EDGE TX power Table 6-4 Operating Mode Currents	
2.4.1	09-Apr-21	Table 1-1 remove n25 from NA, n28 in CN (tentative), remove NAX Table 4-7 add external GNSS isolation	
2.5	08-Jun-21	Section 4.5.1 Power Supply bulk capacitance "design in progress" removed from title page Figure 4-2 antenna terminal positions Table 1-1 Remove C-V2X variants Table 6-4 Max Current values added Section 1.2.1 Air Interface descriptions updated Table 4-2 pull-down resistor for VREG_PX2_DUAL "LTE Antenna" corrected to "Cellular Antenna" throughout	

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2.6	26-Jul-21	Table 4-6 Antenna functions clarified Table 4-39 PCIE_CLKREQ_N, PCIE_WAKE_N pull-ups => 10Kohms Section 1.2.1 Air Interface items updated Figure 2-1 Block Diagram Updated Section 6.8.3 Tune Up Procedure added Table 12-1 Thermal Mitigation description updated Section 6.8.1 GSM Multi-slot power corrected, WCDMA TX limit tightened Section 1.4.3 ENDC support referenced to external document	
2.7	13-Jan-22	Section 7.4 Irrelevant dimensions removed Section 1.3 Package dimension updated for cover-in-place Table 1-1 Updated, specify NSA/SA 5G bands Section 4.5.2 Band, PowerClass (HPUE) Support each antenna Section 6.8.1 RF power specifications updated Table 5-2 unit corrected "ms" => "s" Section 5.1 Module ON Timing updated Section 5.2 Module OFF Timing updated Section 8.2 Module Placement and Component Spacing Guidelines GNSS antenna 2 noted as unused throughout document Section 4.5.6 RESET_N description updated	
2.8	08-Feb-22	Table 7-1 pretinning and height tolerance Section 5 Power Up/Down moved from section 4.5.4, & added uncontrolled power down Table 3-1 Model Names Matrix updated Table 4-7 Antenna Port Usage updated for clarity C-V2X function references nullified or removed throughout "LTE_ANT_#" changed to "ANT_#" in most instances Table 1-1 China variant removed Section 3.2 NAD label fields updated	
2.9	27-May-22	Table 4-7 n41 NSA and UHB PC2/SA antenna usage detail added, non-POR bands removed Table 1-1 NR SA/NSA detail removed, GNSS bands corrected Figure 8-3 Reflow Profile updated Figure 4-2 Unused antennas removed GNSS band L5 references removed throughout Figure 11-6 Power Supply capacitance rec'd change Figure 11-3 Updated layout image to reflect newer footprint Section 12.1 Thermal Mitigation updated Section 3.2 NAD label detail	
3.0	2022-10-19	Section 4.5.8 Corrected VUC2NAD_HOLDUP description ("rising edge") Table 4-8 PC2 and stand-alone config info added Table 1-3 band ranges table added Table 4-7 and Table 4-8 simplified, redundancies removed	
3.1	2022-11-11	Table 1-1 CN variant added Table 4-8 CN variant added, add'l SA band support Section 4.5.2 Separate Tables for TX config, RX config, PC2 per antenna	
3.2	2023-03-24	Table 4-27 External MIPI Antenna Interface detail Table 4-8 CN variant UL MIMO detail Table 1-1 Footnote 1 clarified to apply to SA operation only Section 10 Reliability info updated	
3.3	2023-04-18	Table 1-1 Separate SA/NSA columns in band table Table 1-2 MDB/MNO band mask tables	
3.4	2024-03-19	Section 1.2.1 GNSS air interface support Table 1-1 LTE B06 and B09 not supported, correction Section 10 AEC-Q104 Table 1-1 Band n8 SA on CN variant Table 4-9 Note 4 corrected to include UHB Table 6-1 Storage Temperature updated Section 1.4.1 GNSS L5 support only for CN Table 4-4 Clarified 3V3_PMX detail Table 1-1 Scrub of HW Band Tables Table 1-2 Scrub of MDB/MNO band mask tables 	

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Terms and Acronyms

• ANT_n	Cellular Antenna (may be shown as "LTE_ANT_n" in some instances)
• BB	Baseband
• CA	Carrier Aggregation
• DCM	Data Connectivity Module (also, "TCU")
• DRX	Discontinuous Reception
• DRx	Diversity (2 nd /3 rd /4 th) Receive
• EN-DC	E-UTRAN New Radio – Dual Connectivity (LTE and NR simultaneously)
• ES	Engineering Sample
• FDD	Frequency Division Duplex
• GLONASS	GLObalnaya NAVigatsionnaya Sputnikovaya Sistema
• GNSS	Global Navigation Satellite System
• GPIO	General Purpose Input Output
• GSM	Global System for Mobile (2G)
• HORXD	High Order RX Diversity (using 4 RX antennas)
• HSIC	High Speed Inter-Chip
• HB	Cellular High Band (~2 to 3GHz). See Table 1-3
• HU	Head Unit
• LB	Cellular Low Band (~<1GHz). See Table 1-3
• LTE	Long Term Evolution (4G)
• LTE_ANT	Cellular antennas (not only LTE, may also support 2G,3G & 5G)
• MB	Cellular Mid Band (~1 to 2GHz). See Table 1-3
• MDB	Modem Database Framework (for regional band restrictions)
• MNO	Mobile Network Operator
• MP	Mass Production
• NAD	Network Access Device
• NR	New Radio (5G)
• NSA	Non-Stand-Alone 5G operation (also EN-DC)
• OEM	Original Equipment Manufacturer
• PC2	Power Class 2 (also called "HPUE" on the NAD, nominally +26 dBm)
• PC3	Power Class 3 (standard 3G/4G/5G, nominally +23 dBm)
• PCB	Printed Circuit Board
• PCIe	Peripheral Component Interconnect Express
• PHY	Physical Layer
• PMIC	Power Management IC
• PRx	Primary Receive
• SA	Stand-Alone 5G operation
• SBAS	Satellite-Based Augmentation System
• SIM	Subscriber Identity Module
• SoC	System-On-a-Chip (refers to the Qualcomm SA515M IC)
• TCU	Telematics Control Unit (also, "DCM")
• TDD	Time Division Duplex
• TRx	Transmit and Receive
• TSP	Telematics Service Provider

- UHB Cellular Ultra-High Band (>3GHz). See Table 1-3
- UMTS Universal Mobile Telecommunication System (3G)
- WCDMA Wideband Code Division Multiple Access (3G)

5G NAD Module

1 5G NAD Module

The 5G NAD Module incorporates 5G New Radio technology with 4x4 MIMO Antenna technology. The NAD is part of a family of proprietary embedded 5G wireless modules designed by Continental Automotive Systems, Inc. The modules are intended to be integrated into Data Connectivity Modules (DCMs) or Head Units (HUs) designed and produced by Continental or by a 3rd party for use by automotive OEMs. DCMs will be installed into vehicles during the OEM's factory assembly process and will not be accessible without use of special tools. Primary use-cases are data-centric with data and voice connections to Telematics Service Providers (TSP).

1.1 Hardware Revision Change Tracking

Any relevant changes or deviations will appear in the HW Release Notes.

1.2 Key Features

1.2.1 Air Interface Support

- 5G NR: 3GPP Release 15
- Sub-6 GHz 5G NR bands, including n77, n78, n79 (depending on variant)
- LTE FDD/TDD: 3GPP Rel. 15 Category 19
- LTE UL CA 2 CC (intra-band)
- UMTS: HSUPA CAT6, HSDPA CAT24 3GPP Rel. 9
- GSM: EGPRS Rel-12
- VoLTE – HD Voice
- Embedded Qualcomm GNSS Sub-system
- GNSS Bands: GPS-L1,L5; GLONASS-L1; Galileo-E1,E5a; Beidou-B1 (depending on variant)
- SBAS supported: EGNOS/MSAS/QZSS/WAAS/GAGAN

1.2.2 Processing and Memory Support

• Apps Processor	• Cortex-A7 – 1.5 GHz • ARMv7 Instruction Set • 32-bit architecture • 32 KB L1 I/D Cache • 256 KB L2 Cache
• Modem Processor	• Hexagon QDSP6 – 1.5 GHz • 2MB L2 Cache • Low-Power Audio Post-Processing supported in the modem system
• AOSS (Always On Sub-System) • RPM	• Cortex M3 – 300 MHz • MPM is the only master • MPM coordinates shutdown/wakeup, clock rates, and VDDs
• Internal Memory	• 176 KB Boot ROM • 24 KB IMEM
• Module RAM	• 1x16 LPDDR4x – 1.8 GHz • 8 Gbit (1024 MBytes)
• Module NAND	• 8 Gbit (1024 MBytes) • x8 SLC

1.2.3 Electrical Interface Support

• Antenna Ports	• MIMO: ANT1 - ANT4 • GNSS: ANT7 (Internal diplexer for multiple bands)
• PCIe	• 2x Lane (Gen3) • End Point and Root Complex support

Ethernet	- RGMII Integrated MAC 4 Rx bits 4 Tx bits 1 Gbps
USB	1 port - Supports USB3.1 or USB 2.0
QLINK	- QLINK 2.0 4 lanes - DL0, UDL0, UDL1, UL0
I2S	Up to two ports
UIM	2 available UIM1/UIM2 1.8V/3V support
BLSP Interfaces	4 BLSP Ports (I2C, SPI, UART) 4 bits each; - Muxed serial interfaces - UART: 4 Mbps - I2C: Yes - SPI: Yes (Master)
GPIOs	Up to 29 free GPIO
ADC	2 ADC ports

Note: The NAD supports interfaces to WLAN or Bluetooth ICs, but the NAD itself does not contain WiFi or Bluetooth ICs.

1.3 Package

- 725-pin LGA module of size 52 x 52 x 3.3 mm
- Refer to Figure 4-1 for ball assignment
- Refer to Table 4-4 for overall pin description
- Refer to Section 7 for mechanical specifications.

1.4 Band Support

1.4.1 RF Bands

Hardware Band support for each variant is shown in **Table 1-1**. MDBs (regional operation) and Mobile Network Operators will restrict usable bands to **Table 1-2**. Roaming partners may further restrict band capability.

For a more comprehensive description of the supported Hardware configurations and antenna usage breakdown for each band, see **Table 4-8** (SA/NSA/Antennas), **Table 4-9** (RX Config), or **Table 4-10** (Power Class). These additional tables do **not** take into account MDB and MNO band restrictions.

Table 1-1 Bands Available in Hardware (subject to Band Masking shown in **Table 1-2**)

Model	Region	NR (5G) Bands NSA	NR (5G) Bands SA	LTE (4G) Bands	UMTS (3G) Bands	GSM (2G) Bands	GNSS
FE5CN0020	China	n1, n3, n5, n8, n41, n77, n78, n79	n1, n3, n5, n8, n28, n41, n77, n78, n79	1, 3, 5, 7, 8, 34, 38, 39, 40, 41	1, 3, 8	3, 8	L1, L5
FE5EU0020	EU	n1, n3, n5, n7, n8, n28, n77, n78	n1, n28, n77, n78	1, 3, 5, 7, 8, 20, 28A, 28B, 32Rx, 38, 40	1, 3, 5, 8	3, 8	L1, L5 ³
FE5NA0020	NA	n2, n5, n7, n41, n66, n71, n77, n78	n41, n66, n71, n77, n78	1, 2, 3, 4, 5, 7, 12, 13, 14, 17, 25, 26, 29Rx, 30Rx, 41, 66, 71	1, 2, 3, 4, 5	2, 3, 5, 8	L1, L5 ³
FE5RW0020	RoW	<u>n3</u> , <u>n28</u> , <u>n77</u> , <u>n78</u> , <u>n79</u>	<u>n28</u> , <u>n41</u> , <u>n77</u> , <u>n78</u> , <u>n79</u>	<u>1</u> , <u>2</u> , <u>3</u> , <u>4</u> , <u>5</u> , <u>7</u> , <u>8</u> , <u>18</u> , <u>19</u> , <u>20</u> , <u>25</u> , <u>26</u> , <u>28A</u> , <u>28B</u> , 32Rx, 38, <u>39</u> , <u>40</u> , <u>41</u>	<u>1</u> , <u>3</u> , <u>5</u> , <u>6</u> , <u>8</u> , <u>9</u> , <u>19</u>	2, 3, 5, 8	L1, L5 ³

- (1) Qualcomm POR (plan-of-record) for 5G **Stand-Alone** band support at 2022 SOP: n41, n66, n71, n78, n79. Any 5G bands not in POR may be factory calibrated and tested, but not fully validated and certified.
- (2) RoW: RoW variant with all supported Row 4G/5G bands populated/tested, only Japan bands (underlined) are certified
- (3) Hardware exists to support L5 on all variants, but feature is unlicensed and therefore disabled in SW on all except CN

Table 1-2 Band Support based on Region and MNO

Model	MNO	NR (5G) Bands NSA	NR (5G) Bands SA	LTE (4G) Bands	UMTS (3G) Bands	GSM (2G) Bands	GNSS
FE5CN0020	China Unicom (China)	n78	n1, <u>n3, n5, n8, n28⁴</u> , n78	1, 3, 5, 8	1, 8	3, 8	L1, L5
FE5EU0020	Voda- phone (Germany)	n1, n3, n5, n7, n8, n28, n77, n78	n1, n28, n78	1, 3, 5, 7, 8, 20, 28A, 28B, 32Rx, 38, 40	1, 3, 5, 8	3, 8	L1 ³
FE5EU0020	Telstra (Australia)	n5, n7, n28, n78	---	1, 3, 5, 7, 8, 20, 28A, 28B, 32Rx, 38, 40	1, 3, 5, 8	3, 8	L1 ³
FE5NA0020	Verizon (U.S.)	n2, n5, n41, n66, n71, n77	n41, n71	2, 4, 5, 12, 13, 14, 17, 25, 26, 29Rx, 30Rx, 41, 66, 71	1, 2, 3, 4, 5	2, 3, 5, 8	L1 ³
FE5RW0020	Cubic (Japan)	n77, n78, n79	---	1, 3, 8, 18, 19, 26, 28A, 28B, 41	1, 3, 5, 6, 8, 9, 19	2, 3, 5, 8	L1 ³

(1) The band masking is based on software LE2.0 / R00180.1 and OTP_VERSION=otp-sa515m-le-2-1-3.1.16.57/SA515M.LE.2.1-00160-NBOOT.NEFS.PROD-1.542633.1.568271.2

(2) The above table represents MNO configs as of 2023-04-12. Subject to change with Software updates or changes in MNO requirements/restrictions.

(3) Hardware exists to support L5 on all variants, but feature is unlicensed and therefore disabled in SW on all except CN

(4) Underlined bands, support planned for SOP2 only

1.4.2 RF Band Ranges Defined

Within this document, references may be made to cellular band groups as “Low”, “Mid”, “High” or “Ultra-High”. **Table 1-3** defines which bands are included in these groupings.

Table 1-3 Cellular Band Ranges

Range	Cellular Bands in Range
Low Bands (LB)	5,6,8,12,13,14,17,18,19,20,26,28A,28B,29Rx,71,GSM850,GSM900 n5,n8,n28,n71
Mid Bands (MB)	1,2,3,4,9,25,32Rx,39,66,GSM1800,GSM1900 n1,n2,n3,n66
High Bands (HB)	7,30Rx,38,40,41 n7,n41
Ultra-High Bands (UHB)	n77, n78, n79

1.4.3 5G NAD Band Combinations (EN-DC and CA)

Where the required bands are supported in the hardware of a given variant, the 5G NAD will support EN-DC and CA operation as agreed between customer, mobile network operator, and Continental. EN-DC and CA support is variant-specific and may be subject to frequent updates based on cellular providers and other factors. The details of such support are made available in a separate Continental document which has been shared with the customer.

It should be noted that while the Hardware may be capable of supporting the stated Band Combinations (i.e. RF Configuration and relevant RF path calibration are present in the device), the delivered Software may not support such operation without an update.

1.5 Recommended NAD Interfaces

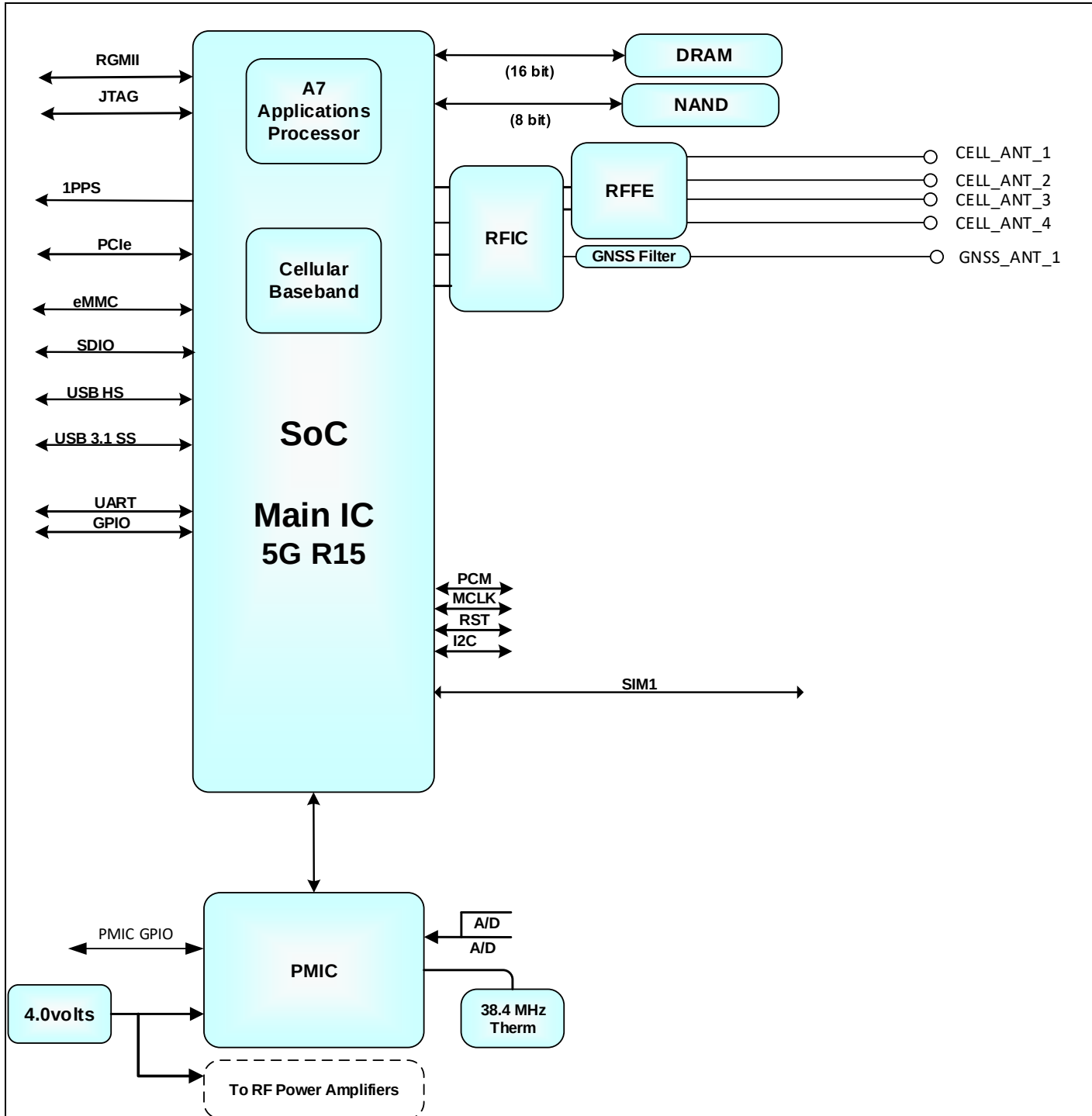
Integrators are strongly recommended to provide access to the following NAD communication ports to be used for debugging, certification, or other developmental activity.

- HS-USB 2.0
- 2-wire UART
- JTAG
- **RF Ports:** If any on-board antennas are used by the product, provisions should be made to support conducted RF measurements on all antenna interfaces
- **SIM Interface:** Electrical performance of the SIM interface is always evaluated during certification testing of the final product. Product teams should insure that the SIM interface can be accessed for testing without degrading its integrity.

2 Block Diagrams for 5G NAD

2.1 System Level Block Diagram for 5G NAD

Figure 2-1 5G NAD - System Block Diagram



3 Part Ordering Information

3.1 Part Numbers

The Conti HW Part Number should be referenced when requesting NAD modules via purchase orders.

Table 3-1: Model Names Matrix

Region	NAD Model	H/W Part Number
North America	FE5NA0020	TRC00168100
EU	FE5EU0020	TRC00111400
RoW	FE5RW0020	TRC00111100
China	FE5CN0020	TRC0804600000

NOTE: Check with Continental Automotive Sales team for availability of model numbers shown in [Table 3-1](#).

3.2 NAD Module Label

Figure 3-1 and **Figure 3-2** depict label dimensions and intended placement of the various NAD identifiers, plus regulatory and certification artwork that may apply to a specific variant. The content shown in each figure is a superset, and a given NAD may only contain a subset of the items displayed. Details of the content of each QR code are in **Table 3-2** and **Table 3-3**.

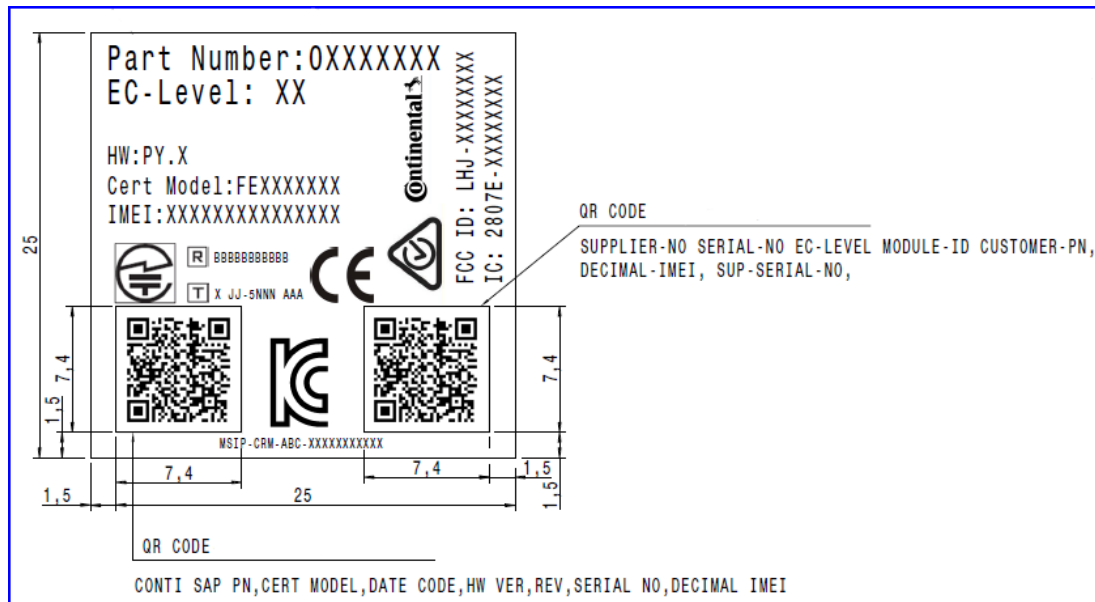


Figure 3-1: NAD Label Example (EU w/regulatory icon superset)

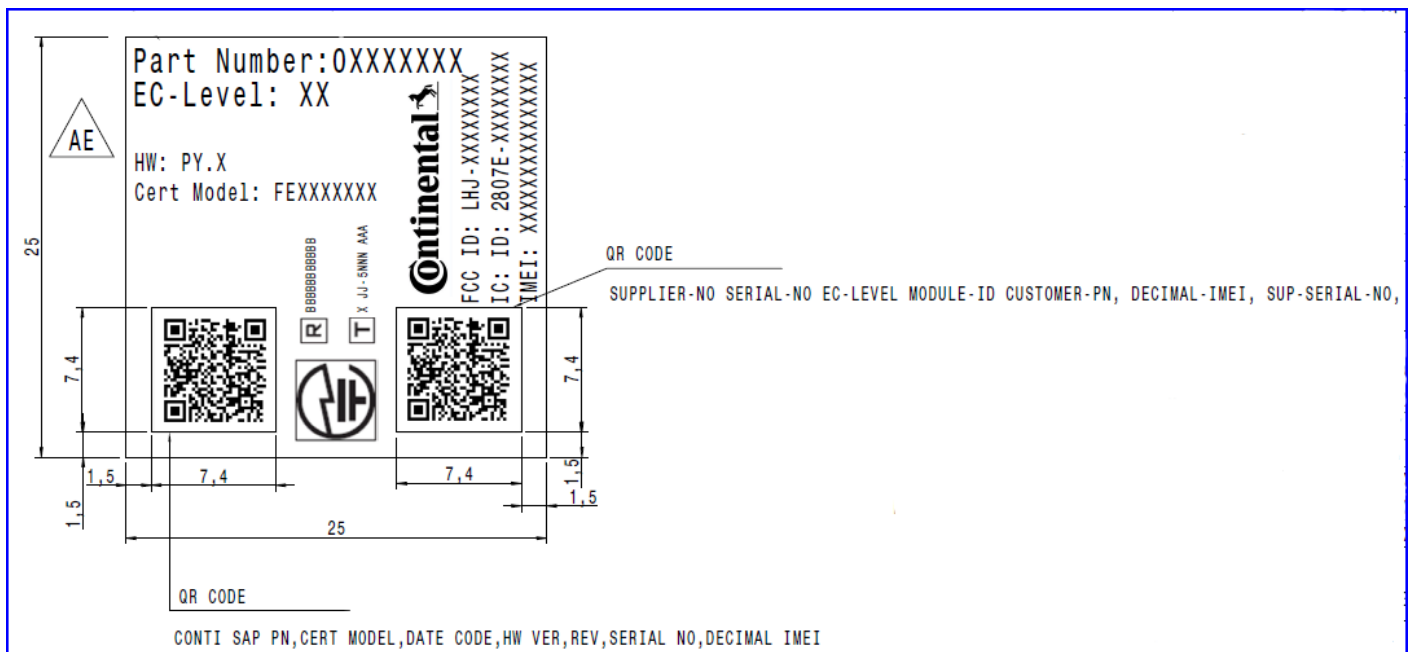


Figure 3-2 NAD Label Example (NA & RW w/regulatory icon superset)

Type Name	Characters	Content
Supplier No.	3	620
Serial No.	9	XXXXXXXXXX
EC-Level	2	XX
Module ID	2	01
Customer PN	8	0XXXXXXXX
Decimal IMEI	19	IMEIXXXXXXXXXXXXXXXXXX
Supplier Serial No	22	LICHRG#XXXXXXXXXXXXXXXXXX

Table 3-2 Harman QR Code Content (right side of label)

Type Name	Characters	Content
Conti SAP PN	11	A2CXXXXXXXXXX
CERT Model	9	FXXXXXXXXXX
Date Code	6	YYMMDD
HW Version	4	PY.X
Rev	1	X (0-F) Hexadecimal
Serial No.	15	XXXXXXXXXXXXXXXXXX
Decimal IMEI	15	XXXXXXXXXXXXXXXXXX

Table 3-3 Conti QR Code Content (left side of label)

4 Pin Assignments and Descriptions

4.1 Pin Number Assignment Grid Array

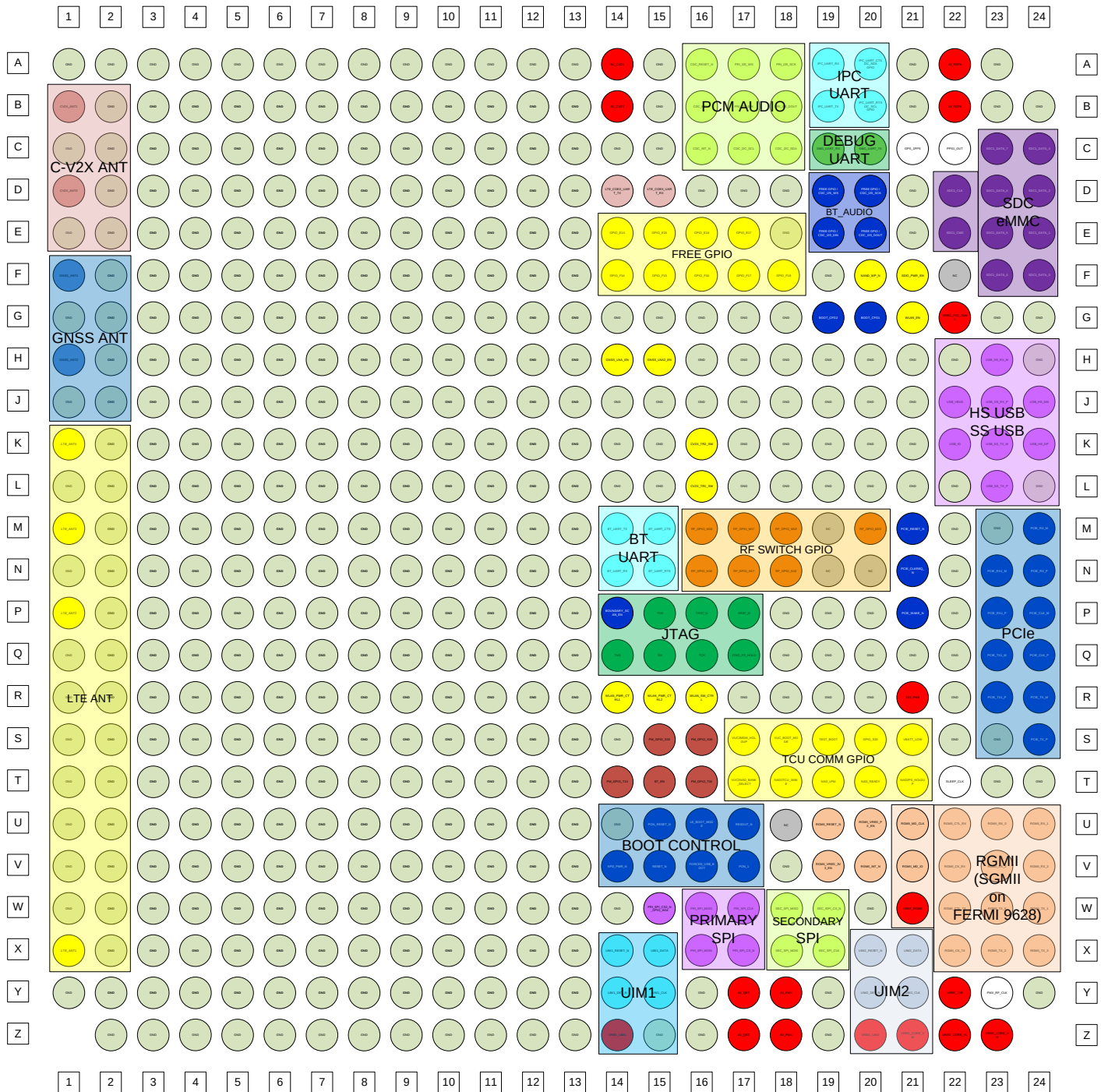


Figure 4-1: Pin number assignments (top view)

4.2 Pin Type Definition

Table 4-1: Pin Type Definition

Symbol	Description
Pad attribute	
AI	Analog input (does not include pad circuitry)
AO	Analog output (does not include pad circuitry)
B	Bidirectional digital with CMOS input
DI	Digital input (CMOS)
DO	Digital output (CMOS)
W	Wakeup interrupt during deep-sleep mode
H	High-voltage tolerant
R	Slew-rate limited
S	Schmitt trigger input
Z	High-impedance (high-Z) output
PSO	Power supply, output
PSI	Power supply, input
Pad pull details for digital I/O	
nppdpukp	Programmable pull resistor. The default pull direction is indicated using capital letters and is a prefix to other programmable options. NP: pdpukp = default no-pull with programmable options following the colon (:) PD: nppukp = default pull-down with programmable options following the colon (:) PU: nppdkp = default pull-up with programmable options following the colon (:) KP: nppdpu = default keeper with programmable options following the colon (:)
KP	Contains an internal weak keeper device (keepers cannot drive external buses)
NP	Contains no internal pull
PU	Contains an internal pull-up device
PD	Contains an internal pull-down device

4.3 Pad Voltage Groupings for Baseband Circuits

Table 4-2: Pad Voltage Groupings for Baseband Circuits

Main IC	Functions	Voltage(s)
P2	SDC1 – SDIO & eMMC Interface System	1.8V or 2.85V (1)
P3	Most Peripherals – Generic GPIOs and Functional Pins, Shared Power with P7	1.8V
P4	UIM1 – Primary UIM Interface for SIM cards	1.8V or 2.95V
P5	UIM2 – Secondary UIM Interface for SIM cards	1.8V or 2.95V
P8	RGMII - Ethernet	1.8V (2)
P12	Peripherals – Generic GPIOs and Functional Pins, Shared Power with P3	1.8V
P71	Peripherals – Generic GPIOs and Functional Pins, Shared Power with P3	1.8V
P72	Peripherals – Generic GPIOs and Functional Pins, Shared Power with P3	1.8V
P73	Peripherals – Generic GPIOs and Functional Pins, Shared Power with P3	1.8V
P74	Peripherals – Generic GPIOs and Functional Pins, Shared Power with P3	1.8V

- (1) P2 pad voltage is supplied by external LDO, VREG_PX2_DUAL, which should be pulled to GND through a 1Kohm resistor if the SDC circuit is not used.
- (2) P8 pad voltage is supplied by external LDO, VREF_RGMII

PM IC	Functions	Voltage(s)
LV	Low voltage input variant	1.2V/1.8V
MV	Medium voltage input variant	

The range of expected voltages on Hi and Lo level output signals, as well as the acceptable Hi and Lo level voltages on input signals, is defined in [Table 4-3](#).

Table 4-3 Pad Voltage Input/Output levels

Pad Type	parameter	Min	Nom	Max	unit
GPIO 1.8V	V_{dd}	1.65	1.80	1.95	V
	V_{ih}	$V_{dd} \times 0.65$	---	$V_{dd} + 0.3$	V
	V_{il}	-0.3	---	$V_{dd} \times 0.35$	V
	V_{oh}	$V_{dd} - 0.45$	---	Vdd	V
	V_{ol}	0.0	---	0.45	V
SDC 3V	V_{dd}	2.70	2.95	3.05	V
	V_{ih}	$V_{dd} \times 0.625$	---	$V_{dd} + 0.3$	V
	V_{il}	-0.3	---	$V_{dd} \times 0.25$	V
	V_{oh}	$V_{dd} \times 0.75$	---	Vdd	V
	V_{ol}	0.0	---	$V_{dd} \times 0.125$	V
SDC 1.8V	V_{dd}	1.65	1.80	1.95	V
	V_{ih}	1.27	---	2	V
	V_{il}	-0.3	---	0.58	V
	V_{oh}	1.4	---	---	V
	V_{ol}	---	---	0.45	V
UIM 3V	V_{dd}	2.70	2.95	3.05	V
	V_{ih}	$V_{dd} \times 0.7$	---	$V_{dd} + 0.3$	V
	V_{il}	-0.3	---	$V_{dd} \times 0.2$	V
	V_{oh}	$V_{dd} \times 0.8$	---	Vdd	V
	V_{ol}	0.0	---	0.4	V
UIM 1.8V	V_{dd}	1.65	1.80	1.95	V
	V_{ih}	$V_{dd} \times 0.7$	---	$V_{dd} + 0.3$	V
	V_{il}	-0.3	---	$V_{dd} \times 0.2$	V
	V_{oh}	$V_{dd} \times 0.8$	---	Vdd	V
	V_{ol}	0.0	---	0.4	V
RGMII 1.8V	V_{dd}	1.65	1.80	1.95	V
	V_{ih}	$V_{dd} \times 0.65$	---	---	V
	V_{il}	---	---	$V_{dd} \times 0.35$	V
	V_{oh}	$V_{dd} - 0.45$	---	---	V
	V_{ol}	---	---	0.45	V

4.4 Module Pinout and Internal Connections – Power, Hardware Functions, GPIO and ADC Assignment

Table 4-4 5G NAD Hardware Functions and GPIO Assignment

Pad	Functional Group	LGA Pad Name	LGA Signal Description	Boot Config	IC GPIO Pin ¹	Wakeable Interrupts	Voltage Pad Type	Pad type
B1	Antennas	CV2X_ANT_1	C-V2X Antenna 1 (not used)					
D1	Antennas	CV2X_ANT_2	C-V2X Antenna 2 (not used)					
F1	Antennas	GNSS_ANT_1	GNSS Antenna 1					
H1	Antennas	GNSS_ANT_2	GNSS Antenna 2 (not used)					
X1	Antennas	LTE_ANT_1	Cellular Antenna 1 (supports 2G, 3G, 4G & 5G)					
P1	Antennas	LTE_ANT_2	Cellular Antenna 2 (supports 2G, 3G, 4G & 5G)					
M1	Antennas	LTE_ANT_3	Cellular Antenna 3 (supports 2G, 3G, 4G & 5G)					
K1	Antennas	LTE_ANT_4	Cellular Antenna 4 (supports 2G, 3G, 4G & 5G)					
G20	BOOT	BOOT_CFG1	Boot_Config_1 and 2 are used for boot device selection (leave unconnected)	BOOT_CONFIG[1]	GPIO_58		P3	B-PD:nppukp B DI
G19	BOOT	BOOT_CFG2	Boot_Config_1 and 2 are used for boot device selection (leave unconnected)	BOOT_CONFIG[2]	GPIO_24	▶	P3	
P14	BOOT	BOUNDARY_SCAN_EN	Pulled to VREF_1V8 during power-up to enable Boundary Scan, otherwise left floating to boot normally					
V16	BOOT	FORCED_USB_BOOT	Do Not Connect		GPIO_42		P3	B-PD:nppukp
V14	BOOT	PM_KPD_PWR_N	Pulled low for ~300ms to power the module on, otherwise left floating. Do not pull high; Internal pull-up exists.					PU
V15	BOOT	RESET_N	Pulled low for ~300ms to reset the module, otherwise left floating. Do not pull high; Internal pull-up exists.					PU
V17	BOOT	PON_1	NAD Power On pin – leave no connect (unused option)					
U16	BOOT	SDX_LE_BOOT_MODE	NAD Test Mode		GPIO_61	▶	P72	B-PD:nppukp
U15	BOOT	SDX_PON_RESET_N	Reset signal for Modem IC – <u>Not</u> the module RESET- leave unconnected					
U17	BOOT	SDX_RESOUT_N	Output that Indicates when SDX is in reset				P3	O
T15	BT	BT_EN	Also configurable as ADC if WIFI is not needed		PM_GPIO_06		MV	Default in-10uA PD
E19	BT_AUDIO	BT_I2S_DIN/GPIO_E19	Bluetooth I ² S Data in / alternate as GPIO		GPIO_17	▶	P12	
E20	BT_AUDIO	BT_I2S_DOUT/GPIO_E20	Bluetooth I ² S Data out / alternate as GPIO		GPIO_18	▶	P12	
D20	BT_AUDIO	BT_I2S_SCK/GPIO_D20	Bluetooth I ² S clock / alternate as GPIO		GPIO_19	▶	P12	
D19	BT_AUDIO	BT_I2S_WS/GPIO_D19	Bluetooth I ² S WS / alternate as GPIO		GPIO_16	▶	P12	
M15	BT_UART	BT_UART_CTS/GPIO_M15	Bluetooth UART CTS / alternate as GPIO		GPIO_65	▶	P3	B-PD:nppukp
N15	BT_UART	BT_UART_RTS/GPIO_N15	Bluetooth UART RTS / alternate as GPIO		GPIO_66		P3	B-PD:nppukp
N14	BT_UART	BT_UART_RX/GPIO_N14	Bluetooth UART RX / alternate as GPIO		GPIO_64	▶	P3	B-PD:nppukp
M14	BT_UART	BT_UART_TX/GPIO_M14	Bluetooth UART TX / alternate as GPIO	BOOT_CONFIG[7]	GPIO_63		P3	B-PD:nppukp
D15	BT-WLAN	LTE_COEX_UART_RX	Bluetooth, Wifi, LTE Co-existence UART RX		GPIO_45	▶	P73	B-PD:nppukp DI
D14	BT-WLAN	LTE_COEX_UART_TX	Bluetooth, Wifi, LTE Co-existence UART TX		GPIO_44	▶	P73	B-PD:nppukp DO
G21	BT-WLAN	WLAN_EN	WLAN enable signal		GPIO_52	▶	P72	B-PD:nppukp
R14	BT-WLAN	WLAN_PWR_CTRL1	Bluetooth / WLAN Power Control signal 1		PM_GPIO_02		LV	Default in-10uA PD
R15	BT-WLAN	WLAN_PWR_CTRL2	Bluetooth / WLAN power control signal 2		PM_GPIO_01		MV	Default in-10uA PD
R16	BT-WLAN	WLAN_SW_CTRL	Bluetooth / WLAN Switch control line		PM_GPIO_10		LV	Default in-10uA PD
Y23	BT-WLAN / CLOCKS	PMX_RF_CLK	RF Clock output from PMX, optionally for BT-WLAN					
T22	CLOCKS	SDX_SLEEP_CLK	Sleep clock				P3	I
L16	C-V2X	CV2X_TR1_SW	Compensator enable C-V2X PA active signal (not used)		W_GRFC_3		P3	
K16	C-V2X	CV2X_TR2_SW	Compensator enable C-V2X PA active signal (not used)		W_GRFC_19		P3	
C19	DEBUG	DBG_UART_RX	Debug UART RX		GPIO_09	▶	P71	
C20	DEBUG	DBG_UART_TX	Debug UART TX		GPIO_08		P71	
E14	FREE GPIO	GPIO_E14	Free GPIO		GPIO_87	▶	P3	B-PD:nppukp
E16	FREE GPIO	GPIO_E16	Free GPIO		GPIO_104	▶	P3	PD:nppukp
E17	FREE GPIO	GPIO_E17	Free GPIO		GPIO_105		P3	PD:nppukp

Pad	Functional Group	LGA Pad Name	LGA Signal Description	Boot Config	IC GPIO Pin ¹	Wakeable Interrupts	Voltage Pad Type	Pad type
F14	FREE GPIO	GPIO_F14	Free GPIO	BOOT_CONFIG[0]	GPIO_75		P73	B-PD:nppukp
F15	FREE GPIO	GPIO_F15	Free GPIO		GPIO_107		P3	PD:nppukp
F16	FREE GPIO	GPIO_F16	Free GPIO		GPIO_78	►	P74	B-PD:nppukp
F17	FREE GPIO	GPIO_F17	Free GPIO		GPIO_79	►	P3	B-PD:nppukp
F18	FREE GPIO	GPIO_F18	Free GPIO	BOOT_CONFIG[11]	GPIO_31		P73	B-PD:nppukp
S20	RESERVED	RESERVED	Do Not Connect		GPIO_33		P12	B-PD:nppukp
S15	FREE GPIO	PM_GPIO_S15	Free PMIC GPIO		PM_GPIO_03		MV	Default in – Hi-Z
T14	FREE GPIO	PM_GPIO_T14	Free PMIC GPIO		PM_GPIO_04		LV	Default in-10uA PD
T16	FREE GPIO	PM_GPIO_T16	Free PMIC GPIO		PM_GPIO_08		LV	Default in-10uA PD
S16	FREE GPIO/ADC	PM_GPIO_S16	Free PMIC GPIO / Configurable as ADC to do antenna diagnostics		PM_GPIO_11		MV	Default in-10uA PD
H14	GNSS / DR	GNSS_LNA_EN	enables external LNA for GNSS L1		W_GRFC_0			
H15	GNSS / DR	GNSS_LNA2_EN	enables external LNA2 for GNSS L5 (L5 is not used)		W_GRFC_1			
E15	Ethernet Audio Video Bridge	GPIO_E15 / 1PPS_OUT_0	Generated from internal EMAC and are used for EAVB functionality or free GPIO		GPIO_106		P3	PD:nppukp DO
C21	GNSS / DR	GNSS_1PPS	PPS clock output from internal GNSS engine. A PPS clock in sync with GPS time that is used for Dead Reckoning.		GPIO_32	►	P73	B-PD:nppukp
C22	Ethernet Audio Video Bridge	GPIO_C22 / 1PPS_OUT_1	Generated from internal EMAC and are used for EAVB functionality or free GPIO		GPIO_95	►	P3	B-PD:nppukp
J24	HS_USB	USB_HS_DM	High Speed USB data minus				P3	AIO
K24	HS_USB	USB_HS_DP	High Speed USB data plus				P3	AIO
K22	HS_USB	USB_ID	USB ID		PM_GPIO_05		MV	Default in-10uA PD
J22	HS_USB	USB_VBUS_DET	USB Vbus Detect					
A20	IPC_UART	IPC_UART_CTS	If this UART is not needed, these can only be GPIO		GPIO_22	►	P12	
B20	IPC_UART	IPC_UART_RTS	If this UART is not needed, these can only be GPIO		GPIO_23	►	P12	
A19	IPC_UART	IPC_UART_RX	If this UART is not needed, these can only be GPIO		GPIO_21	►	P12	
B19	IPC_UART	IPC_UART_TX	If this UART is not needed, these can only be GPIO		GPIO_20		P12	
Q17	JTAG	JTAG_PS_HOLD	Driven by JTAG to keep processor powered up				P3	Z
P17	JTAG	JTAG_SRST_N	JTAG SRST				P3	I
Q16	JTAG	JTAG_TCK	JTAG TCK				P3	I
Q15	JTAG	JTAG_TDI	JTAG TDI				P3	I
P15	JTAG	JTAG_TDO	JTAG TDO				P3	Z
Q14	JTAG	JTAG_TMS	JTAG TMS				P3	B
P16	JTAG	JTAG_TRST_N	JTAG TRST				P3	I
P24	PCle	PCIE_CLK_M	PCle differential clock minus				P3	AIO
Q24	PCle	PCIE_CLK_P	PCle differential clock plus				P3	AIO
N21	PCle	PCIE_CLKREQ_N	PCle clock request (active low)		GPIO_56	►	P72	B-PU:nppukp DO
M21	PCle	PCIE_RESET_N	PCle reset (active low)		GPIO_57	►	P72	B-PD:nppukp DO
M24	PCle	PCIE_RX0_M	PCle differential receive channel 0 minus				P3	AIO
N24	PCle	PCIE_RX0_P	PCle differential receive channel 0 plus				P3	AIO
N23	PCle	PCIE_RX1_M	PCle differential receive channel 1 minus				P3	AIO
P23	PCle	PCIE_RX1_P	PCle differential receive channel 1 plus				P3	AIO
R24	PCle	PCIE_TX0_M	PCle differential transmit channel 0 minus				P3	AIO
S24	PCle	PCIE_TX0_P	PCle differential transmit channel 0 plus				P3	AIO
Q23	PCle	PCIE_TX1_M	PCle differential transmit channel 1 minus				P3	AIO
R23	PCle	PCIE_TX1_P	PCle differential transmit channel 1 plus				P3	AIO
P21	PCle	PCIE_WAKE_N	PCle wake (active low)		GPIO_53	►	P72	B-PU:nppukp B DO DO
R21	POWER IN	3V3_PMX	Apply 4V (3.3 to 4.2V) to this input to be subregulated down to 1.8V within the NAD					

Pad	Functional Group	LGA Pad Name	LGA Signal Description	Boot Config	IC GPIO Pin ¹	Wakeable Interrupts	Voltage Pad Type	Pad type
A22	POWER IN	4V_DSDA	for 5G NAD DSDA 4V input					
B22	POWER IN	4V_DSDA	for 5G NAD DSDA 4V input					
Y18	POWER IN	4V_PMX	4V in for PMX					
Z18	POWER IN	4V_PMX	4V in for PMX					
Y17	POWER IN	4V_QET	4V in for QET					
Z17	POWER IN	4V_QET	4V in for QET					
A14	POWER IN	5V_CV2X	5V in for C-V2X PA - 2A (not used)					
B14	POWER IN	5V_CV2X	5V in for C-V2X PA - 2A (not used)					
Y22	POWER OUT	VREF_1V8	1.8V output supply reference, also used to turn on eMMC supply					
U18	RESERVED	RESERVED	Do Not Connect					
F22	NO CONNECT	No Connect	NO CONNECT					
G22	POWER IN	VREG_PX2_DUAL	External LDO input 1.8V/2.85V for SDC					
W21	POWER IN - RGMII	VREF_RGMII	External LDO input 1.8V					
Z23	POWER OUT-BT-WLAN	VREG_CORE_VH	VDD_CORE_VH (QCA6696) output					
Z22	POWER OUT-BT-WLAN	VREG_CORE_VL	VDD_CORE_VL(0.95) output					
Z21	POWER OUT-BT-WLAN	VREG_CORE_VM	VDD_CORE_VM (1.35V) output					
C17	PRI_I2S	CDC_I2C_SCL	Codec I ² C clock line		GPIO_11	►	P3	
C18	PRI_I2S	CDC_I2C_SDA	Codec I ² C data line		GPIO_10	►	P3	
C16	PRI_I2S	CDC_INT1_N	Codec Interrupt (active low)		GPIO_103	►	P3	B-PD:nppukp
A16	PRI_I2S	CDC_RESET_N	Codec Reset (active low)		GPIO_92		P3	B-PD:nppukp
B17	PRI_I2S	PRI_I2S_DIN	Primary I ² S Data in		GPIO_13	►	P12	
B18	PRI_I2S	PRI_I2S_DOUT	Primary I ² S Data out		GPIO_14	►	P12	
B16	PRI_I2S	PRI_I2S_MCLK	Primary I ² S Master clock		GPIO_62		P71	B-PD:nppukp
A18	PRI_I2S	PRI_I2S_SCK	Primary I ² S bit clock		GPIO_15	►	P12	
A17	PRI_I2S	PRI_I2S_WS	Primary I ² S word select		GPIO_12	►	P12	
M16	RF_SW_CONTROL	RF_GPIO_M16	RF control – GRFC1 or RFFE0_DATA		GPIO_36		P12	B-PD:nppukp B DO
M17	RF_SW_CONTROL	RF_GPIO_M17	RF control – GRFC3 or RFFE1_DATA		GPIO_38		P12	B-PD:nppukp B DO
M18	RF_SW_CONTROL	RF_GPIO_M18	RF control – GRFC5 or RFFE2_DATA		GPIO_40		P12	B-PD:nppukp B DO
M19	NO CONNECT	No Connect	NO CONNECT				[see note (3)]	
M20	NO CONNECT	No Connect	NO CONNECT				[see note (3)]	
N16	RF_SW_CONTROL	RF_GPIO_N16	RF control – GRFC0 or RFFE0_CLK		GPIO_37		P12	B-PD:nppukp Z DO
N17	RF_SW_CONTROL	RF_GPIO_N17	RF control – GRFC2 or RFFE1_CLK		GPIO_39		P12	B-PD:nppukp Z DO
N18	RF_SW_CONTROL	RF_GPIO_N18	RF control – GRFC4 or RFFE2_CLK		GPIO_41		P12	B-PD:nppukp Z DO
N19	NO CONNECT	No Connect	NO CONNECT				[see note (3)]	
N20	NO CONNECT	No Connect	NO CONNECT				[see note (3)]	
V22	RGMII	RGMII_CK_RX	RGMII Receive clock				P8	I
X22	RGMII	RGMII_CK_TX	RGMII Transmit clock				P8	B

Pad	Functional Group	LGA Pad Name	LGA Signal Description	Boot Config	IC GPIO Pin ¹	Wakeable Interrupts	Voltage Pad Type	Pad type
U22	RGMII	RGMII_CTL_RX	RGMII data received / data error				P8	I
W22	RGMII	RGMII_CTL_TX	RGMII transmitter enable / transmitter error				P8	O
V20	RGMII	RGMII_INT_N	RGMII Interrupt (active low)		GPIO_90	►	P3	B-PD:nppukp
U21	RGMII	RGMII_MD_CLK	RGMII management interface clock				P8	O
V21	RGMII	RGMII_MD_IO	RGMII management interface I/O				P8	B
U19	RGMII	RGMII_RESET_N	RGMII reset (active low)		GPIO_91		P3	B-PD:nppukp
U23	RGMII	RGMII_RX_0	RGMII receive data[0]				P8	I
U24	RGMII	RGMII_RX_1	RGMII receive data[1]				P8	I
V23	RGMII	RGMII_RX_2	RGMII receive data[2]				P8	I
V24	RGMII	RGMII_RX_3	RGMII receive data[3]				P8	I
W23	RGMII	RGMII_TX_0	RGMII transmit data[0]				P8	O
W24	RGMII	RGMII_TX_1	RGMII transmit data[1]				P8	O
X23	RGMII	RGMII_TX_2	RGMII transmit data[2]				P8	O
X24	RGMII	RGMII_TX_3	RGMII transmit data[3]				P8	O
V19	RGMII	RGMII_VREG_3V3_EN	RGMII 3.3V regulator enable		GPIO_102		P3	B-PD:nppukp
U20	RGMII	RGMII_VREG_PX_EN	RGMII regulator enable		GPIO_89	►	P3	B-PD:nppukp
D22	SDIO/eMMC	SDC_CLK	SD Card / eMMC clock				P2	B
E22	SDIO/eMMC	SDC_CMD	SD Card / eMMC command				P2	B
F24	SDIO/eMMC	SDC_DATA_0	SD Card / eMMC data[0]				P2	B
E24	SDIO/eMMC	SDC_DATA_1	SD Card / eMMC data[1]				P2	B
D24	SDIO/eMMC	SDC_DATA_2	SD Card / eMMC data[2]				P2	B
C24	SDIO/eMMC	SDC_DATA_3	SD Card / eMMC data[3]				P2	B
F23	SDIO/eMMC	SDC_DATA_4 / SDC1_PWR_EN	eMMC data[4]		GPIO_98		P3	B-PD:nppukp
E23	SDIO/eMMC	SDC_DATA_5	eMMC data[5]		GPIO_99	►	P3	B-PD:nppukp
D23	SDIO/eMMC	SDC_DATA_6 / SDC1_VSE	eMMC data[6]		GPIO_100	►	P3	B-PD:nppukp
C23	SDIO/eMMC	SDC_DATA_7	eMMC data[7]		GPIO_101		P3	B-PD:nppukp
F21	SDIO/eMMC	SDIO_PWR_EN	SD card power enable		GPIO_88	►	P3	B-PD:nppukp
W17	SPI	SPI_CLK	Primary SPI Clock		GPIO_07		P3	
X17	SPI	SPI_CS_N	Primary SPI Chipselect 1 (active low)		GPIO_06	►	P3	
W15	SPI	SPI_CS2_N/GPIO_W14	Primary SPI Chipselect 2		GPIO_71	►	P72	B-PD:nppukp
W16	SPI	SPI_MISO	Primary SPI Master In Slave Out		GPIO_05	►	P3	
X16	SPI	SPI_MOSI	Primary SPI Master Out slave In		GPIO_04		P3	
X19	SPI_2	SPI2_CLK/GPIO_X19	Secondary SPI clock		GPIO_83	►	P3	B-PD:nppukp
W19	SPI_2	SPI2_CS_N/GPIO_W19	Secondary SPI chipselect 1		GPIO_82		P3	B-PD:nppukp
W18	SPI_2	SPI2_MISO/GPIO_W18	Secondary SPI Master in Slave out		GPIO_81		P3	B-PD:nppukp
X18	SPI_2	SPI2_MOSI/GPIO_X18	Secondary SPI Master out Slave in		GPIO_80		P3	B-PD:nppukp
H23	SS_USB	USB_SS_RX_M	Superspeed USB differential Receive minus				P3	AIO
J23	SS_USB	USB_SS_RX_P	Superspeed USB differential Receive plus				P3	AIO
K23	SS_USB	USB_SS_TX_M	Superspeed USB differential transmit minus				P3	AIO
L23	SS_USB	USB_SS_TX_P	Superspeed USB differential transmit plus				P3	AIO
Y15	UIM1	UIM1_CLK	SIM 1 clock		GPIO_70		P4	B-PD:nppukp
X15	UIM1	UIM1_DATA	SIM 1 data		GPIO_67		P4	B-PD:nppukp
Y14	UIM1	UIM1_PRESENT	SIM 1 presence detect		GPIO_68	►	P3	B-PD:nppukp
X14	UIM1	UIM1_RESET_N	SIM 1 reset (active low)		GPIO_69		P4	B-PD:nppukp
Z14	UIM1	VREG_UIM1	Dual Voltage UIM1 Supply					
Y21	UIM2	UIM2_CLK	SIM 2 clock		GPIO_03		P5	
X21	UIM2	UIM2_DATA	SIM 2 data		GPIO_0		P5	
Y20	UIM2	UIM2_PRESENT	SIM 2 presence detect		GPIO_01	►	P3	
X20	UIM2	UIM2_RESET_N	SIM 2 reset (active low)		GPIO_02	►	P5	
Z20	UIM2	VREG_UIM2	Dual Voltage UIM2 Supply					
T20	FREE GPIO	GPIO_T20	Free GPIO		GPIO_97		P3	B-PD:nppukp

Pad	Functional Group	LGA Pad Name	LGA Signal Description	Boot Config	IC GPIO Pin ¹	Wakeable Interrupts	Voltage Pad Type	Pad type
T19	VUC_IO	NAD_LPM	NAD is suspended in low power mode when active.		GPIO_93	►	P3	B-PD:nppukp
T21	VUC_IO	NAD2PS_HOLDUP	Keeps the 4V supply alive when the VUC is off, set low when the NAD is OK to have 4V removed.		GPIO_94	►	P3	B-PD:nppukp
T18	VUC_IO	NAD2TCU_WAKE	Wake up the TCU, or keep the TCU up		GPIO_96	►	P3	B-PD:nppukp
F20	VUC_IO	NAND_WP_N	Pull low when power is going away to halt any writes to NAND		-			
S19	RESERVED	RESERVED	Do Not Connect					
S21	FREE GPIO	GPIO_S21	Free GPIO		GPIO_76	►	P73	B-PD:nppukp
S18	FREE GPIO	GPIO_S18	Free GPIO		GPIO_85	►	P3	B-PD:nppukp
S17	VUC_IO	VUC2NAD_HOLDUP	Used to keep the Linux core up, also used to wake the Linux core		GPIO_77	►	P3	B-PD:nppukp
T17	VUC_IO	VUC2NAD_BANK_SELECT	Indicates from which Storage bank the NAD should operate (0=Primary, 1=Secondary)		GPIO_84	►	P3	B-PD:nppukp

GND	A01; A02; A03; A04; A05; A06; A07; A08; A09; A10; A11; A12; A13; A15; A21; A23; B02; B03; B04; B05; B06; B07; B08; B09; B10; B11; B12; B13; B15; B21; B23; B24; C01; C02; C03; C04; C05; C06; C07; C08; C09; C10; C11; C12; C13; C14; C15; D02; D03; D04; D05; D06; D07; D08; D09; D10; D11; D12; D13; D16; D17; D18; D21 E01; E02; E03; E04; E05; E06; E07; E08; E09; E10; E11; E12; E13; E18; E21; F02; F03; F04; F05; F06; F07; F08; F09; F10; F11; F12; F13; F19; G01; G02; G03; G04; G05; G06; G07; G08; G09; G10; G11; G12; G13; G14; G15; G16; G17; G18; G23; G24 H02; H03; H04; H05; H06; H07; H08; H09; H10; H11; H12; H13; H16; H17; H18; H19; H20; H21; H22; H24 J01; J02; J03; J04; J05; J06; J07; J08; J09; J10; J11; J12; J13; J14; J15; J16; J17; J18; J19; J20; J21; K02; K03; K04; K05; K06; K07; K08; K09; K10; K11; K12; K13; K14; K15; K17; K18; K19; K20; K21 L01; L02; L03; L04; L05; L06; L07; L08; L09; L10; L11; L12; L13; L14; L15; L17; L18; L19; L20; L21; L22; L24 M02; M03; M04; M05; M06; M07; M08; M09; M10; M11; M12; M13; M22; M23; N01; N02; N03; N04; N05; N06; N07; N08; N09; N10; N11; N12; N13; N22; P02; P03; P04; P05; P06; P07; P08; P09; P10; P11; P12; P13; P18; P19; P20; P22; Q01; Q02; Q03; Q04; Q05; Q06; Q07; Q08; Q09; Q10; Q11; Q12; Q13; Q18; Q19; Q20; Q21; Q22; R01; R02; R03; R04; R05; R06; R07; R08; R09; R10; R11; R12; R13; R17; R18; R19; R20; R22 S01; S02; S03; S04; S05; S06; S07; S08; S09; S10; S11; S12; S13; S14; S22; S23; T01; T02; T03; T04; T05; T06; T07; T08; T09; T10; T11; T12; T13; T23; T24; U01; U02; U03; U04; U05; U06; U07; U08; U09; U10; U11; U12; U13; U14; V01; V02; V03; V04; V05; V06; V07; V08; V09; V10; V11; V12; V13; V18 W01; W02; W03; W04; W05; W06; W07; W08; W09; W10; W11; W12; W13; W14; W20; X02; X03; X04; X05; X06; X07; X08; X09; X10; X11; X12; X13; Y01; Y02; Y03; Y04; Y05; Y06; Y07; Y08; Y09; Y10; Y11; Y12; Y13; Y16; Y19; Y24; Z02; Z03; Z04; Z05; Z06; Z07; Z08; Z09; Z10; Z11; Z12; Z13; Z15; Z16; Z19
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- (1) GPIO pin is either from the main BB processor IC (noted as GPIO_XX), PMIC (noted as PM_GPIO_XX), or RF Interface IC (noted as W_GRFC_XX)
- (2) Wakeable interrupts, indicates that signal can be used as an interrupt to the BB processor IC.

4.5 Pin Descriptions and Application Notes

Warning: If the NAD is in OFF or Sleep mode, ensure all input signals are at 0V to prevent latch-up.

4.5.1 Main Module Supply Pins

The NAD module is powered by one main supply at a voltage of 4V. This supply is required to be applied before the NAD module can be operated.

System Integrator must provide bulk capacitors near the 4V_QET and 4V_PMX pins of the NAD.

Table 4-5: Main Module Supply Pins

Name	Direction	Function	If Unused
4V_PMX ¹	Input	PMX Input Voltage	Must Be Used
4V_QET ¹	Input	QET Input Voltage	Must Be Used
5V_CV2X	Input	5V for C-V2X PA (not used)	Leave open
4V_DSDA	Input	4V for 5G DSDA (not used)	Leave open
3V3_PMX	Input	3.3V Supply to be subregulated	Must Be Used

- (1) Although these inputs are separated on the NAD LGA pins, they are not intended to be sequentially powered up. All NAD development is done with these supplies tied together externally.

4.5.2 Antennas

The SA515 NAD has eight antenna pins, only five of which are used by any of the NAD variants. The general functions served by each antenna pin are described in [Table 4-6](#). Not all variants will use every TX/RX function listed.

The antenna pins are physically located on the PC board – and in the order shown – as in [Figure 4-2](#).

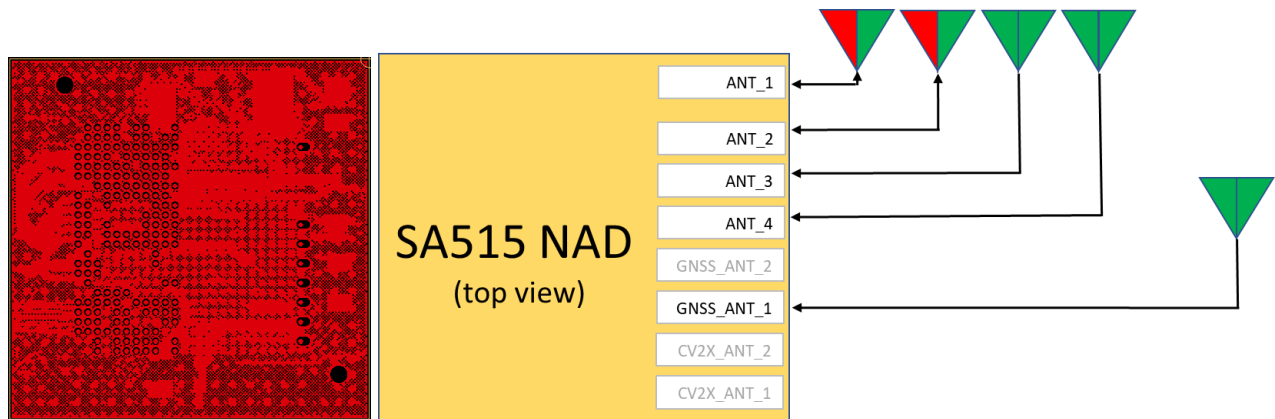


Figure 4-2 Antenna PAD Positions on NAD

Table 4-6: Antenna Pin Usage

Name	Direction	Function	If Unused
ANT_1	Bidirectional	Cellular Antenna 1 (Table 4-8 and Table 4-9 for details)	Must Be Used
ANT_2	Bidirectional	Cellular Antenna 2 (Table 4-8 and Table 4-9 for details)	Must Be Used
ANT_3	Input	Cellular Antenna 3 (Table 4-8 and Table 4-9 for details)	Leave open
ANT_4	Input	Cellular Antenna 4 (Table 4-8 and Table 4-9 for details)	Leave open
GNSS_ANT_1	Input	GNSS Receive Only	Must Be Used
GNSS_ANT_2	Input	GNSS Receive Only (not used)	Leave open
CV2X_ANT_1	Bidirectional	C-V2X Antenna 1 (not used)	Leave open
CV2X_ANT_2	Bidirectional	C-V2X Antenna 2 (not used)	Leave open

- (1) Antennas should be separated from each other to create sufficient RF isolation between them, where at least one antenna in the pair supports RF transmission. Additionally, if the vehicle system will incorporate other functions outside of the NAD, such as WiFi or SDARS, isolation from those systems' antennas is also advisable. The recommended isolation between specific pairs of antennas is given in [Table 4-7](#).

Table 4-7 Recommended Antenna Isolation

<u>Antenna Isolation Performance</u>	<u>Isolation</u>
Cellular ANT1 or ANT2 to Cellular ANT1, ANT2, ANT3, ANT4 (617-960MHz)	20 dB min
Cellular ANT1 or ANT2 to Cellular ANT1, ANT2, ANT3, ANT4 (1710MHz-5GHz)	20 dB min
Cellular ANT1 or ANT2 to GNSS (Internal GNSS engine)	20 dB min
Cellular ANT1 or ANT2 to GNSS (External GNSS engine)	30 dB min
Cellular ANT1 or ANT2 to SDARS	20 dB min
Cellular ANT1 or ANT2 to 2.4GHz WiFi	30 dB min
Cellular ANT1 or ANT2 to 5GHz WiFi	30 dB min

Routing guidelines are provided in [Section 11, Layout and Routing Recommendations](#).

4.5.2.1 TX Band/Antenna Configuration

Transmit modes for each RAT, band, and EN-DC configuration vary with each Antenna. The associations are shown in [Table 4-8](#). **These are representative of Hardware Capability – band usage may be further restricted by operating region or carrier-mandated configurations (MBN)**

Table 4-8 TX Band Configuration by Antenna

Model Name	ANT1 (LTE_ANT_1)			ANT2 (LTE_ANT_2)		
	2G/3G/4G	5G NSA	5G SA	2G/3G/4G	5G NSA	5G SA
FE5CN0020 ¹	per Table 1-1	n1, n3, n5, n8, n41	n1, n28, n41, n77, n78, n79	---	n77, n78, n79	n41, n77, n78, n79
FE5EU0020	per Table 1-1	n1, n3, n5, n7, n8, n28	n1, n28	---	n77, n78	n77, n78
FE5NA0020	per Table 1-1	n2, n5, n7, n66, n71	n66, n71	---	n41, n77 n78	n41, n77 n78
FE5RW0020	per Table 1-1	n3, n28	n28	---	n77, n78, n79	n41, n77, n78, n79

Note 1: TX support on both antennas means 2x2 Uplink MIMO is supported for that band.

4.5.2.2 RX Band/Antenna Configuration

Receiver modes for each RAT and band group will vary with each Antenna. The RX support by antenna and band group is shown in [Table 4-9](#). The listed band groups are a superset; for a given variant, the bands actually supported are limited to those in [Table 1-1](#)

Table 4-9 RX Band Configuration by Antenna

Model Name	ANT1 (LTE_ANT_1)				ANT2 (LTE_ANT_2)				ANT3 (LTE_ANT_3)		ANT4 (LTE_ANT_4)	
	GSM	UMTS	LTE	NR	GSM	UMTS	LTE	NR	LTE	NR	LTE	NR
FE5##0020	LB, MB	LB, MB	LB, MB, HB	LB, MB, HB, UHB	LB, MB	LB, MB	LB, MB, HB	LB, MB, HB, UHB	LB, MB, HB	MB, HB, UHB	LB, MB, HB	MB, HB, UHB

Note 1: For cellular band group definitions (LB/MB/HB/UHB), see [Table 1-3](#).

Note 2: For LTE and NR, RX support on 2 antennas means 2x2 Downlink MIMO is supported for the indicated band group.

Note 3: For LTE, RX support on 4 antennas means 4x4 Downlink MIMO is supported for the indicated band group (MB or HB), or that HORXD is supported (LB).

Note 4: For NR, RX support on 4 antennas means 4x4 Downlink MIMO is supported for the indicated band group (MB, HB & UHB only). NR does not support 4 antenna RX on LB.

4.5.2.3 Transmitter Power Class 2 (HPUE) Support

Cellular 4G and 5G operation is Power Class 3 by default. For select bands and configurations, Power Class 2 is supported and these are shown in [Table 4-10](#).

Table 4-10 TX Power Class 2 Support

Model Name	5G NR (NSA)		5G NR (SA)		4G LTE	
	Ant1	Ant2	Ant1	Ant2	Ant1	Ant2
FE5CN0020		n77, n78, n79		n77, n78, n79	---	
FE5EU0020		---		n77, n78	---	
FE5NA0020		n77		n41, n77, n78	---	
FE5RW0020		---		n41, n77, n78, n79	---	

4.5.3 Boot Configuration Pins

The NAD has boot configuration pins. It is critical that these pins are not biased at power up. If these signals are used as GPIO, they must be used as output only, not as inputs.

Table 4-11: Boot Config Pins

Name	Direction	Function	If Unused
BOOT_CFG2	output	BOOT_CONFIG[2]	Leave open
BOOT_CFG1	output	BOOT_CONFIG[1]	Leave open
BT_UART_TX	output	BOOT_CONFIG[7]	Leave open
GPIO_F14	output	BOOT_CONFIG[0]	Leave open
GPIO_F18	output	BOOT_CONFIG[11]	Leave open
FORCED_USB_BOOT	Input	Do Not Connect	Leave open

The NAD has boot configuration pins to control from which device it will boot.

FORCED_USB_BOOT	BOOT_CFG2	BOOT_CFG1	BT_UART_TX	GPIO_F14	GPIO_F18	State
Float	X	X	X	X	X	Emergency Download Mode - USB
Float	0	0	Float	Float	Float	NAND
Float	0	1	Float	Float	Float	Not supported
Float	1	0	Float	Float	Float	Not supported
Float	1	1	Float	Float	Float	Not Supported

4.5.4 PM_KPD_PWR_EN

The NAD supports this ACTIVE LOW INPUT to power on and power off the module. It should be noted, that although this could be used to power off the module, it is not the preferred method. Please refer to Section 5 "Power Up and Power Down Sequence" for more details.

Table 4-12: PM_KPD_PWR_EN Control Pin

Name	Direction	Function	If Unused
PM_KPD_PWR_EN	Input	ON/OFF Control	Must Be Used

The PM_KPD_PWR_EN pin is pulled up high internally to the module and must not be pulled high externally. The voltage will be approximately 0.8V when unloaded.

Table 4-13: PM_KPD_PWR_EN Internal Pull-Up

Signal	Parameter	MIN	TYP ¹	MAX	UNIT
PM_KPD_PWR_EN	Internal Pull-Up	150	200	250	kΩ

(1) Typical value provided as design reference only. This value is not guaranteed.

This pin should not be driven high external to the NAD embedded module.

Figure 4-3 illustrates the recommended external circuit to properly interface to the NAD.

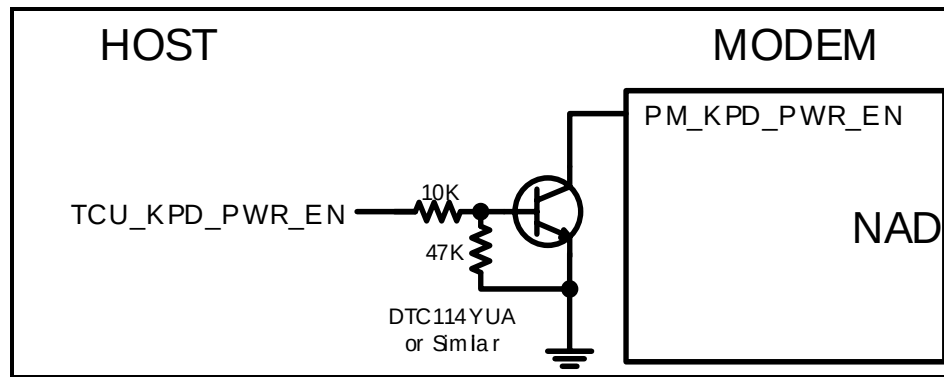


Figure 4-3: Recommended External Interface Circuit for PM_KPD_PWR_EN

Note: Prior to powering up the NAD, the main module supply should be applied to unit.

4.5.5 NAD Low Power Mode

Table 4-14: NAD Low Power Mode

Name	Direction	Function
NAD_LPM	Output	When driven active high, NAD is in low power mode.

VUC2NAD_HOLDUP will take NAD Module out of Low Power Mode. This implies that VUC2NAD_HOLDUP is defined as a wakeup signal when the NAD Module is in Low Power Mode.

4.5.6 RESET_N

The NAD module supports this ACTIVE LOW INPUT to reset the NAD in case communication is lost, but the NAD has been determined to still be powered on. This signal is also used to enter Fastboot mode during boot.

Table 4-15: RESET_N Control Pin

Name	Direction	Function	If Unused
RESET_N	Input	External Reset input to reset module	Leave Open

The RESET_N pin is pulled up high internally to the module and must not be pulled high externally.

Table 4-16: RESET_N Parameters

Signal	Parameter	MIN	TYP ¹	MAX	UNIT
RESET_N	Internal Pull-Up Inside Power Management ASIC		40		kΩ

(1) Typical value provided as design reference only. This value is not guaranteed.

Figure 4-4 illustrates the recommended external circuit to properly interface to NAD.

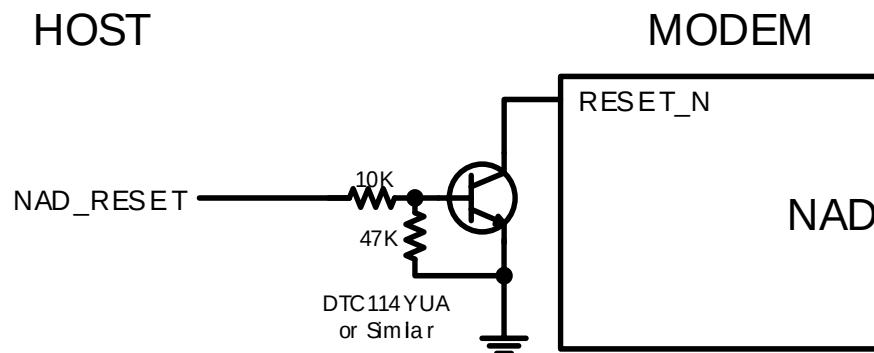


Figure 4-4: Recommended External Interface Circuit for RESET_N

Table 4-17 shows the planned timing that is configured by software, to be applied to Figure 4-5.

To prevent extraneous interrupts to NAD processor, a pulse on RESET_N of less than 50-ms will be ignored.

To initiate a **software controlled reboot**, RESET_N should be held low for tRESET-1. **THIS IS THE PREFERRED METHOD**, and should be attempted first.

If the software controlled reboot is unsuccessful, then a **hard reset** may be required. To initiate a hard reset, RESET_N should be held low for tRESET-2.

If the NAD is still not responsive, a NAD power cycle may be required. See section 5.2.

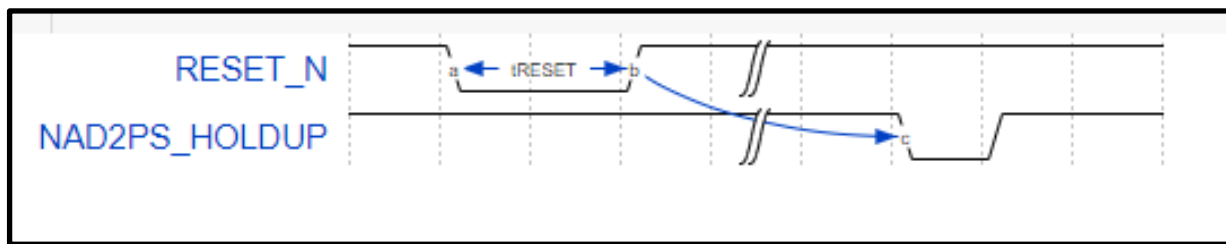


Figure 4-5: Module Reset Timing

Table 4-17: RESET_N Timing Sequence Definitions

Parameter		MIN	TYP	MAX	UNIT
tRESET-1	Pulse duration for software controlled reboot	50	-	2050	ms
tRESET-2	Pulse duration for hard reset	2050	-	8000	ms

4.5.7 NAD2PS_HOLDUP

The NAD module supports this ACTIVE HIGH OUTPUT to indicate power latching from NAD software. This signal holds power logic on. It is used by DCM to monitor boot sequence.

Active state means NAD has powered up and boot sequence has started, typically 190ms after falling edge of PM_KPD_PWR_EN.

Active state means NAD requests power supply logic to latch on.

Inactive state means NAD has fully powered down and it is okay to release the NAD power supply latch.

Table 4-18: NAD2PS_HOLDUP Control Pin

Name	Direction	Function
NAD2PS_HOLDUP	Output	NAD2PS_HOLDUP

4.5.8 VUC2NAD_HOLDUP

The primary function of this signal is to wake up the NAD. A rising edge on this input will cause the NAD to wake up.

When executing a software-controlled power down, this signal should be kept low to ensure the power off sequence is fully executed.

Table 4-19: VUC2NAD_HOLDUP Control Pin

Name	Direction	Function
VUC2NAD_HOLDUP	Input	When NAD is in Low Power Mode (A7 asleep), a high pulse on this signal will wake up the NAD

4.5.9 VUC2NAD_BANK_SELECT

A high indicates that the NAD should boot from the primary bank.

A low indicates that the NAD should boot from the secondary bank.

The NAD module supports this input to select the memory bank from which the NAD will boot. This input should be set to the desired state before starting or resetting the NAD. It is sampled shortly after the NAD starts booting (in SBL). This input has no further impact on the selected bank after it is sampled.

The transition to high of the output signal NAD2PS_HOLDUP can be monitored as an external indication that VUC2NAD_BANK_SELECT input has been sampled.

If the NAD fails to boot on one bank, this input can be changed to select the other bank and then reboot the NAD.

This signal is pulled up internally to 1.8V, so if unused, leave open. An open collector buffer can be used to pull the signal low.

Table 4-20: VUC2NAD_BANK_SELECT Control Pin

Name	Direction	Function	If Unused
VUC2NAD_BANK_SELECT	Input	Bank Select for NAD (internally pulled up)	Leave Open

4.5.10 ADC

The NAD module supports up to two ADC input for Antenna Diagnostics.

Free GPIO can be used to control an external analog switch to sample multiple antennas.

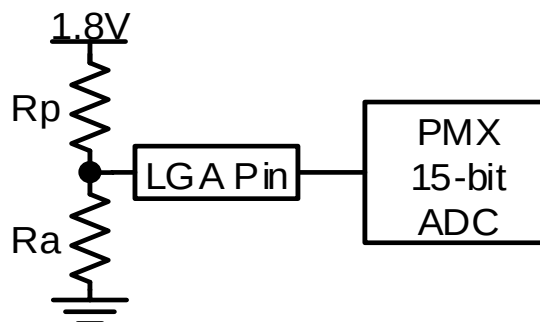


Figure 4-6: Example external resistor network for ADC input

Table 4-21: State ADC Table

State	R_a min	R_a max
Open	$R_p * 4$	[Open Ckt]
Normal	$R_p * 0.25$	$R_p * 4$
Short	0 ohm	$R_p * 0.25$

- (1) Choice of R_p value (to be used on parent module detection circuit) will dependent on customer antenna resistance (R_a). R_a limit values are only examples here, the range will ultimately depend on the expected variation in resistance of the customer antenna. Customer SW will need to accomodate these parameters for ADC detection.

Table 4-22: ADC1 Analog Inputs

Name	Direction	Function	If Unused
S16 (PM_GPIO_S16)	Input	ADC1	Leave Open
T15 (BT_EN)	Input	GPIO/ADC (note 1)	

- (1) Available only if BT/WLAN is not used

If additional ADC's are needed, a mux controlled by free GPIO can be used to switch in the various antenna ports to the remaining ADC.

4.5.11 VREF_1V8

The NAD module supports this 1.8V reference, VREF_1V8, to be used to turn on or enable 1.8V interfaces. **It is not to be used to power external devices.**

The maximum current which may be drawn at this pin is 10 mA.

This supply is generated from main NAD 1.8V to allow for proper detection at boot up of NAD.

This supply cannot be a reference to any pull ups before power on is initiated.

Table 4-23: VREF_1V8 Voltage Output

Name	Direction	Function	If Unused
VREF_1V8	Output	1.8V Voltage Reference – 10 mA maximum	Leave Open

4.5.12 GNSS_1PPS

The NAD module supports this ACTIVE HIGH PULSE TRAIN OUTPUT to generate a 1PPS signal for GNSS dead reckoning. This signal is also needed by the NAD to support any other high accuracy position and time synchronization requirements. Alternate functions aren't available on this pin when used for high accuracy mode.

Table 4-24: GPS_1PPS Control Pin

Name	Direction	Function	If Unused
GNSS_1PPS	Output	PPS clock output from internal GNSS engine. A PPS clock in sync with GPS time that is used for Dead Reckoning.	Leave Open

4.5.13 GNSS_LNA1/2_EN

The NAD module supports two ACTIVE HIGH OUTPUT signals to generate a GNSS antenna LNA enable. *Active state means external LNA1/2 is enabled. Inactive state means external LNA1/2 is disabled.*

Table 4-25: GNSS_LNA1_EN Control Pin

Name	Direction	Function	If Unused
GNSS_LNA_EN	Output	GNSS LNA 1 Enable	Leave Open
GNSS_LNA2_EN	Output	GNSS LNA 2 Enable ¹	Leave Open

(1) This pin must not be arbitrarily pulled high at boot.

4.5.14 External MIPI RFFE bus

The NAD module can support external RF device control via RFFE MIPI. These two signals should be routed as isolated 50 ohm traces.

Table 4-26: External Antenna Switch Control Pins

Name	Direction	Function	If Unused
RF_GPIO_N16	Output	RFFE_CLK	Leave Open
RF_GPIO_M16	Bidirectional	RFFE_DATA	Leave Open

The RFFE0 interface lines in [Table 4-26](#) will provide communication to the external antenna switches on the parent device, using identifier codes as in [Table 4-27](#).

Table 4-27 External MIPI Antenna Switch IDs

For QPC1251Q Switch	NA	EU	RW	CN
	FE5NA0020	FE5EU0020	FE5RW0020	FE5CN0020
RFC HWID	664	662	661	666
DEVICE_MODULE_TYPE_INSTANCE (logical dev)	39	37	45	45

4.5.15 SDX_SLEEP_CLK

The NAD module has the option to support this WLAN 32-kHz sleep clock signal from the Sleep clock output from the PMIC, to external WLAN module.

Active state is a 32-kHz logic clock signal. No Inactive state exists in the 5G NAD Module.

Table 4-28: SLEEP Clock Output

Name	Direction	Function	If Unused
SDX_SLEEP_CLK	Output	SLEEP_CLK ¹	Leave Open

- (1) Platform software does not inherently support the use of the SDX_SLEEP_CLK alternate function or GPIO on this pin; therefore to be used, software resources will be required to implement function.

4.5.16 Bluetooth and WLAN Control

The NAD module has the option to support this WLAN enable signal, to external WLAN module to be controlled.

Active state means external WIFI device is in operational mode.

Inactive state means external WIFI device is in standby mode.

Table 4-29: WLAN_EN Control Pin

Name	Direction	Function	If Unused
WLAN_EN	Output	Active high enables WLAN_EN ¹	Leave Open
WLAN_PWR_CTRL1	Output	BT-WLAN Power Control signal	Leave Open
WLAN_PWR_CTRL2	Output	BT-WLAN Power Control signal 2	Leave Open
WLAN_SW_CTRL	Output	BT-WLAN Switch Control signal	Leave Open
BT_EN	Output	Active high enables Bluetooth	Leave Open

- (1) Platform software does not inherently support the use of the WLAN_EN alternate function or GPIO on this pin; therefore to be used, software resources will be required to implement function.

4.5.17 LTE_COEX_UART

The NAD module supports this two wire LTE coexistence signaling system to allow for communications for LTE coexistence with other standards, such as BT and WIFI.

The signaling is based on IEEE 802.15.2, using a standard 2-wire UART interface.

These pins can optionally be configured as standard GPIO.

Table 4-30: LTE_COEX_UART Control Pins

Name	Direction	Function	If Unused
LTE_COEX_UART_TX	Output	TX signal for coexistence UART ^{1,2}	Leave Open
LTE_COEX_UART_RX	Input	RX signal for coexistence UART ¹	Leave Open

- (1) Platform software does not inherently support the use of GPIO alternate function on these pins; therefore software resources will be required to implement this alternate function.
- (2) This pin must not be arbitrarily pulled high at boot.

4.5.18 UIM1 / UIM2 Interfaces

The 5G NAD embedded module supports two USIM/CSIM devices for LTE, WCDMA, and GSM. eUICC / RUIM / CSIM cards and devices are regulated by ISO/IEC 7816-3:2006.

Table 4-31: UIM1 / UIM2 Interfaces

Name	Direction	Function	If Unused
VREG_UIM1	Output	UIM1 Supply	Leave Open
UIM1_RESET_N	Output	UIM1 Reset	Leave Open
UIM1_PRESENT	Input	UIM1 Card Detect ²	Leave Open
UIM1_CLK	Output	UIM1 Clock	Leave Open
UIM1_DATA	Bidirectional	UIM1 Data ²	Leave Open
VREG_UIM2	Output	UIM2 Supply	Leave Open
UIM2_RESET_N	Output	UIM2 Reset (alternate GPIO and other functions) ¹	Leave Open
UIM2_PRESENT	Input	UIM2 Card Detect (alternate GPIO and other functions) ^{1, 2}	Leave Open
UIM2_CLK	Output	UIM2 Clock (alternate GPIO and other functions) ¹	Leave Open
UIM2_DATA	Bidirectional	UIM2 Data (alternate GPIO and other functions) ^{1, 2}	Leave Open

- (1) Platform software does not inherently support the use of GPIO or other alternate function on these pins; therefore software resources will be required to implement this alternate function.
- (2) For proper UICC operation, external pull-ups are required outside the module and described below.

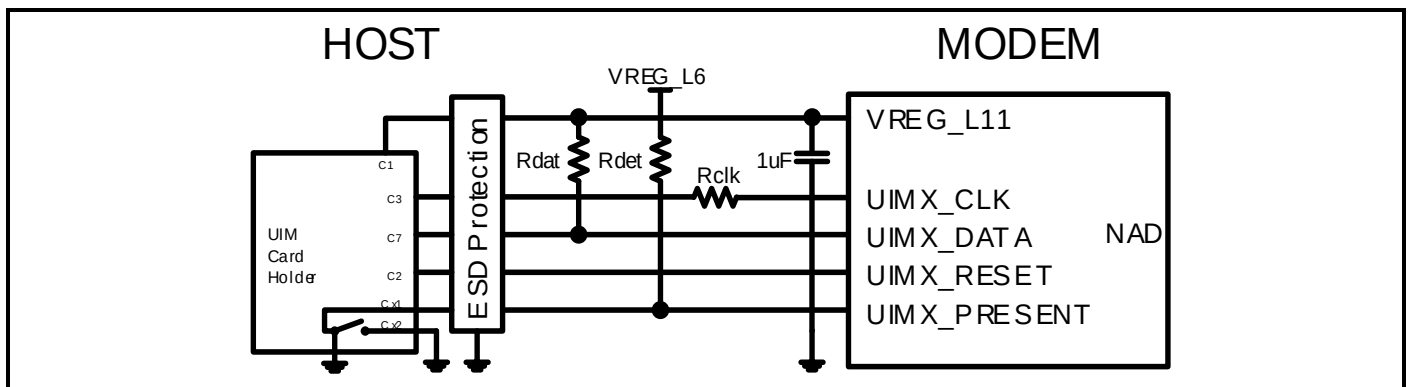
Figure 4-7 shows proper connection and external devices required for each UIM connections.

UIM_DETECT requires a pull-up resistor (Rdet) of 100 kΩ to VREG_L11_1P8, external to NAD. It should be set to GND if a SIM/UIM is present.

UIM_DATA requires a pull-up resistor (Rdat) of 10 kΩ to VREG_UIM external to the NAD.

The 1 uF capacitor and Rclk resistor should be added as placeholders to compensate for potential layout issues. UIM_DAT trace should be routed away from the UIM_CLK trace. Distance between NAD to UIM-Holder should be as short as possible.

An ESD device specifically designed for SIM/UIM cards is recommended for all signals.

**Figure 4-7: Recommended External Interface Circuit for UIM1 and UIM2**

See guidelines as provided in Section 11.3.

4.5.19 USB VBUS Detection Pin

The module requires USB VBUS to be provided for detection of USB 2.0.

The supply is turned on when USB VBUS is provided by the USB host, whether the NAD is on or off.

Table 4-32: USB VBUS Supply Pin

Name	Direction	Function	If Unused
USB_VBUS_DET	Input	USB VBUS Input Voltage	Must Be Used

An example of a circuit to provide an interface from the parent device to the USB_VBUS_DET pin is shown in [Figure 4-8](#).

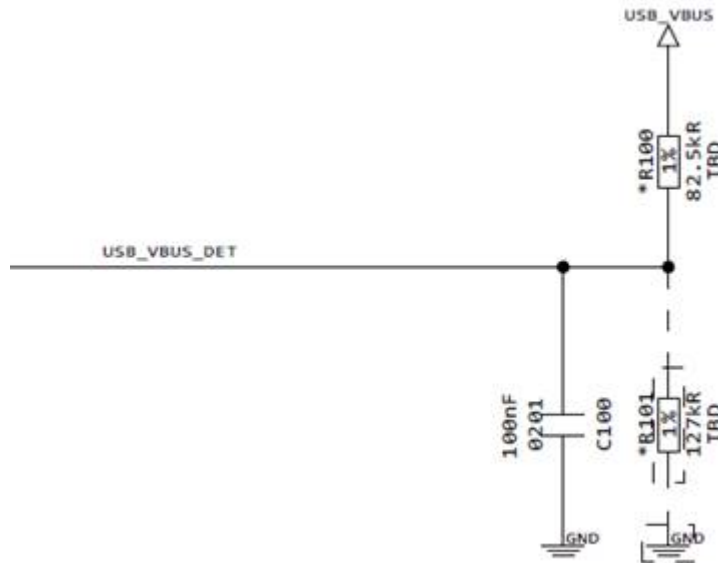


Figure 4-8 Example detection circuit on TCU

Table 4-33 VBUS detection specifications

Parameter	Comments	Min	Typ	Max	Units
Detection Threshold *		1.0	1.1	1.2	V
Pull-down resistor	Always on	50	120	150	kohm

* If the voltage is higher than the detection threshold, VBUS presence will be detected by the PMIC.

4.5.20 USB 2.0 High-Speed

The NAD module supports USB 2.0 HS for USB host communication or debugging.

USB 2.0 HS is a half-duplex serial bus regulated by the USB.org.

Link Speed up to 480-Mbps (60-MBps)

Practical Data Throughput up to 280-Mbps (35-MBps)

Table 4-34: USB 2.0 High Speed Bus Interface

Name	Direction	Function	If Unused
USB_HS_DP	Bidirectional	USB 2.0 HS D+, Differential Pair	Leave Open
USB_HS_DM	Bidirectional	USB 2.0 HS D-, Differential Pair	Leave Open
USB_VBUS_DET	Input	USB VBUS Input, Shared with USB 3.1	Leave Open

The diagram below illustrates the recommended external circuit to properly interface to Gen2 NAD.

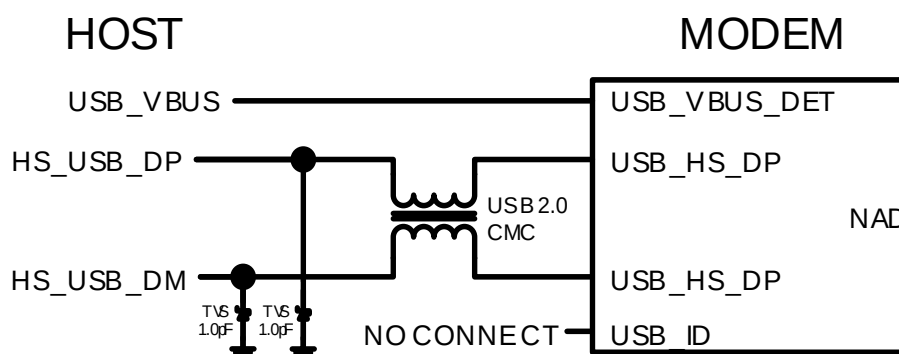


Figure 4-9: Recommended External Interface Circuit for USB 2.0

Routing guidelines are provided in Layout and Routing Recommendations Section.

4.5.21 USB3.1 Super-Speed Gen1

The NAD module supports USB 3.1 SS Gen1 for USB host communication or debugging.

USB 3.1 SS Gen1 is a full-duplex serial bus regulated by the USB.org.

Link Speed of 5-Gbps and 8b10b encoding has Raw Throughput of up to 4-Gbps (500-MBps)

Practical Data Throughput up to 3.6-Gbps (450-MBps)

Table 4-35: USB 3.1 Super Speed Gen1 Bus Interface

Name	Direction	Function	If Unused
USB_SS_RX_P	Input	USB 3.1 SS Gen1 RX+, Differential Pair	Leave Open
USB_SS_RX_M	Input	USB 3.1 SS Gen1 RX-, Differential Pair	Leave Open
USB_SS_TX_P	Output	USB 3.1 SS Gen1 TX+, Differential Pair	Leave Open
USB_SS_TX_M	Output	USB 3.1 SS Gen1 TX-, Differential Pair	Leave Open
USB_VBUS_DET	Input	USB VBUS Input, Shared with USB 2.0	Leave Open

Routing guidelines are provided in Section 11.5.

4.5.22 CODEC I2C

The NAD module supports a CODEC I2C port.

The CODEC I2C is shared with both Primary and Auxiliary Digital Audio interfaces.

I2C is a half-duplex serial bus created by Philips/NXP. The I2C can transfer data at up to 400-kbps.

Table 4-36: CODEC_I2C Bus Interface

Name	Direction	Function	If Unused
CDC_I2C_SCL	Output	CODEC I2C Clock ¹	Leave Open
CDC_I2C_SDA	Input / Output	CODEC I2C Data ²	Leave Open

- (1) For proper I2C operation, I2C_CDC_SCK requires a 2.2-kOhm to 4.7-kOhm pull-up to VOUT1 outside the module. The exact value required depends on the capacitive load. Rise times should be met with the chosen resistor pull-up.
- (2) For proper I2C operation, I2C_CDC_DATA requires a 2.2-kOhm to 4.7-kOhm pull-up to VOUT1 outside the module. The exact value required depends on the capacitive load. Rise times should be met with the chosen resistor pull-up.

4.5.23 Primary Digital Audio

The NAD module supports primary digital audio interface, providing stereo audio.

The primary PCM provides stereo audio for both transmit and receive.

The primary PCM interface can be master or slave.

This primary PCM could be used for primary CODEC.

Table 4-37: Primary Digital Audio Interface

Name	Direction	Function	If Unused
PRI_I2S_DOUT	Output	Primary PCM audio data out	Leave Open
PRI_I2S_SCK	Output	Primary PCM audio clock	Leave Open
PRI_I2S_DIN	Input	Primary PCM audio data in	Leave Open
PRI_I2S_WS	Output	Primary PCM audio frame sync	Leave Open
CDC_RESET_N	Output	ACTIVE LOW reset to CODEC	Leave Open
PRI_I2S_MCLK	Output	Master CODEC Clock	Leave Open
CDC_I2C_SCL	Output	CODEC I2C Clock	Leave Open
CDC_I2C_SDA	Bidirectional	CODEC I2C Data	Leave Open

4.5.24 Auxiliary Digital Audio (Bluetooth)

Not currently supported in software.

The NAD module supports this optional auxiliary digital audio interface in but software does not currently support.

The auxiliary PCM provides stereo audio for both transmit and receive.

The auxiliary PCM interface can be master or slave.

This auxiliary PCM could be used for Bluetooth audio or optional second CODEC.

If this optional feature is required, the debug UART0 and I2C will be unavailable.

Table 4-38: Auxiliary Digital Audio Interface

Name	Direction	Function	If Unused
BT_UART_CTS	Output	Auxiliary PCM audio data out ¹	Leave Open
BT_UART_RTS	Output	Auxiliary PCM audio clock ¹	Leave Open
BT_UART_RX	Input	Auxiliary PCM audio data in ¹	Leave Open
BT_UART_TX	Output	Auxiliary PCM audio frame sync ¹	Leave Open
CDC_RESET_N	Output	ACTIVE LOW reset to CODEC ¹	Leave Open
PRI_I2S_MCLK	Output	Master CODEC Clock	Leave Open
CDC_I2C_SCL	Output	CODEC I2C Clock	Leave Open
CDC_I2C_SDA	Bidirectional	CODEC I2C Data	Leave Open

(1) Platform software does not inherently support the use of the AUX PCM alternate function on these pins; therefore software resources will be required to implement this alternate function.

4.5.25 PCIe (Ethernet and/or WIFI)

The NAD module supports a PCIe v2.0 single transmit and receive lane. The current system architecture would use this for WIFI or Ethernet, or both if an external packet switch is used.

PCIe is a full-duplex serial bus regulated by the PCI-SIG.

Link Speed of 5-Gbps and 8b10b encoding has Raw Throughput of up to 4-Gbps (500-MBps)

Table 4-39: PCIe Interface

Name	Direction	Function	If Unused
PCIE_TX0_P	Output	PCIE TX+, Differential Pair (ac coupled)	Leave Open
PCIE_TX0_M	Output	PCIE TX-, Differential Pair (ac coupled)	Leave Open
PCIE_RX0_P	Input	PCIE RX+, Differential Pair	Leave Open
PCIE_RX0_M	Input	PCIE RX-, Differential Pair	Leave Open
PCIE_TX1_P	Output	PCIE TX+, Differential Pair (ac coupled)	Leave Open
PCIE_TX1_M	Output	PCIE TX-, Differential Pair (ac coupled)	Leave Open
PCIE_RX1_P	Input	PCIE RX+, Differential Pair	Leave Open
PCIE_RX1_M	Input	PCIE RX-, Differential Pair	Leave Open
PCIE_CLK_P	Output	PCIE CLK+, Differential Pair	Leave Open
PCIE_CLK_M	Output	PCIE CLK-, Differential Pair	Leave Open
PCIE_CLKREQ_N	Output	PCIE CLK Request	Leave Open
PCIE_RESET_N	Output	PCIE Host Reset ¹ (If using RC mode)	Leave Open
PCIE_WAKE_N	Output	PCIE Wake	Leave Open

(1) PCIE_RST may require a 100-kOhm pull-down to ground.

(2) PCIE_WAKE_N requires a 10-kOhm pull-up to VOUT1.

(2) PCIE_CLKREQ_N requires a 10-kOhm pull-up to VOUT1.

Routing guidelines are provided in Section 11.6.

4.5.26 SDC (SDIO/SDC/eMMC)

The NAD module supports a 4-bit SD 3.0 port up to 50-MHz DDR or 200MHz SDR for WIFI, SD, Car2X, etc. or an 8-bit eMMC 4.51 interface up to 50MHz DDR or 100-MHz SDR for eMMC

SD 3.0 and SDIO 3.0 is a half-duplex parallel bus regulated by the SD Card Association.

eMMC 4.51 is a half-duplex parallel bus regulated by JEDEC.

Link Speed up to 800-Mbps is possible.

Table 4-40: SDC Interface

Name	Direction	Function	If Unused
SDC_DATA_0	Bidirectional	SDIO/SDC/eMMC Data bit 0	Leave Open
SDC_DATA_1	Bidirectional	SDIO/SDC/eMMC Data bit 1	Leave Open
SDC_DATA_2	Bidirectional	SDIO/SDC/eMMC Data bit 2	Leave Open
SDC_DATA_3	Bidirectional	SDIO/SDC/eMMC Data bit 3	Leave Open
SDC_DATA_4	Bidirectional	eMMC Data bit 4	Leave Open
SDC_DATA_5	Bidirectional	eMMC Data bit 5	Leave Open
SDC_DATA_6	Bidirectional	eMMC Data bit 6	Leave Open
SDC_DATA_7	Bidirectional	eMMC Data bit 7	Leave Open
SDC_CLK	Output	SDIO/SDC/eMMC Clock	Leave Open
SDC_CMD	Bidirectional	SDIO/SDC/eMMC Command ¹	Leave Open
SDC_DATA_5	Input	SDC Card Detect ^{2,3}	Leave Open

- (1) For proper SD and eMMC operation, CMD requires and external pull up resistor to SDC IO voltage external to the module. This is different than some other NADs.
- (2) For removable SD card, The SDC Card Detect (SDX_SDC_DATA_5) requires a 100-kOhm pull-up to 1.8V outside the module.
- (3) For embedded SDIO or eMMC, SDC_CARD_DET_N is not used.
- (4) The supply for SDC_DATA_[3:0], SDC_CLK, SDC_CMD is VREG_L7, and it needs to be set by SW to 1.8V or 2.95V depending on the interface configuration requirements.

Routing guidelines are provided in Section 11.

4.5.27 Primary SPI (Primary Bulk Data)

The NAD module supports a SPI bus as a bulk data interface between Cortex A7 and the DCM.

The SPI interface is master only.

These pins can optionally be configured as standard GPIO or a 4-wire UART interface for a standard Bluetooth interface.

SPI is a full-duplex serial bus created by Motorola. The SPI can transfer data at up to 50-Mbps.

Table 4-41: SPI Interface

Name	Direction	Function	If Unused
SPI_MISO	Input	SPI Data In (alternate Bulk UART / GPIO) ¹	Leave Open
SPI_MOSI	Output	SPI Data Out (alternate Bulk UART / GPIO) ¹	Leave Open
SPI_CLK	Output	SPI Clock (alternate Bulk UART / I2C / GPIO) ¹	Leave Open
SPI_CS_N	Output	SPI Chip Select 0 (alternate Bulk UART / I2C / GPIO) ¹	Leave Open
SPI_CS2_N	Output	SPI Chip Select 1 ²	Leave Open

- (1) Platform software does not inherently support the use of GPIO or I2C or Bulk UART alternate functions on these pins; therefore software resources will be required to implement this alternate function.
- (2) These pins must not be arbitrarily pulled high at boot.

4.5.28 Secondary SPI (Secondary Bulk Data)

The NAD module supports a SPI bus as a bulk data interface between Cortex A7 and the DCM.

The SPI interface is master only.

These pins can optionally be configured as standard GPIO or a 4-wire UART interface for a standard Bluetooth interface.

SPI is a full-duplex serial bus created by Motorola. The SPI can transfer data at up to 50-Mbps.

Table 4-42: SPI Interface

Name	Direction	Function	If Unused
SPI2_MISO	Input	SPI Data In (alternate BT UART / GPIO) ¹	Leave Open
SPI2_MOSI	Output	SPI Data Out (alternate BT UART / GPIO) ¹	Leave Open
SPI2_CLK	Output	SPI Clock (alternate BT UART / I2C / GPIO) ¹	Leave Open
SPI2_CS_N	Output	SPI Chip Select 0 (alternate BT UART / I2C / GPIO) ¹	Leave Open

- (1) Platform software does not inherently support the use of GPIO or I2C or BT UART alternate functions on these pins; therefore software resources will be required to implement this alternate function.
- (2) These pins must not be arbitrarily pulled high at boot.

4.5.29 UART0 (Debug)

The NAD module supports UART0 as a 2-wire debug interface from Cortex A7 via the Linux Console.

This debug UART can optionally be configured as a 4-wire interface.

The 2-wire debug interface can optionally be configured as standard GPIO.

UART is a full-duplex serial bus regulated by the TIA. The UART can transfer data at up to 4-Mbps.

Table 4-43: Debug UART0 Interface

Name	Direction	Function	If Unused
DBG_UART_TX	Output	Debug Data Output (alternate GPIO) ¹	Leave Open
DBG_UART_RX	Input	Debug Data Input (alternate GPIO) ¹	Leave Open

- (1) Platform software does not inherently support the use of GPIO alternate function on these pins; therefore software resources will be required to implement this alternate function
- (2) Platform software does not inherently support the use of 4-wire Debug UART mode alternate function on these pins; therefore software resources will be required to implement this alternate function

4.5.30 UART1 (IPC)

The NAD module supports UART1 as a 2-wire IPC control interface between Cortex A7 and the DCM, but for maximum throughput, must use all 4 wires.

These pins can optionally be configured as standard GPIO.

UART is a full-duplex serial bus regulated by the TIA. The UART can transfer data at up to 4-Mbps.

Table 4-44: IPC UART1 Interface

Name	Direction	Function	If Unused
IPC_UART_RX	Input	IPC Input (alternate GPIO) ¹	Leave Open
IPC_UART_TX	Output	IPC Output (alternate GPIO) ¹	Leave Open
IPC_UART_CTS	Input	IPC Input (alternate GPIO) ¹	Leave Open
IPC_UART_RTS	Output	IPC Output (alternate GPIO) ¹	Leave Open

- (1) Platform software does not inherently support the use of GPIO alternate function on these pins; therefore software resources will be required to implement this alternate function

4.5.31 UART2 (Optional Bulk Data)

Not currently supported in software. The platform solution for bulk data is the SPI interface.

The NAD module has the option to support UART2 as a 2-wire IPC bulk data interface between Cortex A7 and the DCM. This is optional feature, as SPI is platform bulk data transport.

These pins can optionally be configured as standard GPIO, a secondary SPI interface, a 2-wire UART interface for full Bluetooth interface with an I2C bus or a 4-wire UART interface for Bluetooth Low Energy.

UART is a full-duplex serial bus regulated by the TIA. The UART can transfer data at up to 4-Mbps.

Table 4-45: Data UART2 Interface

Name	Direction	Function	If Unused
UART2_RX	Input	Data Input (alternate BLE UART / BT PCM /GPIO) ¹	Leave Open
UART2_TX	Output	Data Output (alternate BLE UART / BT PCM /GPIO) ¹	Leave Open

(1) Platform software does not inherently support the use of GPIO or I2C or SPI or Bulk UART or BLE UART or BT PCM alternate function on these pins; therefore software resources will be required to implement this alternate function

4.5.32 RGMII (Ethernet Phy)

The NAD module supports an Ethernet PHY using the RGMII (Reduced Gigabit Media Independent Interface).

RGMII is a full duplex 4-bit serial bus that can operate up to 125MHz and clock in data on both edges of the clock for a total 8 bits per clock cycle or 1Gbps.

The RGMII is standardized by IEEE 802.3u.

Table 4-46: RGMII Interface

Name	Direction	Function	If Unused
RGMII_CK_TX	Output	RGMII Transmit clock ¹	Leave Open
RGMII_TX_0	Output	RGMII Transmit data	Leave Open
RGMII_TX_1	Output	RGMII Transmit data	Leave Open
RGMII_TX_2	Output	RGMII Transmit data	Leave Open
RGMII_TX_3	Output	RGMII Transmit data	Leave Open
RGMII_CTL_TX	Output	RGMII Transmit Control	Leave Open
RGMII_CK_RX	Input	RGMII Receive Clock ²	Leave Open
RGMII_RX_0	Input	RGMII Receive data	Leave Open
RGMII_RX_1	Input	RGMII Receive data	Leave Open
RGMII_RX_2	Input	RGMII Receive data	Leave Open
RGMII_RX_3	Input	RGMII Receive data	Leave Open
RGMII_CTL_RX	Input	RGMII Receive Control	Leave Open
RGMII_MD_CLK	Output	RGMII Management interface clock	Leave Open
RGMII_MD_IO	Bidirectional	RGMII Management interface I/O	Leave Open
RGMII_VREG_3V3_EN	Output	External PHY 3.3V regulator enable	Leave Open
RGMII_RESET_N	Output	External PHY reset	Leave Open
RGMII_VREG_PX_EN	Output	External PHY regulator enable	Leave Open

(1) The RGMII_CK_TX output internal delay is 2ns.

(2) The NAD has a programmable internal delay on the RGMII_CK_RX input of 0.9 to 1.5ns. The default is 1.5ns. The tolerance is 0.2ns.

Routing guidelines are provided in Section 11.

4.5.33 UART3 (Bluetooth or Bluetooth Low Energy)

Not currently supported in software.

The NAD module has the option to support a Bluetooth or Bluetooth Low Energy, 4-wire UART interface between Cortex A7 and a Bluetooth Low Energy module. The UART can transfer data at up to 4-Mbps.

If this optional feature is required, the bulk data UART2 and I2C will be unavailable; however bulk data SPI will still be available.

UART is a full-duplex serial bus regulated by the TIA.

Table 4-47: BLE UART3 Interface

Name	Direction	Function	If Unused
BT_UART_TX	Output	BLE UART Data Output (alternate UART2 / GPIO) ¹	Leave Open
BT_UART_RX	Input	BLE UART Data Input (alternate UART2 / GPIO) ¹	Leave Open
BT_UART_RTS	Output	BLE UART RTS (alternate GPIO) ²	Leave Open
BT_UART_CTS	Input	BLE UART CTS (alternate GPIO) ²	Leave Open

(1) Unavailable if UART2 is configured. Platform software does not inherently support the use of the BLE 4-wire UART alternate function on these pins; therefore software resources will be required to implement this alternate function

(2) Unavailable if I2C is configured. Platform software does not inherently support the use of the BLE 4-wire UART alternate function on these pins; therefore software resources will be required to implement this alternate function

4.5.34 Secondary I2C

The NAD module supports a secondary I2C port.

These pins can optionally be configured as standard GPIO, a secondary PCM interface for full Bluetooth interface or a 4-wire UART interface for Bluetooth Low Energy.

I2C is a half-duplex serial bus created by Philips/NXP. The I2C can transfer data at up to 400-kbps.

Table 4-48: Data UART2 Interface

Name	Direction	Function	If Unused
BT_UART_RTS	Output	Secondary I2C Clk (alternate BLE UART / BT PCM /GPIO) ^{1, 2}	Leave Open
BT_UART_CTS	Bidirectional	Secondary I2C Data (alternate BLE UART / BT PCM /GPIO) ^{1, 2}	Leave Open

(1) Platform software does not inherently support the use of GPIO or BLE UART or BT PCM alternate functions on these pins; therefore software resources will be required to implement this alternate function

(2) For proper I2C operation, I2C_SCK and I2C_DATA each require a 2.2-kOhm to 4.7-kOhm pull-up to VOUT1 outside the module. The exact value required depends on the capacitive load. Rise times should be met with the chosen resistor pull-up.

4.5.35 JTAG

JTAG is provided for debugging purposes.

Table 4-49: JTAG Interface

Name	Direction	Function	If Unused
JTAG_PS_HOLD	Input	JTAG Power Supply Hold – ACTIVE HIGH	Leave Open
JTAG_SRST_N	Input	JTAG System Reset	Leave Open
JTAG_TRST_N	Input	JTAG Test Reset	Leave Open
JTAG_TMS	Bidirectional	JTAG Mode Select	Leave Open
JTAG_TDO	Output	JTAG Data Output	Leave Open
JTAG_TDI	Input	JTAG Data Input	Leave Open
JTAG_TCK	Input	JTAG Clock Input	Leave Open

4.5.36 FREE GPIO

These GPIO are not currently assigned to any platform feature. Platform software does not inherently support the use of these GPIO for any purpose; therefore software resources will be required to implement any alternate functions on these GPIO.

Table 4-50: Free GPIOs

Name	Direction	Function	If Unused
GPIO_E14	GPIO	Free GPIO	Leave Open
PM_GPIO_S15	GPIO	PMX GPIO	Leave Open
PM_GPIO_T14	GPIO	PMX GPIO	Leave Open
PM_GPIO_T16	GPIO	PMX GPIO	Leave Open
RF_GPIO_M16	GPIO	GRFC, RFFE or GPIO	Leave Open
RF_GPIO_M17	GPIO	GRFC, RFFE or GPIO	Leave Open
RF_GPIO_M18	GPIO	GRFC, RFFE or GPIO	Leave Open
RF_GPIO_M19	GPIO	GRFC, RFFE or GPIO	Leave Open
RF_GPIO_M20	GPIO	GRFC, RFFE or GPIO ¹	Leave Open
RF_GPIO_N16	GPIO	GRFC, RFFE or GPIO	Leave Open
RF_GPIO_N17	GPIO	GRFC, RFFE or GPIO	Leave Open
RF_GPIO_N18	GPIO	GRFC, RFFE or GPIO	Leave Open
RF_GPIO_N19	GPIO	GRFC, RFFE or GPIO	Leave Open

(1) These pins must not be arbitrarily pulled high at boot.

4.5.37 Antenna Diagnostics

Antenna detection is limited on the current platform due to the limited number of ADC's available. A solution that was adopted by one TCU (as an example) was to use an external Analog mux to route the different antennas' detection circuits to the ADC. Below is the current assignments and logic to implement detection.

Note: If Antenna Diagnostics needs to be read quickly, it is suggested to read the external Mux in the following order (0, 1, 3, 2, 6, 7, 5, 4 or in "gray code" sequence). This allows for just one GPIO to be changed at a time to select the next port on the mux. This will minimize intermediate values to appear on the port.

Table 4-51: Signal to pin assignments

Name	Direction	Pin assignment
NAD_AD_MUX_SEL_B0	Output	RF_GPIO_N16
NAD_AD_MUX_SEL_B1	Output	RF_GPIO_N17
NAD_AD_MUX_SEL_B2	Output	RF_GPIO_N18
NAD_ADC	Input	PM_GPIO_S16

Table 4-52: Antenna Diagnostic Truth Table

NAD_AD_MUX_SEL			NAD_ADC
B2	B1	B0	
0	0	0	LTE_ANT_1
0	0	1	LTE_ANT_3
0	1	0	LTE_ANT_4
0	1	1	LTE_ANT_2
1	0	0	GNSS_ANT_1_V
1	0	1	GNSS_ANT_1_ISENSE
1	1	0	Thermister
1	1	1	open

4.5.38 GPIO Power Up State

App Note

The GPIO pin state is unknown before the GPIO power domain is 90% of its target value. This could be a concern when using a GPIO to control a LED or FLASH, or other interfaces. See [Figure 4-10](#).

The default pull state is defined in [Table 4-1](#). The diagram below illustrates the GPIO power up sequence.

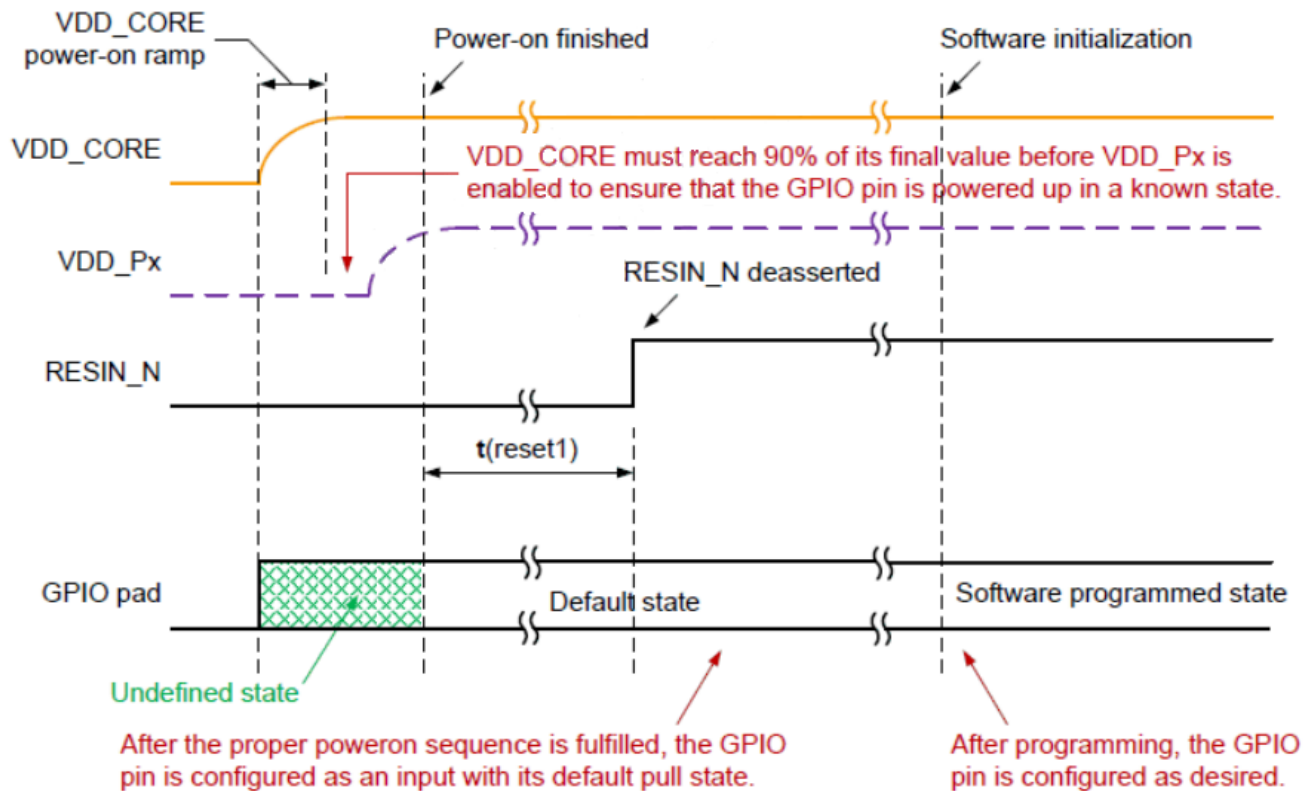


Figure 4-10: GPIO Pin Power Up

4.5.39 C-V2X SA2150 EAP Interface (not used)

The NAD supports the interface to the SA2150 EAP for C-V2X.

The SA2150 EAP interface consists of an Ethernet interface, plus 5 sideband signals, and the 1PPS_CLK

This interface reassigns one of the SPI interfaces to the SA2150 sideband signals and can be enabled with software.

Table 4-53 SA2150 Interface Signals

SA2150 AP Pin	SA515M NAD Signals		
	Pin	GPIO	SA515M direction
AP2NAD_STATUS	W15	GPIO_71 (INT)	Input
AP2NAD_WAKEUP	X17	GPIO_06 (INT)	Input
NAD2AP_ERR_FATAL	W17	GPIO_07	Output
NAD2AP_STATUS	W16	GPIO_05	Output
NAD2AP_WAKEUP	X16	GPIO_04	Output
1PPS_CLK	C21	GPIO_32	Output

5 Power Up and Power Down Sequence

This section describes proper power up and power down sequences. Note that this documents from strictly a hardware point of view. Please consult with the TP_F101U (User Guide 5G NAD Tier1 Power Management) in order to get a viewpoint restrictions from software and system.

5.1 Module Power-Up (ON) Timing

To power up the NAD using the PM_KPD_PWR_EN pin, drive the pin low per [Figure 5-1](#).

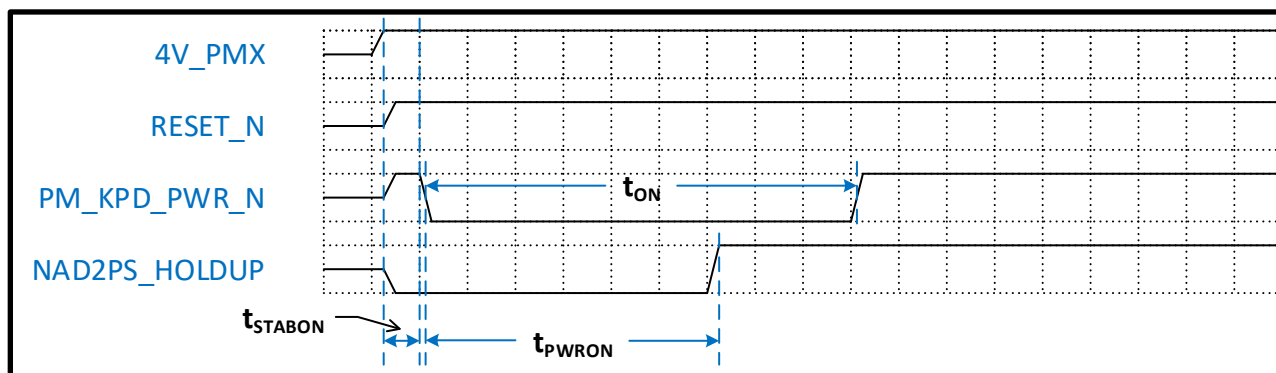


Figure 5-1: Power Up Timing

Parameter	Description	MIN	TYP	MAX	Units
t_{STABON}	Power applied stabilization (delay before power up pulse)	100	-	-	ms
t_{PWRON}	Turn on delay to module boot start	150	-	1000	ms
t_{ON}	Turn on pulse duration	300	-	1000	ms

Table 5-1: Power Up Timing Parameters

5.2 Module Power-Down (OFF) Timing (Controlled)

Please refer to the TP_F101U (User Guide 5G NAD Tier1 Power Management) for more details.

The recommended primary method to turn off the NAD is to perform a Linux controlled power down, and is depicted in [Figure 5-2](#).

The time required to complete a graceful power down will be software-dependent. The NAD2PS_HOLDUP pin will go low during the last phase of the NAD power down sequence.

Note that a delay of $t_{STABOFF}$ after NAD2PS_HOLDUP goes low is mandatory before the 4V NAD power supply can safely be removed from the NAD. This is to cover for the case that the NAD had an unexpected error and is recovering from the error through a reboot. In which case the system should wait to see if the NAD2PS_HOLDUP goes back high again. If it does, then power removal should not happen, then system must handle the NAD reboot and manage a new system power down again.

While powering down the NAD, the input VUC2NAD_HOLDUP must be driven low before and during the shutdown process.

The system must ensure that the 4V supply has dropped below 1.0V for at least 100ms before turning the 4V supply back on.

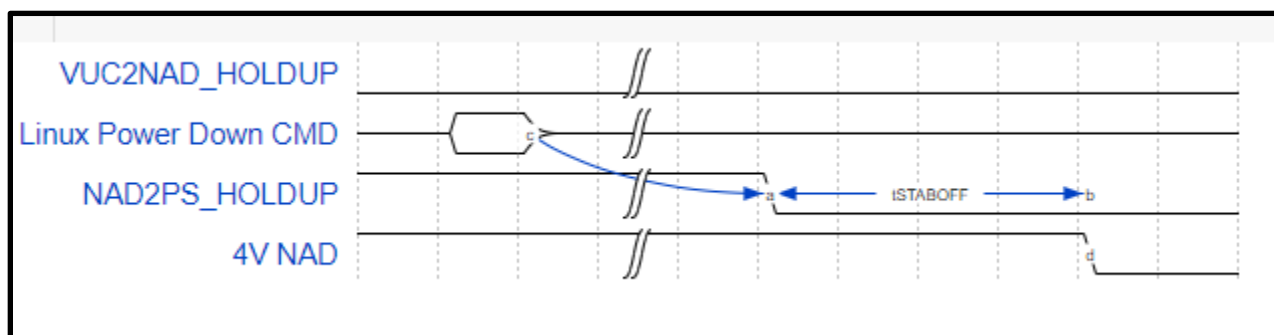


Figure 5-2: Primary Power Down Method Timing

Table 5-2: Primary Power Down Timing Parameters

Parameter	Description	MIN	TYP	MAX	Units
tSTABOFF	Delay from NAD2PS_HOLDUP to when 4V supply can be removed	1000	-	-	ms

If the primary method to turn off the NAD fails, a secondary method using the KPD_PWR_N signal is available, as shown in Figure 5-3. Pulsing PM_KPD_PWR_N low for tOFF will turn off the NAD. After time tSTABOFF following the NAD2PS_HOLDUP pin going low, the 4V NAD power supply can be safely removed from the NAD.

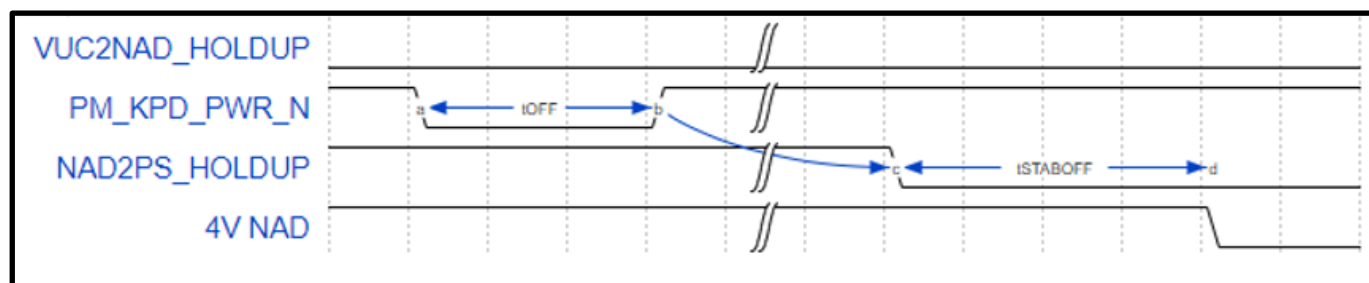


Figure 5-3: Secondary Power Down Method Timing

Table 5-3: Secondary Power Down Timing Parameters

Parameter	Description	MIN	TYP	MAX	Units
tOFF	Turn off pulse duration	3		7	S
tSTABOFF	Refer to Table 5-2				

If both the primary and secondary methods are unsuccessful at powering down the NAD, as a last resort an uncontrolled power down may be performed by removing the 4V supply from the NAD. See section 5.3 for details and requirements for recovery after an uncontrolled power down.

5.3 Uncontrolled Power Down

Every effort should be made to power down the NAD in a controlled manner as described above. Sometimes it is unavoidable that the main 4V power supply to the NAD is removed. (e.g. the car battery is disconnected without warning). In these uncontrolled cases, the NAD behaves in the following manner. When the main 4V supply drops below the low voltage threshold (default setting is 2V by QC

SBL software, 2.4V reset hardware default), the PMIC will reset the SOC within 3 μ s. As a result of the reset, the NAD2PS_HOLDUP signal will be pulled low.

The system must ensure that the 4V supply has dropped below 1.0V for at least 100ms before turning the 4V supply back on.

6 Specifications

6.1 Absolute Maximum Ratings ⁽¹⁾

Table 6-1: Absolute Maximum Ratings

Parameter		MIN	MAX	UNIT
4V_PMX	Power management input supply voltage	-0.5	4.60	V
4V_QET	Average power tracking input supply voltage	-0.5	4.60	V
V_IN (3)	Voltage on any “nonpower supply” pin – see below			
	P3 Pad type signals voltage (3)	-0.3	Pin F3 + 20% (3)	V
	P7 Pad type signals voltage (3)	-0.3	Pin F3 + 20% (3)	V
	P4 Pad type signals voltage (3)	-0.3	Pin B9 + 20% (3)	V
	P5 Pad type signals voltage (3)	-0.3	Pin M9 + 20% (3)	V
	Digital input voltage to GND	-0.3	2.16	V
	Analog input voltage to GND	-0.3	2.16	V
	Power dissipation (2)		20.0	W
	Storage temperature, T _{stg}	-40	+90	°C

- (1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the module. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under Recommended Operating Conditions. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) Power dissipation assumes maximum operating current of 2.5A and nominal operating voltage of 4.0V.
- (3) Signal pins shall be referenced to their respective Pad Type Supply (see [Table 4-4](#)). No voltage shall be applied to signal pins when NAD is powered off.

6.2 Recommended Operating Conditions

- Over operating temperature range (unless otherwise stated)

Table 6-2: Recommended Operating Conditions

Parameter		MIN	TYP	MAX	UNIT
4V_PMX_V	Power management input supply voltage ⁽¹⁾	3.8	4.0	4.2	V
3V3_PMX	Power management input supply voltage ⁽¹⁾	-	3.3	4.2	V
4V_QET_V	Average power tracking input supply voltage ⁽¹⁾	3.8	4.0	4.2	V
5V_CV2X (not used)	Average power tracking input supply voltage ⁽¹⁾	4.75	5.0	5.25	V
VREG_PX2_DUAL	Input voltage for SD card signals – 1.8V or 2.85V		1.8 / 2.85		V
VREF_RGMII			1.8		V
Operating Temperature (ambient):					
Fully Compliant Operation (Ta) ⁽²⁾		-30		+75	°C
Extended Operation (Ta) ⁽³⁾		-40		+85	°C
Junction Temperature (Tj) of Key Qualcomm ICs					
SA515M (Tj) ⁽⁴⁾					
eCALL temperature				+115	°C
Operating temperature				+105	°C
SDR867AQ (Tj) ⁽⁴⁾					
eCall temperature				+115	°C
Operating temperature				+105	°C
QPM5670 (Tj) ⁽⁴⁾					
eCall temperature				135	°C
Operating temperature				135	°C

(1) Voltage range includes both DC and AC components, including ripple, overshoot, undershoot or sag.

(2) Module is functional and meets all required 3GPP RF performance criteria at Ta (ambient temperature).

(3) Module is functional but may not meet all required RF performance criteria.

(4) Emergency calling allowed for Tj (junction temperature) within this range.

6.3 DC Current Ratings

- Over operating temperature range (unless otherwise stated)

The parent device must be capable of supplying power to the NAD as shown in [Table 6-3](#).

Table 6-3: Peak Current on Power Supply Pins

Parameter	Comments	MIN	TYP	MAX	UNIT
5V_CV2X (not used)	Pins A14, B14 -- C-V2X Power Amplifiers	---	---	2.0	A
4V supply (entire NAD) (Single SIM operation)	5G NSA TDD with 4G anchor at Max output, plus C-V2X, plus GNSS, with Tj=85C (4V_PMX + 4V_QET pins) ⁽¹⁾⁽²⁾⁽³⁾	-	-	5.1	A

(1) Instantaneous peak current with 1μs to 300μs duration.

(2) Does not include NAD-powered external peripherals such as WLAN/BT

(3) Peak module current drain is heavily influenced by GSM pulse, which is on for 0.57-ms and off for 4.03-ms.

(4) DSDA current draw is provided as reference, if designing for its future use in same parent device. This document otherwise only details the SIM1 NAD characteristics

Power required from the parent device's power supply under selected operating conditions is detailed in [Table 6-4](#).

Table 6-4: Current Draw for Specific Operating Modes

Parameter	Comments	MIN	TYP	MAX	UNIT
NAD OFF	Fully Powered Down, waiting for KBD_PWR	---	35	45	μA@4V
Rock Bottom Sleep	Airplane mode, no modem operation	---	1.2	1.5	mA@4V
GSM standby	900MHz band, with 0.48 second paging interval	---	2.3	3.0	mA@4V
WCDMA standby	Band 1, with 0.64 second paging interval	---	2.4	3.0	mA@4V
WCDMA standby	Band 1, with 1.28 second paging interval	---	1.9	2.5	mA@4V
WCDMA standby	Band 1, with 2.56 second paging interval	---	2.0	2.5	mA@4V
LTE standby	Band 1, with 0.32 second paging interval	---	3.8	5.0	mA@4V
LTE standby	Band 1, with 0.64 second paging interval	---	2.7	3.5	mA@4V
LTE standby	Band 1, with 1.28 second paging interval	---	2.1	3.0	mA@4V
LTE standby	Band 1, with 2.56 second paging interval	---	2.2	3.0	mA@4V

(1) Values obtained with otp-thick-3.1.17.8 (r120.1)

(2) Measured at room temperature

6.4 Output Voltage Supplies

Table 6-5: Output Voltage Supplies

Parameter		MIN	TYP	MAX	UNIT
VREF_1V8	Digital Reference – 1.8V	1.70	1.80	1.95	V
VREF_1V8_IMAX	Digital Reference – 1.8V	-	-	10	mA
VREG_CORE_VH	WLAN power supply – 1.8V	1.8	1.8	2	V
VREG_CORE_VH_IMAX				tbd	mA
VREG_CORE_VL	WLAN power supply – 0.8V	0.824	0.824	0.952	V
VREG_CORE_VL_IMAX				tbd	mA
VREG_CORE_VM	WLAN power supply – 1.2V	1.224	1.224	1.3	V
VREG_CORE_VM_IMAX				tbd	mA

6.5 ESD Ratings

Table 6-6: ESD Ratings

Parameter		VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per AEC Q100-002 (1)	±1000
		Charged-device model (CDM), per AEC Q100-011 (for all pins except as noted below)	±250
		USB SS TX pins (K23, L23) (CDM)	±125

(1) AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS001 specification.

6.6 Digital Electrical Characteristics

6.6.1 1.8V Digital I/O Characteristics

- At 25°C, VDD_PX = 1.8 V (unless otherwise noted)

Table 6-7: 1.8V Digital I/O Characteristics

Parameter		Comments	MIN	MAX	UNIT
VIH	High-level input voltage	CMOS/Schmitt	$0.65 * V_{DD_PX}$	$V_{DD_PX} + 0.3$	V
VIL	Low-level input voltage	CMOS/Schmitt	-0.3	$0.35 * V_{DD_PX}$	V
VSHYS	Schmitt hysteresis voltage		100	-	mV
IIH	Input high leakage current	No pull-down	-	1	uA
IIL	Input low leakage current	No pull-up	-1	-	uA
IIHPD	Input high leakage current	With pull-down	27.5	97.5	uA
IILPU	Input low leakage current	With pull-up	-97.5	-27.5	uA
VOH	High-level output voltage	CMOS, at rated drive strength	$V_{DD_PX} - 0.45$	V_{DD_PX}	V
VOL	Low-level output voltage	CMOS, at rated drive strength	0	0.45	V
IOZH	Tri-state leakage current	Logic high out, no pull-down	-	1	uA
IOZL	Tri-state leakage current	Logic low out, no pull-up	-1	-	uA
IOZHPD	Tri-state leakage current	Logic high out with pull-down	27.5	97.5	uA
IOZLPU	Tri-state leakage current	Logic low out with pull-up	-97.5	-27.5	uA
IOZHKP	Tri-state leakage current	Logic high out with keeper	-22.5	-7.5	uA
IOZLKP	Tri-state leakage current	Logic low out with keeper	7.5	22.5	uA
RP	Pull resistance	Pull-up and pull-down	55	390	kΩ
RK	Keeper resistance		30	250	kΩ
CIO	Input capacitance		-	5	pF

6.6.2 SDC Dual Voltage 1.8V/2.85V Digital I/O Characteristics

- At 25°C, VDD_PX = 1.8V (unless otherwise noted)

Table 6-8: SDC 1.8V Digital I/O Characteristics

Parameter		Comments	MIN	MAX	UNIT
VIH	High-level input voltage	CMOS/Schmitt	1.27	2.0	V
VIL	Low-level input voltage	CMOS/Schmitt	-0.3	0.58	V
VSHYS	Schmitt hysteresis voltage		100	-	mV
IIH	Input high leakage current	No pull-down	-	5	uA
IIL	Input low leakage current	No pull-up	-5	-	uA
VOH	High-level output voltage	CMOS, at rated drive strength	1.4	V _{DD_PX}	V
VOL	Low-level output voltage	CMOS, at rated drive strength	0	0.45	V
IOZH	Tri-state leakage current	Logic high out, no pull-down	-	5	uA
IOZL	Tri-state leakage current	Logic low out, no pull-up	-5	-	uA

- At 25°C, VDD_PX = 2.85V (unless otherwise noted)

Table 6-9: SDC 2.85V Digital I/O Characteristics

Parameter		Comments	MIN	MAX	UNIT
VIH	High-level input voltage	CMOS/Schmitt	0.625 * V _{DD_PX}	V _{DD_PX} + 0.3	V
VIL	Low-level input voltage	CMOS/Schmitt	-0.3	0.25 * V _{DD_PX}	V
VSHYS	Schmitt hysteresis voltage		100	-	mV
IIH	Input high leakage current	No pull-down	-	10	uA
IIL	Input low leakage current	No pull-up	-10	-	uA
VOH	High-level output voltage	CMOS, at rated drive strength	0.75 * V _{DD_PX}	V _{DD_PX}	V
VOL	Low-level output voltage	CMOS, at rated drive strength	0	0.125 * V _{DD_PX}	V
RP	Pull resistance	Pull-up and pull-down	55	390	kΩ
RK	Keeper resistance		30	250	kΩ
CIO	Input capacitance		-	5	pF

6.6.3 UIM1 and UIM2 Dual Voltage 1.8V/2.85V Digital I/O Characteristics

- At 25°C, VDD_PX = 1.8V (unless otherwise noted)

Table 6-10: UIM1/UIM2 1.8V Digital I/O Characteristics

Parameter		Comments	MIN	MAX	UNIT
VIH	High-level input voltage	CMOS/Schmitt	$0.70 * V_{DD_PX}$	$V_{DD_PX} + 0.3$	V
VIL	Low-level input voltage	CMOS/Schmitt	-0.3	$0.2 * V_{DD_PX}$	V
VSHYS	Schmitt hysteresis voltage		100	-	mV
IIH	Input high leakage current	No pull-down	-20	20	uA
IIL	Input low leakage current	No pull-up	-	1000	uA
VOH	High-level output voltage	CMOS, at rated drive strength	$0.8 * V_{DD_PX}$	V_{DD_PX}	V
VOL	Low-level output voltage	CMOS, at rated drive strength	0	0.4	V
IOZH	Tri-state leakage current	Logic high out, no pull-down	-	5	uA
IOZL	Tri-state leakage current	Logic low out, no pull-up	-5	-	uA

- At 25°C, VDD_PX = 2.85V (unless otherwise noted)

Table 6-11: UIM1/UIM2 2.85V Digital I/O Characteristics

Parameter		Comments	MIN	MAX	UNIT
VIH	High-level input voltage	CMOS/Schmitt	$0.70 * V_{DD_PX}$	$V_{DD_PX} + 0.3$	V
VIL	Low-level input voltage	CMOS/Schmitt	-0.3	$0.2 * V_{DD_PX}$	V
VSHYS	Schmitt hysteresis voltage		100	-	mV
IIH	Input high leakage current	No pull-down	-20	20	uA
IIL	Input low leakage current	No pull-up	-	1000	uA
VOH	High-level output voltage	CMOS, at rated drive strength	$0.8 * V_{DD_PX}$	V_{DD_PX}	V
VOL	Low-level output voltage	CMOS, at rated drive strength	0	0.4	V
IOZH	Tri-state leakage current	Logic high out, no pull-down	-	10	uA
IOZL	Tri-state leakage current	Logic low out, no pull-up	-10	-	uA

6.6.4 Digital Output Characteristics

In all digital I/O cases, V_{OL} and V_{OH} are linear functions (Figure 6-1) with respect to the drive current per drive table (Table 6-12). They can be calculated using these relationships:

$$V_{OL}[\max] = \left(\frac{\%drive \times 450}{100} \right) mV$$

$$V_{OH}[\min] = V_{DD_{PX}} - \left(\frac{\%drive \times 450}{100} \right) mV$$

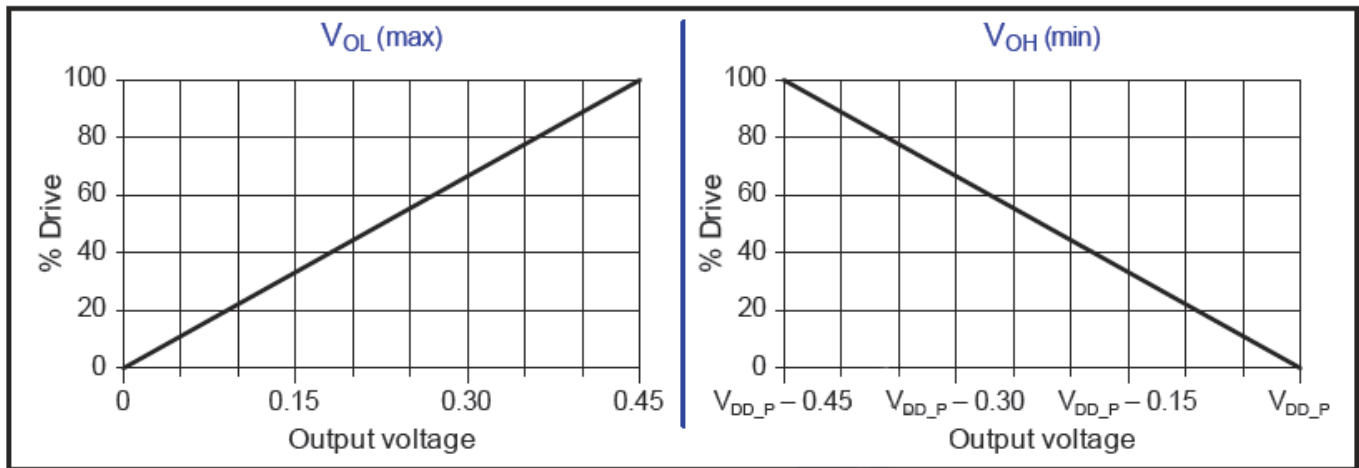


Figure 6-1: IV curve for V_{OL} and V_{OH} (valid for all V_{DD_PX})

Table 6-12: Output Current Drive Strengths

Parameter		Comments	MIN	MAX	UNIT
Iout1	SDC 1.8V	Eight levels, 2-mA steps	2	16	mA
Iout2	SDC 2.85V	Eight levels, 2-mA steps	2	16	mA
Others	All other 1.8V IO	Eight levels, 2-mA steps	2	16	mA

6.7 Timing Characteristics

6.7.1 Secure Digital Interface

Table 6-13: Supported SD Standards and Exceptions

Applicable Standard	Feature Exceptions	Device Variations
<i>Secure Digital: Physical Layer Specification, version 3.0</i>	None	Timing specifications – see Table 6-15
<i>SDIO Card Specification, version 3.0</i>	None	

6.7.2 eMMC Interface

Table 6-14: Supported eMMC Standards and Exceptions

Applicable Standard	Feature Exceptions	Device Variations
<i>Embedded Multimedia Card (eMMC), Electrical Standard 4.51</i>	Clock frequency up to 100-MHz	Timing specifications – see Table 6-15 .

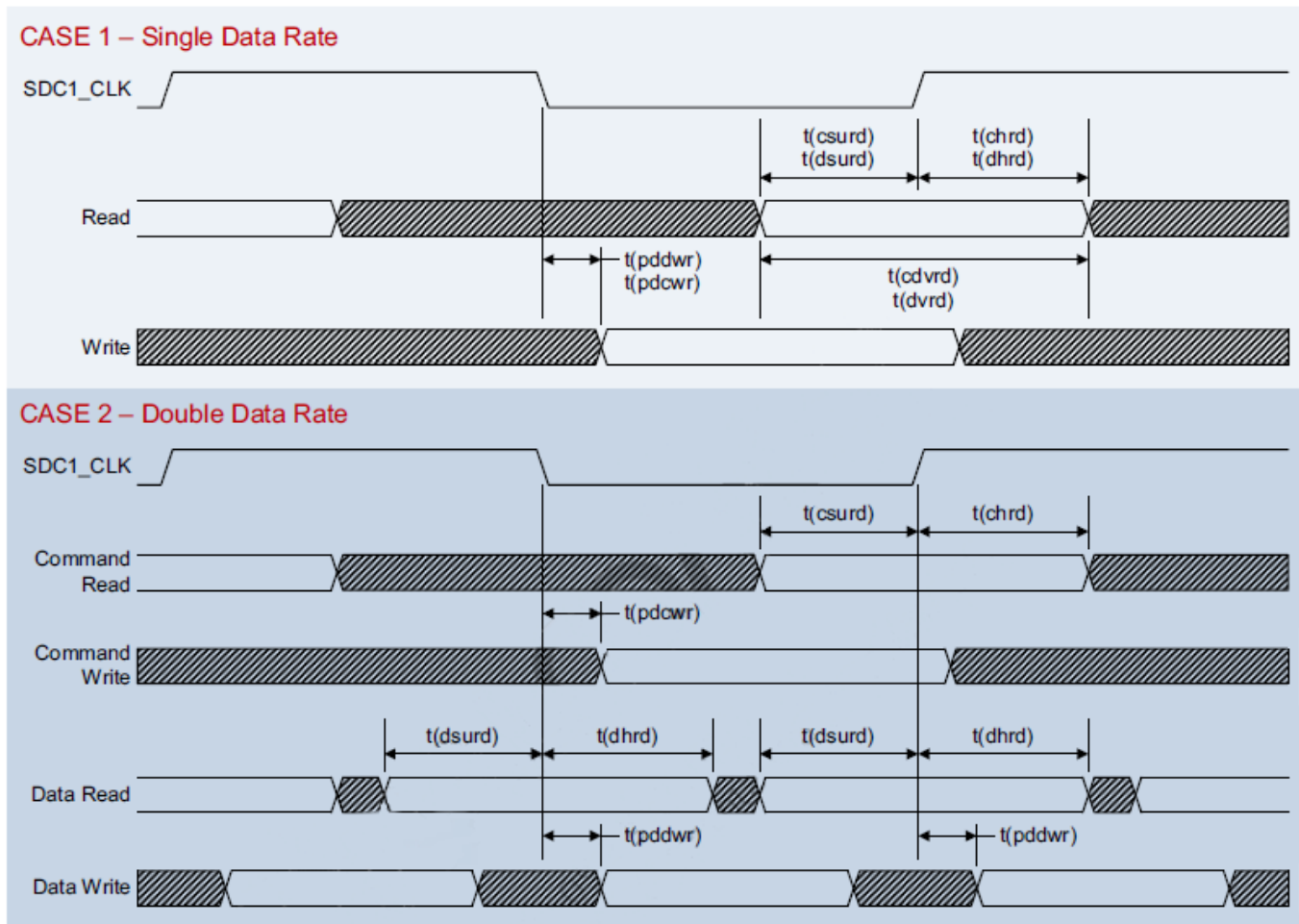


Figure 6-2: Secure Digital Interface Timing Diagram

Table 6-15: SDC Interface Timing Characteristics

Parameter		Comments	MIN	MAX	UNIT
Single data rate (SDR) mode – SDC up to 208-MHz					
t(cvdrd)	Command valid		2.4	-	ns
t(dvdrd)	Data valid		2.4	-	ns
t(pddwr)	Propagation delay on data write		-1.36	0.76	ns
t(pdcwr)	Propagation delay on command write		-1.36	0.76	ns
Single data rate (SDR) mode – SDC up to 100-MHz					
t(chrd)	Command hold		1.5	-	ns
t(csurd)	Command setup		2.5	-	ns
t(dhrd)	Data hold		1.5	-	ns
t(dsurd)	Data setup		2.5	-	ns
t(pddwr)	Propagation delay on data write		-3.7	1.5	ns
t(pdcwr)	Propagation delay on command write		-3.7	1.5	ns
Double data rate (DDR) mode – SDC up to 50-MHz					
t(chrd)	Command hold		1.5	-	ns
t(csurd)	Command setup		6.3	-	ns
t(dhrd)	Data hold		1.5	-	ns
t(dsurd)	Data setup		2.0	-	ns
t(pddwr)	Propagation delay on data write		0.8	6.0	ns
t(pdcwr)	Propagation delay on command write		-8.2	3.0	ns

6.7.3 USB 2.0 Interface

Table 6-16: Supported USB2.0 Standards and Exceptions

Applicable Standard	Feature Exceptions	Device Variations
Universal Serial Bus Specification, Revision 2.0 (April 27, 2000 or later)	Low-speed peripheral mode not supported	None
On-The-Go Supplement to USB 2.0 Specification (June 24, 2003, Revision 1.0A or later)	Supports host mode aspect of OTG Only	None

6.7.4 USB 3.1 Interface

Table 6-17: Supported USB3.1 Standards and Exceptions

Applicable Standard	Feature Exceptions	Device Variations
Universal Serial Bus Specification, Revision 3.0 (June 6, 2011 or later)	Only SuperSpeed Gen1 Supported	None

6.7.5 PCIe Interface

Table 6-18: Supported PCIe Standards and Exceptions

Applicable Standard	Feature Exceptions	Device Variations
PCI Express Specification, Revision 1.1 (January 15, 2005 or later)	None	None

6.7.6 I2C Interface

Table 6-19: Supported I2C Standards and Exceptions

Applicable Standard	Feature Exceptions	Device Variations
I2C Specification, version 5.0, October 2012	None	Multimaster is not supported

6.7.7 UIM Interface

Table 6-20: Supported UIM Standards and Exceptions

Applicable Standard	Feature Exceptions	Device Variations
ISO/IEC 7816-3	None	None

6.7.8 SPI Interface

The NAD module supports SPI as a master only.

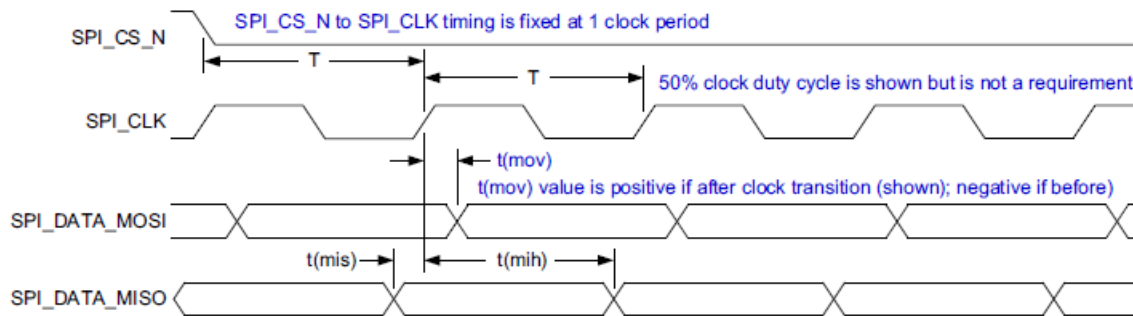


Figure 6-3: SPI Master Timing Diagram

Table 6-21: SPI Master Timing Characteristics

Parameter	Comments	MIN	MAX	UNIT
f	SPI Clock Frequency		50	MHz
T	SPI Clock Period	20.0	-	ns
t(ch)	Clock high	9.0	-	ns
t(cl)	Clock low	9.0	-	ns
t(mov)	Master output valid	-5.0	5.0	ns
t(mix)	Master input setup	5.0	-	ns
t(mih)	Master input hold	1.0	-	ns

6.7.9 I2S Interface

Table 6-22: Supported I2S Standards and Exceptions

Applicable Standard	Feature Exceptions	Device Variations
<i>Phillips I2S Bus Specifications, revised June 5,1996</i>	None	When an external SCK clock is used, a duty cycle of 45% to 55% is required.

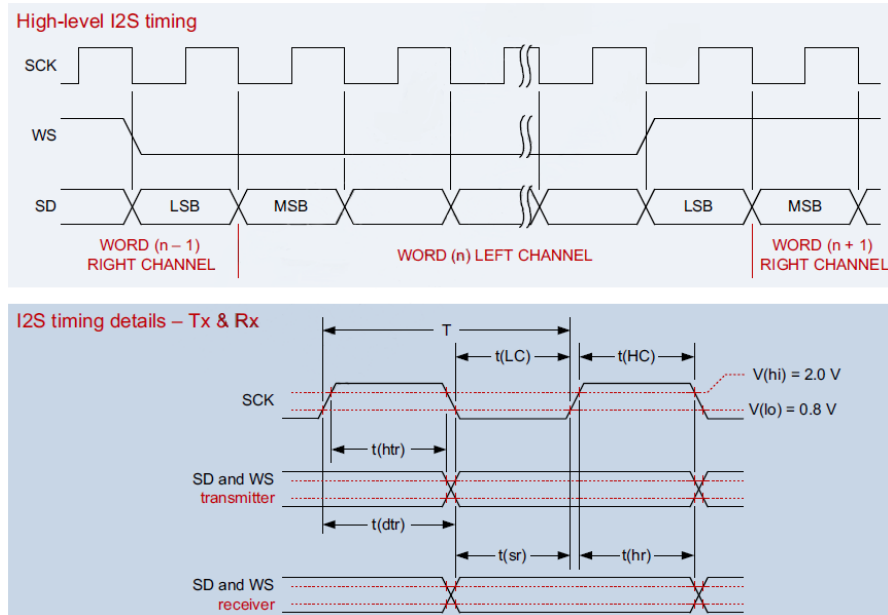


Figure 6-4: I2S Interface Timing Diagram

6.7.9.1 I2S Interface Timing

Table 6-23: 5G NAD I2S Interface Timing Characteristics

Parameter	Comments (1)	MIN	MAX	UNIT
Using NAD SCK				
f	Frequency	-	12.288	MHz
T	Clock period	81.380	-	ns
t(HC)	Clock high	0.45 x T	0.55 x T	ns
t(LC)	Clock low	0.45 x T	0.55 x T	ns
t(sr)	SD and WS input setup time	16.276	-	ns
t(hr)	SD and WS input hold time	0	-	ns
t(dtr)	SD and WS output delay	-	65.100	ns
t(htr)	SD and WS output hold time	0	-	ns
Using external SCK				
f	Frequency	-	12.288	MHz
T	Clock period	81.380	-	ns
t(HC)	Clock high	0.45 x T	0.55 x T	ns
t(LC)	Clock low	0.45 x T	0.55 x T	ns
t(sr)	SD and WS input setup time	16.276	-	ns
t(hr)	SD and WS input hold time	0	-	ns
t(dtr)	SD and WS output delay	-	65.100	ns
t(htr)	SD and WS output hold time	-1.45	0.85	ns

(1) Load capacitance between 10-pF and 40-pF

6.7.10 External CODEC PCM Interface

6.7.10.1 Short sync (Primary) PCM Interface (2048-kHz clock)

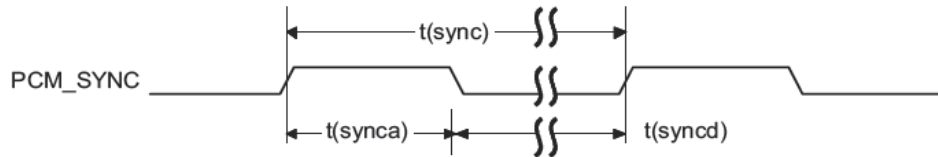


Figure 6-5: PCM_SYNC Timing Diagram

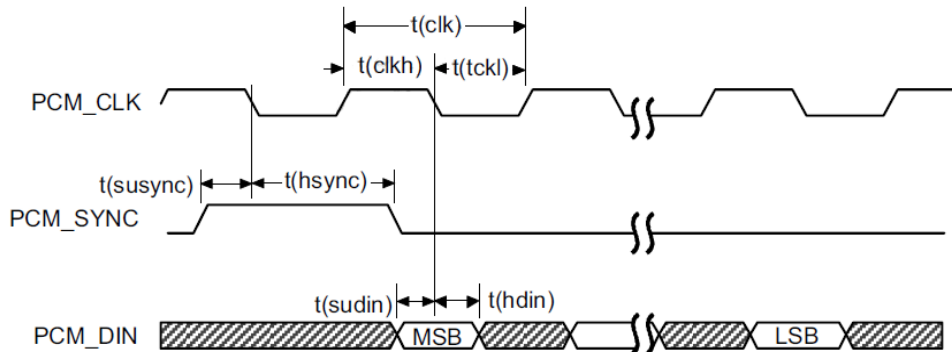


Figure 6-6: PCM_CODEC to NAD Timing Diagram

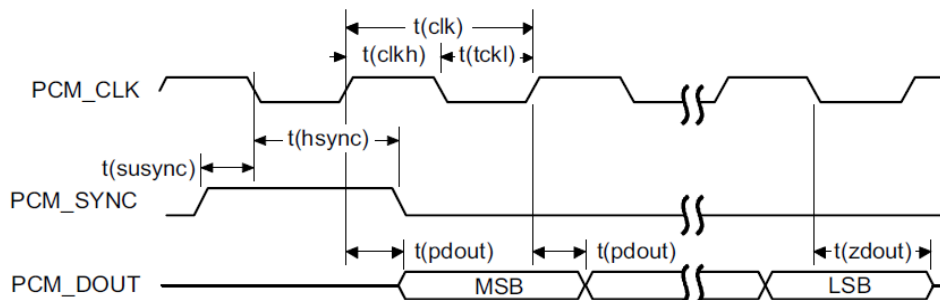


Figure 6-7: NAD to PCM_CODEC Timing Diagram

6.7.10.2 Primary Short Sync PCM Interface Timing

Table 6-24: Primary Short Sync PCM Interface Timing Characteristics

Parameter	Comments	MIN	TYP	MAX	UNIT
t(sync)	PCM_SYNC cycle time	-	125	-	us
t(synca)	PCM_SYNC asserted time	-	488	-	ns
t(syncd)	PCM_SYNC de-asserted time	-	124.5	-	ns
t(clk)	PCM_CLK cycle time	-	488	-	ns
t(clkh)	PCM_CLK high time	-	244	-	ns
t(clkl)	PCM_CLK low time	-	244	-	ns
t(susync)	PCM_SYNC offset time to PCM_CLK falling	-	122	-	ns
t(sudin)	PCM_DIN setup time to PCM_CLK falling	60	-	-	ns
t(hdin)	PCM_DIN hold time after PCM_CLK falling	10	-	-	Ns
t(pdout)	Delay from PCM_CLK rising to PCM_DOUT valid	-	-	60	ns
t(zdout)	Delay from PCM_CLK falling to PCM_DOUT HIGH-Z	-	160	-	ns

6.7.10.3 Long sync (Auxiliary) PCM Interface (128-kHz clock)

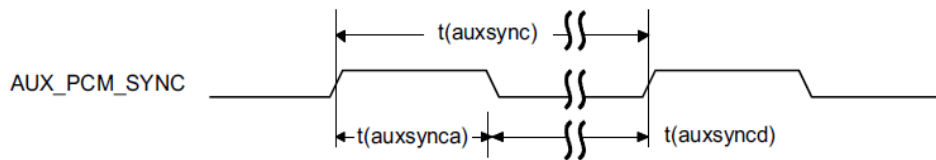


Figure 6-8: AUX_PCM_SYNC Timing Diagram

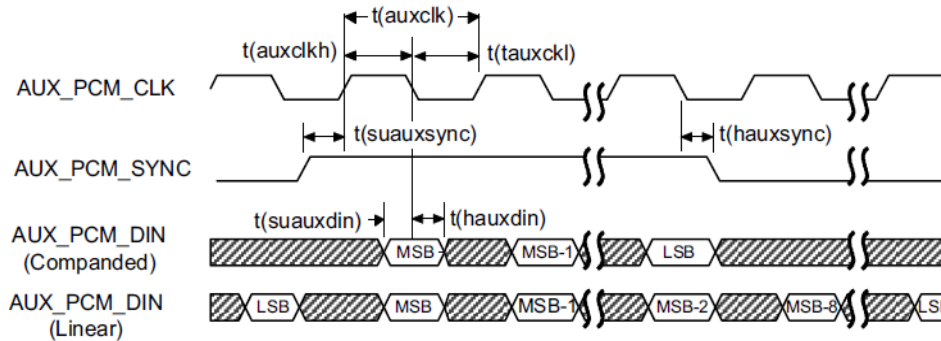


Figure 6-9: AUX_PCM_CODEC to NAD Timing Diagram

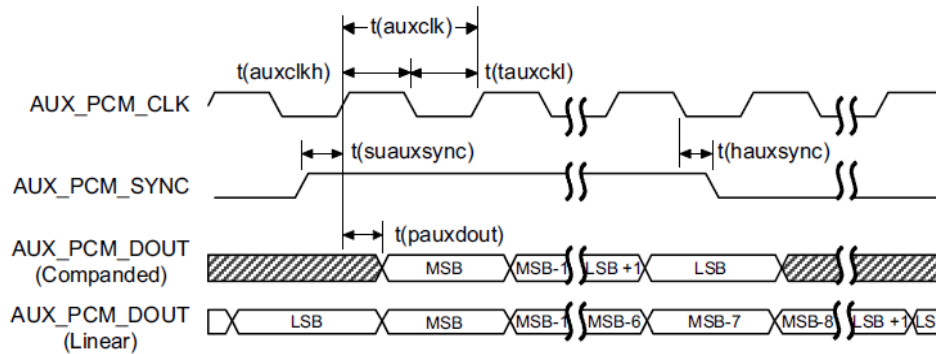


Figure 6-10: NAD to AUX_PCM_CODEC Timing Diagram

6.7.10.4 Auxiliary Long Sync PCM Interface Timing

Table 6-25: Auxiliary Long Sync PCM Interface Timing Characteristics

Parameter		Comments	MIN	TYP	MAX	UNIT
t(auxsync)	AUX_PCM_SYNC cycle time		-	125	-	us
t(auxsynca)	AUX_PCM_SYNC asserted time		62.4	62.5	-	ns
t(auxsyncd)	AUX_PCM_SYNC de-asserted time		-	124.5	-	ns
t(auxclk)	AUX_PCM_CLK cycle time		-	488	-	ns
t(auxclkh)	AUX_PCM_CLK high time		-	244	-	ns
t(auxclkl)	AUX_PCM_CLK low time		-	244	-	ns
t(suauxsync)	AUX_PCM_SYNC offset time to AUX_PCM_CLK falling		-	122	-	ns
t(hauxsync)	AUX_PCM_SYNC offset time to AUX_PCM_CLK falling		-	122	-	ns
t(suauxdin)	AUX_PCM_DIN setup time to AUX_PCM_CLK falling		60	-	-	ns
t(hauxdin)	AUX_PCM_DIN hold time after AUX_PCM_CLK falling		10	-	-	ns
t(pauxdout)	Delay from AUX_PCM_CLK rising to AUX_PCM_DOUT valid		-	-	60	ns
t(zdout)	Delay from AUX_PCM_CLK falling to AUX_PCM_DOUT HIGH-Z		-	160	-	ns

6.8 NAD Module RF Characteristics

All RF parameters are referenced at the antenna terminals *of the NAD*. The RF performance of the parent device (i.e. TCU) can differ depending on the additional line losses – as well as the impedance match – presented to each of the NAD's antenna terminals.

6.8.1 NAD Module RF Transmitter Output Power

The Transmitter Power at the NAD antenna terminal (not the RF connector of the evaluation board or the Telematics/parent module) at Room Temperature:

- GSM low bands (800/900): +32.5 +1.0/-1.5 dBm
- GSM hi bands (1800/1900): +29.5 +1.0/-1.5 dBm
- EDGE low bands (800/900)¹: +26.5 +1.0/-1.5 dBm
- EDGE hi bands (1800/1900)¹: +25.5 +1.0/-1.5 dBm
- WCDMA bands: +23.0 +1.0/-2.0 dBm
- LTE bands (PC3): +23.0 +1.0/-2.0 dBm*
- 5G NR bands (PC3): +23.0 +1.0/-2.0 dBm*
- 5G NR bands (PC2): +26.0 +1.0/-2.0 dBm**

* For most bands. Some exceptions exist, based on 3GPP standard

** For antenna ports that are HPUE capable on specific 5G NR bands. See section 4.5.2.

⁽¹⁾ Single-slot power. For MultiSlot cases, see Table 6-26.

Allowance for reduction in maximum transmitter power is specified in the 3GPP standard for GSM multi-slot operation. The SA515 NAD adjusts the GSM maximum TX power in these cases per Table 6-26.

Table 6-26 GSM Multi-slot power reduction

Band	GSM/EDGE Multi-slot maximum output power reduction		
	2 slots	3 slots	4 slots
GSM	3.0 dB	4.5 dB	6.0 dB
EDGE	2.0 dB	3.0 dB	4.0 dB

6.8.2 NAD Module RF Receiver Sensitivity

The Receiver Sensitivity at the NAD antenna terminal (not the RF connector of the evaluation board or the Telematics/parent module) at Room Temperature:

- GSM low bands (800/900): 3GPP TS 51.010-1 Section 14.2
- WCDMA bands: 3GPP TS 34.121-1 Section 6.2
- LTE bands: 3GPP TS GPP 36.521 Section 7.3
- 5G NR bands: 3GPP TS GPP 38.521 Section 7.3
- GNSS bands: -163 dBm (in Out-Of-Service mode)

6.8.3 Tune-Up Procedure

During manufacturing, each module is individually calibrated. The measurement is done in a fully calibrated setup which is built around a Rohde & Schwarz CMW100 base station simulator (system tester). Furthermore, the highest power level is verified afterwards in a non-signaling mode for each technology and on each frequency band.

Procedure:

1. Supply the FE515 module with operational voltage in GSM, WCDMA, LTE, or 5G-NR operation in a special service mode by means of company proprietary software.
2. Establish connection with the measurement equipment, set to Maximum Transmit Power and measure the transmit power.
3. The maximum transmit power calibration target levels are as specified in Section 6.8.1:
4. The gain factors of each individual device are adjusted until the target value is met. The appropriate gain control settings are stored in each device individually (for each power level).
5. After calibration, the set powers are measured and verified.
6. The user has no possibility to change the settings later on.

7 Mechanical Information

SA515 NAD Module mechanical details are provided in this document as a quick reference only. For complete details on mechanical specifications, as well as the latest updates, please reference Continental Document [#A2C78080700](#) ("Module Assy NAD SA515").

7.1 Module Exploded View

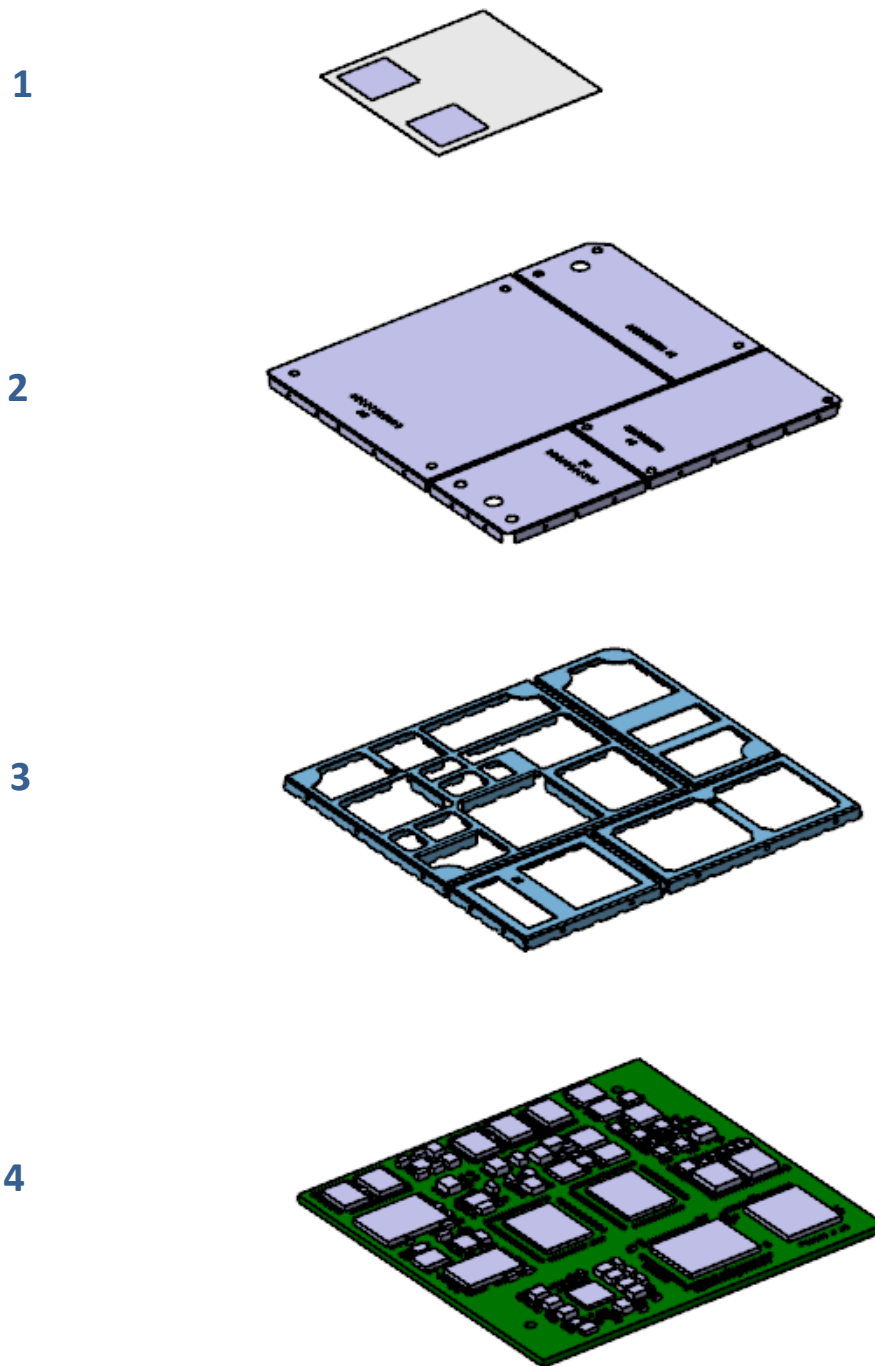


Figure 7-1: Module Exploded View

- 1: Label
- 2: Shield Cover(s)
- 3: Shield Frame
- 4: PCB Assembly

7.2 Module Top View

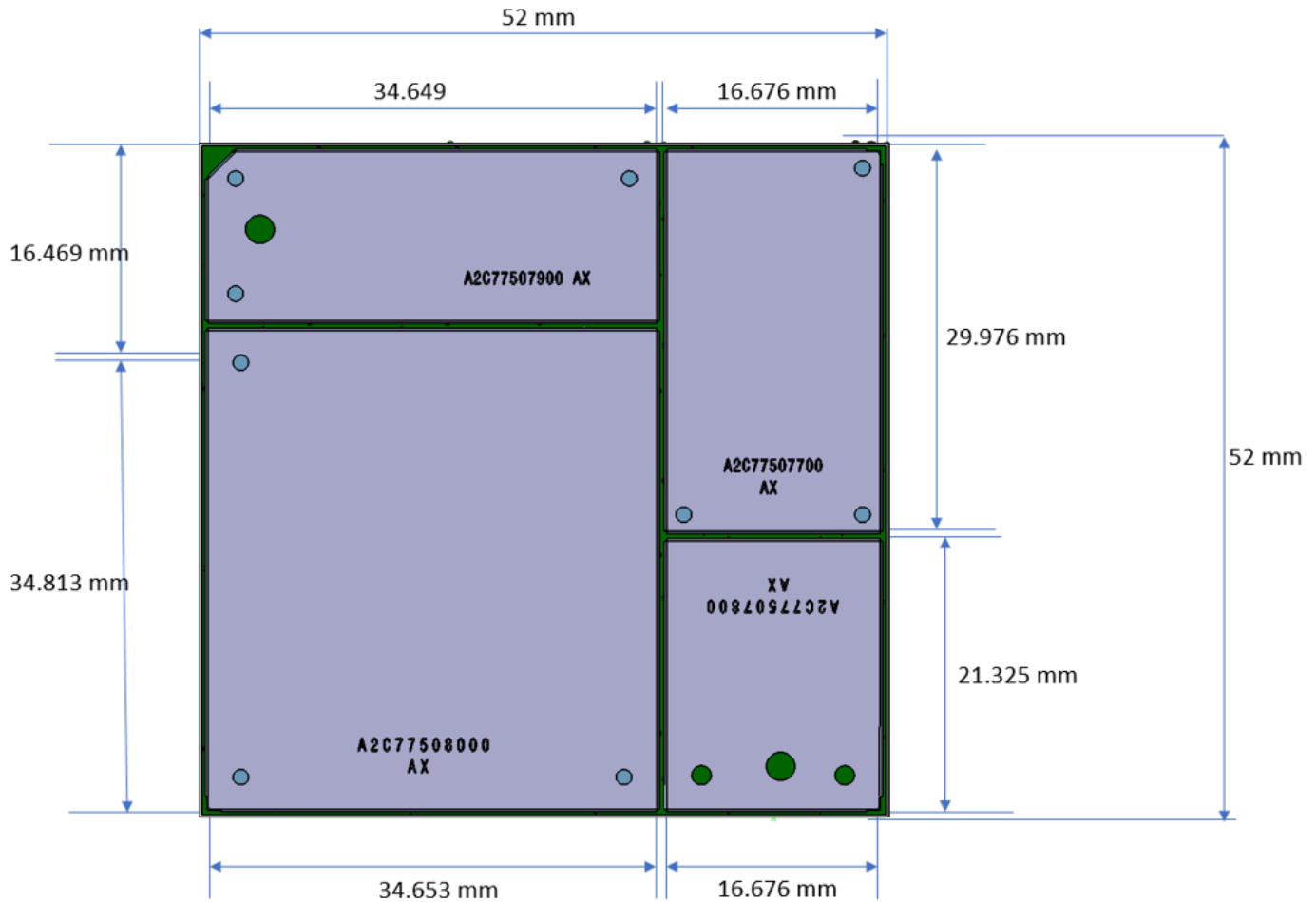


Figure 7-2: Module Top View

All dimensions are in mm.

7.3 Module Side View



Figure 7-3 Module Side View

7.4 Module Height and Tolerance

Figure 7-4 highlights the portion of the NAD which is expanded in detail in Figure 7-5 and Figure 7-6.

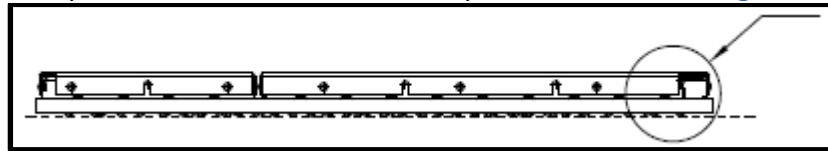


Figure 7-4 Component Thickness detail location

Table 7-1 explicitly lists the dimensions and tolerances of the NAD module, and the individual components contributing to its overall thickness.

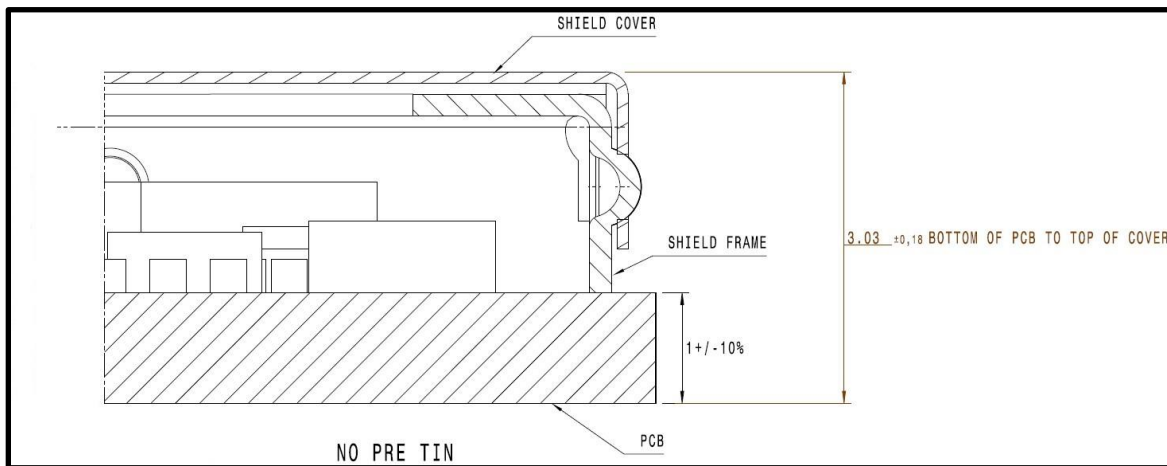


Figure 7-5 Component Thicknesses and Tolerances (not pre-tinned)

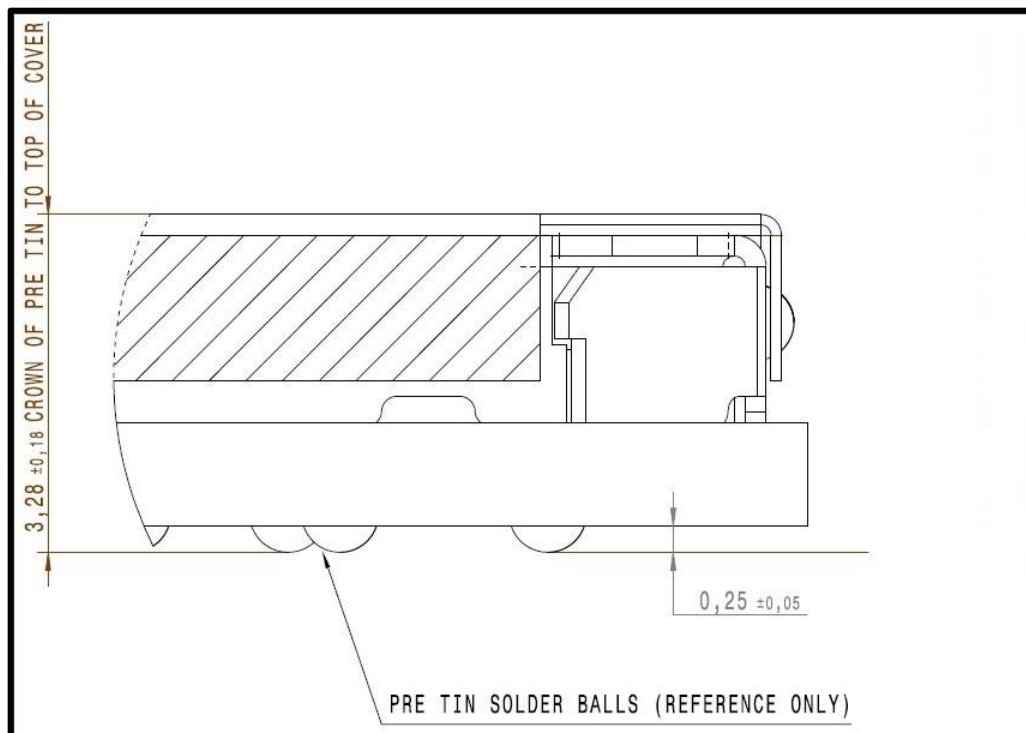


Figure 7-6 Component Thicknesses and Tolerances (pre-tinned)

Table 7-1 Component Thicknesses and Tolerances

Thickness Parameter Description	value	tolerance	units
NAD PCB (no pre-tinning)	1	+/-10%	mm
Pre-tin Solder Balls height	0.25	+/-0.05	mm
NAD with Shield Cover placed (no pre-tinning)	3.03	+/-0.18	mm
NAD with Shield Cover placed (with pre-tinning)	3.28	+/-0.18 ¹	mm

Note 1: Per root-sum-square tolerancing technique employed, pre-tinning adds a negligible amount to the total height tolerance, and the difference is not reflected in the two-decimal-place values.

8 PCB Mounting Guidelines

8.1 Footprint

Figure 8-1 is a view from top of the part, as if looking through the part. Table 8-1 details the pad dimensions.

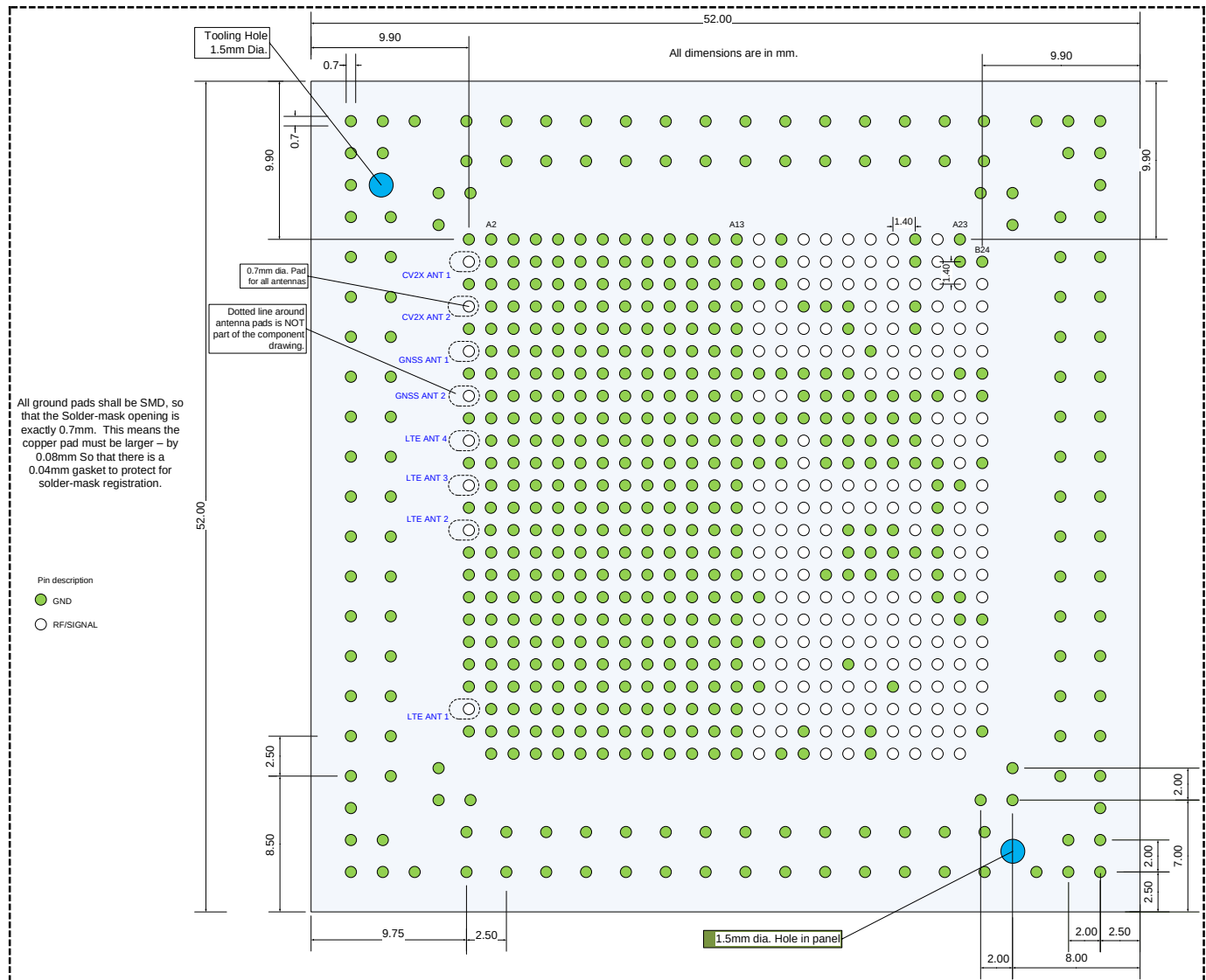


Figure 8-1: Pad location and dimension information (top view)

Table 8-1 Pad Dimensions and Tolerances

Pad Type	Diameter (μm)	Location Tolerance
Non-Solder-Mask Defined (All signal pads)	700+/-xx ¹	+/-zz μm^2
Solder-Mask defined pads (All GND pads)	700+/-yy μm	+/-30 μm^2

(1) Measured at the bottom surface of the copper plating

(2) The same location error is applied to all pads in a group (SMD or NSMD)

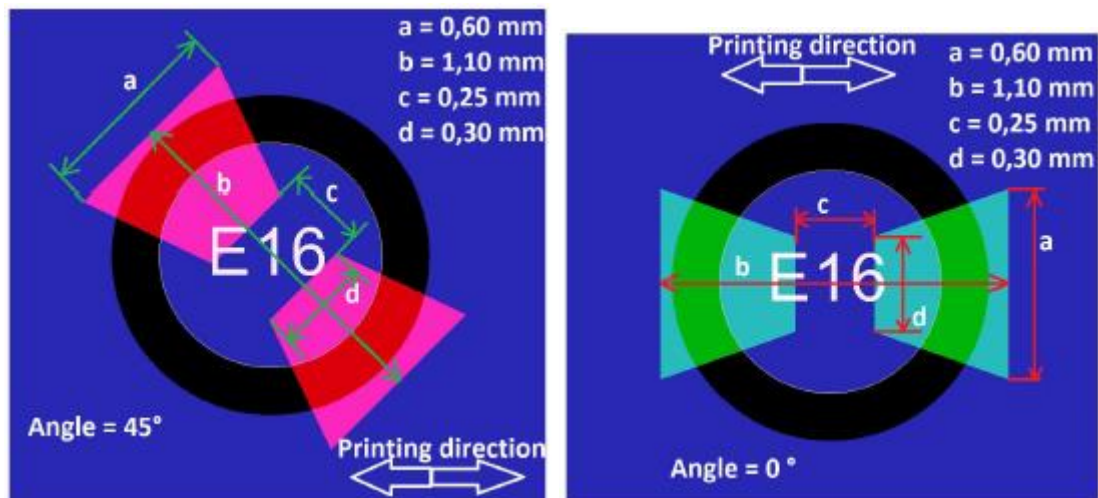
8.2 Manufacturing recommendations

8.2.1 Aperture Design

There are two recommended options for stencil aperture for TCU / motherboard to reduce voids, pictured in **Figure 8-2**:

- Butterfly aperture rotation 0°
- Butterfly aperture rotation 45°

The recommendation depends on the printing direction.

**Figure 8-2 Aperture Details**

8.2.2 NAD Module Placement

While the NAD module is designed to accommodate as many customer requirements as possible, it is also designed to fit a particular reference design, hence placement and orientation on the parent PCB should be similar for all designs. This will facilitate short antenna routes and simpler high-speed busses. Please use the appropriate reference design as a guide.

8.2.3 Component Spacing

When a vision system is used, a 1mm gap between the NAD module and any adjacent components is required. Please consult local manufacturing guidelines to be certain they do not require additional spacing.

If a step stencil is required, the gap must be increased to 5mm.

8.2.4 Soldering

8.2.4.1 Solder Paste recommendation is INDIUM 8.9HF

8.2.4.2 Reflow Profile as in Figure 8-3

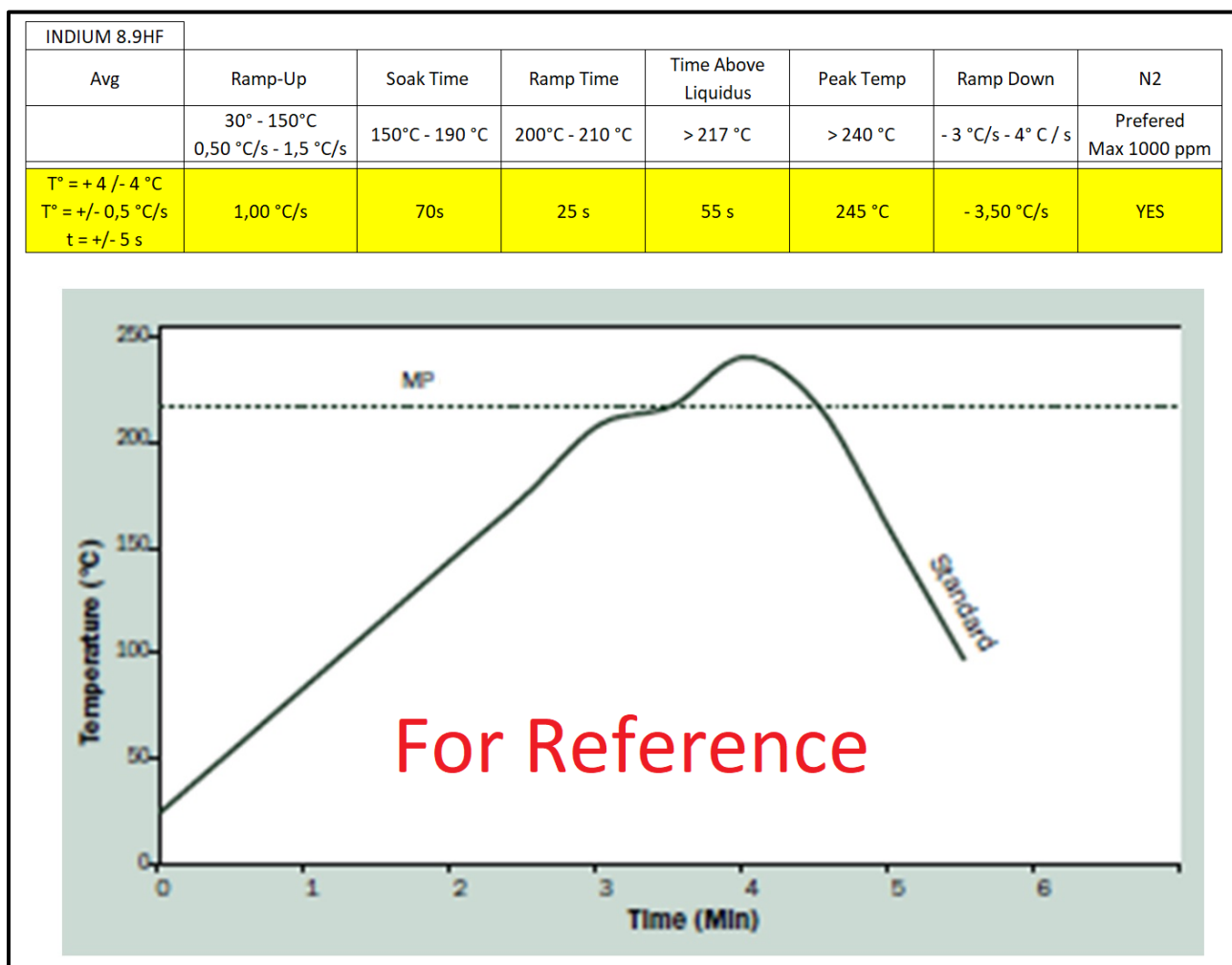


Figure 8-3 Reflow Oven Profile

8.3 Printed Circuit Board Material Requirements

8.3.1 Pre-Tinning

All 5G NADs shipped are pre-tinned with solder on the LGA pads, and the details and ordering codes listed in this HW specification are relevant for pre-tinned NADs. The option is available to order NADs without pre-tinning applied, but a separate part number/ordering code will have to be created.

8.3.2 Warpage / Flatness

The NAD coplanarity is expected to be maintained within $180\mu\text{m}$ across its entire footprint. See [Figure 8-4](#).

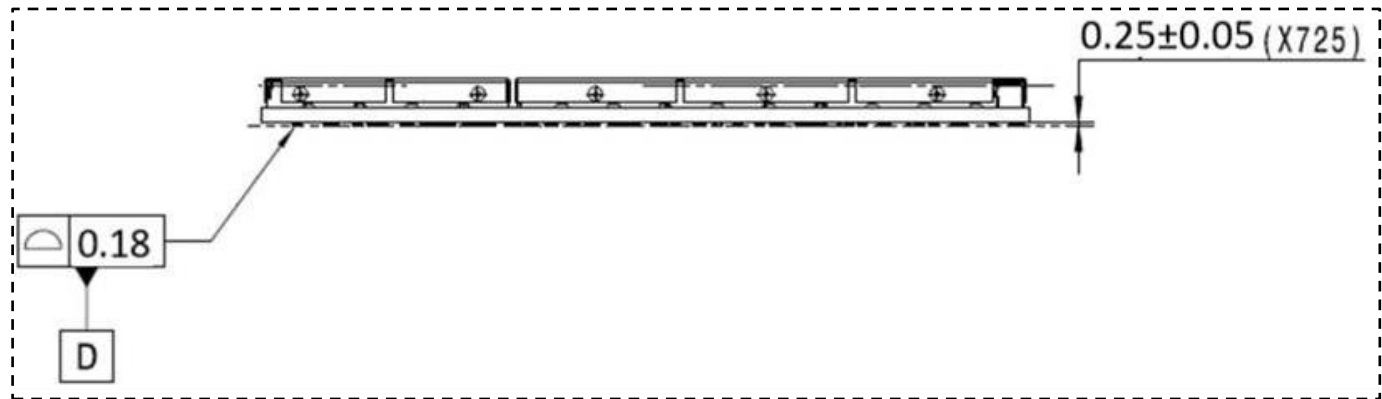


Figure 8-4 NAD PCB Coplanarity

Dimensions of [Figure 8-4](#) are in millimeters

1. Primary Datum "D" is determined by the first order LMS Regression Plane through the crowns of all 725 solder balls on the bottom side of the NAD PCB
2. Dimension measured at the maximum solder ball height from the bottom side of the PCB

To prevent manufacturing issues that might occur with the placement of the NAD module onto a parent device, coplanarity of both devices' PCBs is important. The combination of any warpage of the parent PCB and of the NAD PCB should be within the capability of the solder height used on the parent PCB and (if pre-tinning is applicable) on the NAD to overcome, to allow a good solder joint to be made.

8.3.3 Coefficient of Thermal Expansion (CTE)

To prevent reliability issues that can occur due to a mismatch between the thermal expansion coefficients of the NAD and parent PCBs, Continental recommends to keep the CTE in the X-Y direction of the main PCB between 11 and 13 ppm/°C.

9 Storage and Handling

9.1 Moisture Sensitivity Level (MSL)

All NAD modules are moisture sensitive and should be kept in their sealed moisture resistant bags until ready for assembly onto the DCM via the soldering process. Any parts that are not used immediately should be properly resealed in the same moisture resistant bag using appropriate equipment or placed into a dry box until they are needed again. The moisture sensitivity level (MSL) shown below is the amount of time the NAD modules may be exposed before this action must be taken. If the allowed MSL time elapses, the NAD modules must be baked per standard protocol to remove moisture.

Moisture Sensitivity Level: MSL Level 3 (1 Week)

9.2 Electrostatic Discharge

The NAD is generally well-protected from ESD by conductive full shielding when mounted on the parent device where adequate protections have been taken at the NAD electrical interfaces.

As a stand-alone device, such protection is absent, and care must be taken to protect the NAD from ESD during handling prior to placement onto the parent device. The susceptibility of the external pins of the stand-alone NAD to ESD is detailed in Section 6.5.

10 Part Reliability

The 5G NAD has been Qualified to Automotive Reliability Standards per AEC-Q104.

AEC-Q104 – QUALIFICATION TEST PLAN AND REPORT

TEST GROUP A - ACCELERATED ENVIRONMENT STRESS TESTS

STRESS	ABV	TES T#	TEST METHOD	TEST CONDITIONS	REQUIREMENTS			RESULTS Fails / S.S.
					SAMPL E SIZE (S.S.)	# LOTS	ACCEPTA NCE CRITERIA	
Preconditioning	PC	A1	IPC/JEDEC J-STD-020 JEDEC JESD22-A113	Peak Reflow Temp.= 245 °C Preconditioning = 24h at 125°C, + 192h at 30°C/60% R.H., + reflow profile for Pb-free solder, + reflow to test board	Min. MSL = 3		0 Fails	MSL = 3 0 / 120
Temp-Humidity-Bias	THB	A2	JEDEC JESD22-A101 OR A110	Preconditioning followed by 1000 h at 85°C/85% R.H., with bias	30	1	0 Fails	0 / 30
Temp-Humidity (without Bias)	TH	A3	JEDEC JESD22-A102, OR A118, or A101	Preconditioning followed by 1000 h at 85°C/85% R.H.	30	1	0 Fails	0 / 30
Temperature Cycling	TC	A4	JEDEC JESD22-A104	Preconditioning followed by 1000 cycles between -40°C / 85°C	30	1	0 Fails	0 / 30
Power Temp Cycling	PTC	A5	JEDEC JESD22-A105	Preconditioning followed by 1000 cycles between -40°C / 85°C, with voltage,	30	1	0 Fails	0 / 30
High Temp Storage Life	HTS L	A6	JEDEC JESD22-A103	1000 hours at 85°C	30	1	0 Fails	0 / 30

TEST GROUP B - ACCELERATED LIFETIME SIMULATION TESTS

High Temp Operating Life	HTO L	B1	JEDEC JESD22-A108	1000 hours at 85°C, with voltage	30	1	0 Fails	0 / 30
Early Life Failure Rate	ELF R	B2	See AECQ-104, Appendix 2	48 hours at 85°C, with voltage	231	1	0 Fails	0 / 231

TEST GROUP C - PACKAGE ASSEMBLY INTEGRITY TESTS

Solderability NAD External Leads	SD	C3	JEDEC J-STD-002	Preconditioning step: 150°C for 16 hours	15	1	>95% ext. lead coverage of the NAD	0 / 15
Physical Dimensions	PD	C4	JEDEC JESD22-B100 and B108 AEC Q003	All dimensions shall conform to the part drawing.	10	1	Cpk > 1.67	0 / 10

TEST GROUP E - ELECTRICAL VERIFICATION TESTS

Pre and Post-Stress Funct/Parameter	TES T	E1	Product Test Specification	All electrical tests before and after the qualification stresses were performed at -40°C, RT, and 85°C..	All	1	0 Fails	0 / 266
Electrostatic Discharge Human Body Model	HBM	E2	AEC Q100-002 or ANSI/ESDA JEDEC JS-001	TEST before and after ESD at room and hot temperature. NAD shall be classified according to the maximum withstand voltage level. HBM < 1000V require customer notification.	12	1	Target: 0 Fails; ≥ 1000 V	0 / 9
Electrostatic Discharge Charged Device Model	CDM	E3	AEC Q100-011 or ANSI/ESDA JEDEC JS-002	TEST before and after ESD at room and hot temperature. NAD shall be classified according to the maximum withstand voltage level. CDM < 500V require customer notification.	3	1	Target: 0 Fails; ≥ 125 V	0 / 3
Electromagnetic Compatibility	EMC	E8	CISPR25 Radiated Emissions	CISPR25 Radiated Emissions	2	1	CISPR25 Class 5	0 / 2
Lead (Pb) Free	LF	E10	AEC-Q005 / JESD201	Test 1: 1000 hours at 85°C/85% R.H. Test 2: 1000 cycles between -40°C / 85°C	60	1	Tin whisker length < 40 µm	0 / 60

TEST GROUP H - MODULE SPECIFIC TESTS

Low Temperature Storage Life	LTS L	H2	JEDEC JESD22-A119	1000 hours at -40°C	30	1	0 Fails	0 / 30
Start Up and Temperature Steps	STE P	H3	ISO 16750-4	Start up at cold and hot temperature and ramp in 10°C increments. Confirm functionality at each step, from -40°C to 85°C.	5	1	0 Fails	0 / 5
NAD Drop Test	NAD DRO P	H4	JEDEC JESD22-B111	Condition B (1500 G, 0.5 ms half-sine pulse, equivalent drop height 112 cm), as listed in JESD22-B110B. Limit each NAD to two drops.	6	1	0 Fails	0 / 6

11 Layout and Routing Recommendations

11.1 ANTENNAS

The NAD has five functional antenna pins.

- **LTE/5G NR:** ANT_1 – ANT_4 (Cellular antennas)
- **GNSS:** GNSS_ANT_1

11.1.1 NAD Antenna Breakout

The 5G NAD should be oriented on the main board to minimize the length of the primary Cellular TX/RX antenna (ANT1). This 50ohm line should be as short as possible to the external RF connector or internal antenna feed point.

The RF traces on the main board connecting from the NAD antenna pins can be either stripline or microstrip, but the microstrip routing must be on the layer opposite from the NAD, since the bottom layer of the NAD will be mostly ground and it would become a near RF short to any trace that runs on the main board's top layer, while still underneath the NAD. For a given line length, the stripline approach will tend to be more lossy, so it is generally not recommended. Thus, microstrip antenna lines are favored in most cases.

11.1.1.1 NAD Antenna Pad Ground Cutout

It is likely that the parent device's PCB will use only thru-hole vias, and so the antenna pads on the parent PCB may need to be offset slightly from their vias for manufacturability reasons. The NAD bottom layers have been designed to accommodate this need, and ground cutout in these layers has been extended to avoid shorting to those vias on the parent PCB's top layer, where it mates with the NAD. **Figure 11-3** details the ground cutout provided on the bottom six layers of the NAD. The PCB of the parent device should utilize this cutout area for placement of any thru vias which serve the antenna pads:

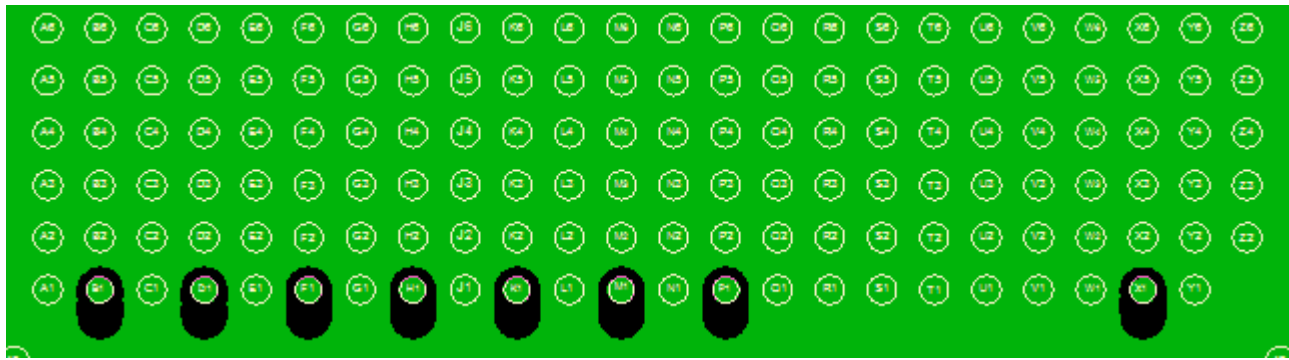


Figure 11-1: Antenna Pad Ground Cutouts (NAD bottom layer)

Figure 11-4 provides a more detailed view of the cutout around each antenna pad on the NAD PCB.

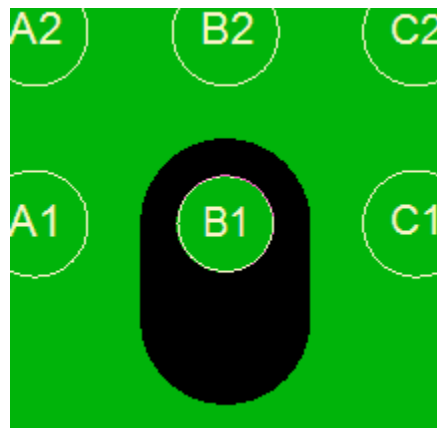


Figure 11-2 Antenna Pad Ground Cutout x8 (detail view)

11.1.1.2 Parent PCB Antenna Pad Design Recommendation

The main board on which the 5G NAD will be mounted should have ground cut out under the RF antenna pins of the NAD as shown in Figure 11-3 and Figure 11-4:



Figure 11-3: Antenna Pad Ground Cutout Recommended (Main Board, top layer/NAD side)

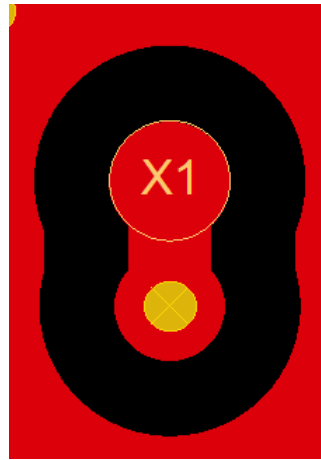


Figure 11-4 Antenna Pad Ground Cutout x8 (detail view)

The dimensions of the RF trace on the Main Board will be dependent on the layer stack-up of the board, their thicknesses and how many layers are used. As an example, the main PCB shown in Figure 11-5 is a 6 layer stack up with **Ground cut out on layer 5** so the microstrip lines on the bottom (layer 6) **reference Ground on layer 4**. For this example, the dielectric thickness from L6 to L4 is 21.2 mils.

		Thickness in microns	Tolerance in microns	Thickness in mils
	Solder Mask	30 micron	+11 micron	1.17 mil
Plated 0.5oz Cu	ML1	49 micron	+/-10 micron	1.91 mil
7628 x 1+1080 x 1 Prepreg	DL1	253 micron	+/-38 micron	9.88 mil
1oz Cu	ML2	36 micron	+/-10 micron	1.41 mil
2116 x 2 Core	DL2	254 micron	+/-38 micron	9.92 mil
1oz Cu	ML3	36 micron	+/-10 micron	1.41 mil
2165 x 2 Prepreg	DL3	277 micron	+/-38 micron	10.82 mil
1oz Cu	ML4	36 micron	+/-10 micron	1.41 mil
2116 x 2 Core	DL4	254 micron	+/-38 micron	9.92 mil
1oz Cu	ML5	36 micron	+/-10 micron	1.41 mil
7628 x 1+1080 x 1 Prepreg	DL5	253 micron	+/-38 micron	9.88 mil
Plated 0.5oz Cu	ML6	49 micron	+/-10 micron	1.91 mil
	Solder Mask	30 micron	+11 micron	1.17 mil
Total board thickness:		1593 micron	+272 mu -250 mu	62.23 mil

Figure 11-5: Recommended PCB Stack-up

Using any common microstrip impedance calculation tool, the result for a 50-ohm microstrip line (in typical dielectric constant of 4.3) with the height in the above example yields a width [w] of 37.7mils (967micron).

The line width will of course vary, particularly depending on the stack up and use of layers on the main board. In the end, the characteristics of the RF traces connecting to the NAD on the main board should remain within the guidelines of Table 11-1.

Table 11-1: RF Antenna Layout Parameters

Type of Guidance	Requirement
Trace impedance	50±10% Ohms, single-ended
Total route length	< 100 mm
Ground between signals	> 1 x [RF line width] with stitched VIA to ground layer
Spacing to other signals	> 3 times RF line width, to any non-RF traces
VSWR for all Antenna ports	< 3:1

- Trace impedances apply to either microstrip or stripline.
- Length for all antenna traces should be kept to a minimum, with priority on cellular antennas 1 and 2.

11.1.2 Integrated Device RF Insertion Loss

The integrated device TX and RX performance must comply with 3GPP and MNO RF requirements at the RF connectors. In addition, in some use cases, such as on-board antenna, TRP and TIS performance requirements must be met. It is a Product Team's responsibility to insure the integrated device complies with these requirements.

The Insertion Loss between NAD's RF and the parent module antenna terminal (RF connector of the evaluation board or the Telematics/parent module) must be kept at minimum and not to exceed these values:

- RF Loss < 0.6dB from 700 - 1000MHz
- RF Loss < 1.2dB from 1.7 - 2.7GHz
- RF Loss < X.XdB from 2.7 – 6.0GHz

11.2 SDC / eMMC Routing

Table 11-2: SDC / eMMC Layout Parameters

Type of Guidance	Requirement
Trace impedance	50-Ohms ± 10% single-ended
Total route length	< 50-mm recommended, but 75-mm is acceptable
Trace matching (SDC application)	< 1.6-mm
Trace matching (eMMC application)	< 5 - mm
Trace spacing	> 0.508-mm / > 20-mils
Number of vias per trace	< 5
Spacing to other signals	>= 2 times line width

- Trace impedances should follow table, either as microstrip or stripline.
- Trace lengths should follow table.
- Number of vias per trace should follow table.
- No stubs allowed on traces.
- No test points allowed on traces.
- Trace lengths should follow table.
- Spacing to ground or other signals on outside of bundled signals should follow table.

11.3 UIM Routing

Table 11-3: UIM1/UIM2 Layout Parameters

Type of Guidance	Requirement
Trace impedance	50-Ohms $\pm$ 10% single-ended
Total route length	< 100-mm
Inter-trace matching	< 5-mm
Inter-trace spacing	> 0.508-mm / > 20-mils
Spacing to other signals	> 3 times line width
External UIMX_DATA pull-up	10-kOhms to UIM_VCC near connector or embedded SIM
External UIMX_DET_N pull-up	10-kOhms to UIM_VCC near connector or embedded SIM

- A 1uF capacitor should be placed on the UIMX_VCC supply as close as possible to the eUICC contacts on the main board. UIMX_VCC should be at least 500-um (20-mils) wide.
- Internal pull ups are not provided for on the 5G NAD for UIMX_DATA and UIMX_DET_N. UIMX_DATA requires a 10-kOhm pull-up resistor tied to UIMX_VCC on the main board, near the connector or embedded SIM. UIMX_DET_N requires a 10-kOhm pull-up resistor tied to UIMX_DET_N on the main board, near the connector, if card is removable. If it is embedded SIM, then UIMX_DET_N should be tied to ground and no pull-up is needed.
- All UIMX routes on the main board should be no longer than 10cm. UIMX_CLK and UIMX_DATA should not be routed next to UIMX_RESET. Make sure there is sufficient GND between UIMX_RESET and UIMX_CLK and UIMX_DATA. (UIMX_CLK and UIMX_DATA can route together)
- The UIMX interface can be shared with another SIM device by adding a switch on the UIMX_DATA line.

11.4 USB2.0 Routing

Table 11-4: USB 2.0 Layout Parameters

Type of Guidance	Requirement
Pair impedance	90-Ohms $\pm$ 10% differential pair
Pair impedance	30-Ohms $\pm$ 10% common-mode impedance pair
Trace impedance	45-Ohms $\pm$ 10% single-ended
Total route length	< 150-mm
Intra-pair matching	< 2.0-mm / < 15 ps
Number of vias per trace	< 5
Spacing to other signals	$\geq$ 2 times line width

- Differential pairs should be routed as microstrip or stripline, using solid ground planes.
- Trace impedances should follow table, either as microstrip or stripline.
- Intra-pair lengths should follow table.
- Number of vias per trace should follow table.
- No stubs allowed on traces.
- No test points allowed on traces.
- Inter-pair lengths should follow table.
- Spacing to ground or other signals on outside of bundled signals should follow table.

11.5 USB3.1 Routing

Table 11-5: USB 3.1 Layout Parameters

Type of Guidance	Requirement
Pair impedance	90-Ohms $\pm$ 15% differential pair
Pair impedance	30-Ohms $\pm$ 15% common-mode impedance pair
Trace impedance	45-Ohms $\pm$ 15% single-ended
Total route length	< 142-mm
Tx Intra-pair matching	< 0.7-mm / < 5 ps
Rx Intra-pair matching	< 0.7-mm / < 5 ps
Number of vias per trace	< 3
Spacing to other signals	$\geq$ 2 times line width

- Differential pairs should be routed as microstrip or stripline, using solid ground planes.
- Inter-pairs can be routed next to each other per spacing in table.
- Trace impedances should follow table, either as microstrip or stripline.
- Intra-pair lengths should follow table.
- Number of vias per trace should follow table.
- No stubs allowed on traces.
- No test points allowed on traces.
- Inter-pair lengths should follow table.
- Spacing to ground or other signals on outside of bundled signals should follow table.

11.6 PCIe Routing

Table 11-6: PCIe Layout Parameters

Type of Guidance	Requirement
Pair impedance	90-Ohms $\pm$ 10-Ohms differential pair
Trace impedance	50-Ohms $\pm$ 15% single-ended
Total route length	< 300-mm
Tx Intra-pair matching	< 0.2-mm
Rx Intra-pair matching	< 0.2-mm
Number of vias per trace	< 4
Spacing to data lane-to-lane	$\geq$ 3 times line width
Spacing to other signals	$\geq$ 4 times line width
Series capacitor values	75-nF to 200-nF

- Differential pairs should be routed as microstrip or stripline, using solid ground planes.
- Inter-pairs can be routed next to each other per spacing in table.
- Trace impedances should follow table, either as microstrip or stripline.
- Intra-pair lengths should follow table.
- Number of vias per trace should follow table.
- No stubs allowed on traces.
- No test points allowed on traces.
- Inter-pair lengths should follow table.
- Spacing to ground or other signals on outside of bundled signals should follow table.
- Series capacitors should be on each pair, placed closest to the transmitter.

11.7 RGMII Routing

Table 11-7: RGMII Layout Parameters

Type of Guidance	Requirement
Trace impedance	50-Ohms $\pm$ 20% single-ended
TX-CLK to TX_CTL, TX_DATA length match	< 1.0 mm/ +/- 7 ps
RX-CLK to RX-CTL, RX-DATA length match	< 1.0 mm/ +/- 7 ps
Spacing within TX or RX lanes	$\geq$ 2.0 x line width ($\geq$ 3x for longer channel lengths)
Spacing between TX and RX lanes	$\geq$ 2.5 x line width
Spacing to other signals	$\geq$ 3 times line width

- Inter-pairs can be routed next to each other per spacing in table.
- Intra-pair lengths should follow table.
- No stubs allowed on traces.
- No test points allowed on traces.
- Inter-channel lengths should follow table.

11.8 4V Module Supply Layout

The **Figure 11-6** shows suggested method for routing power to the NAD module. Note that darker red is top layer and lighter red is inner layer.

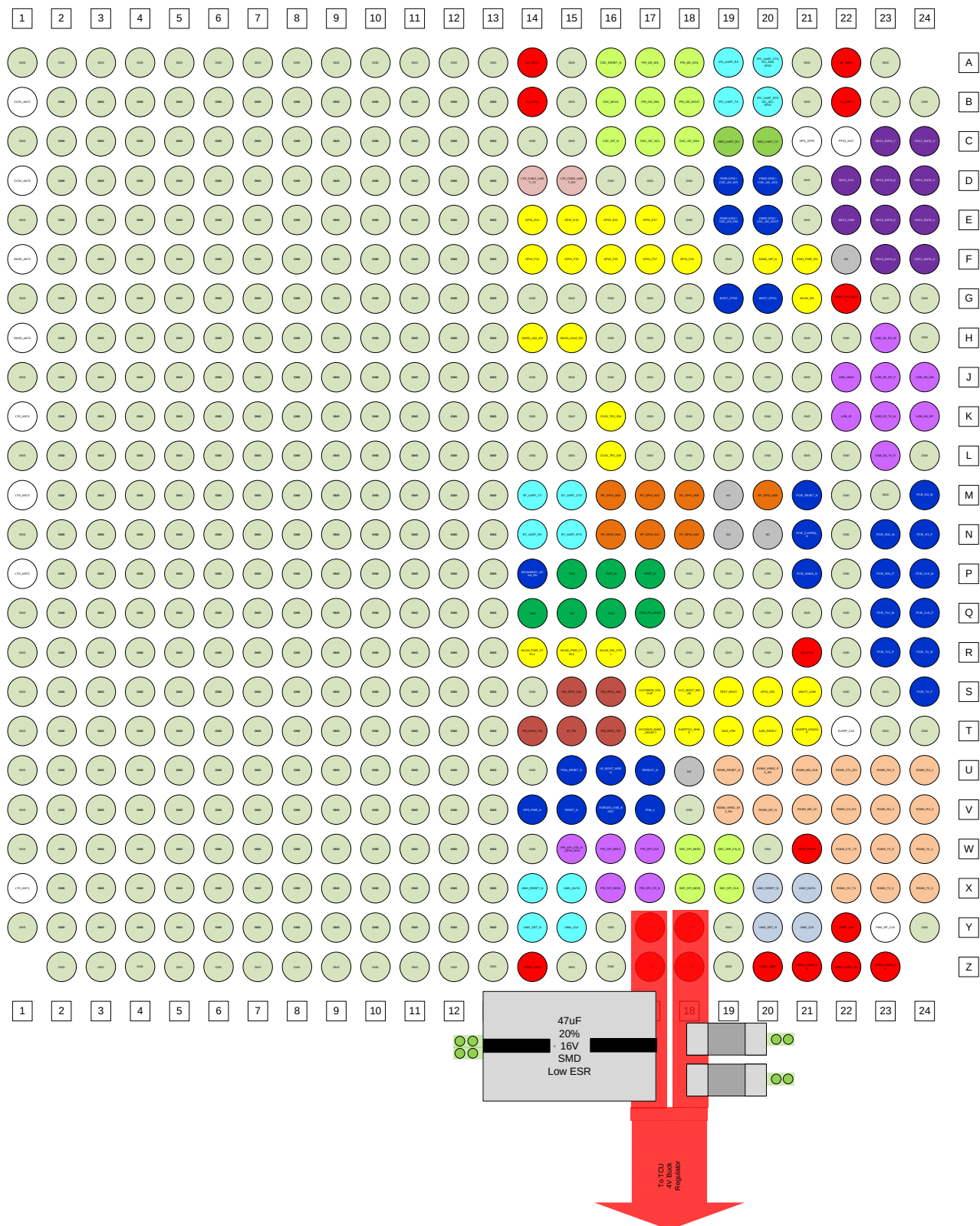


Figure 11-6: 4V Power Routing Recommendations (top view)

12 Thermal Considerations

12.1 Thermal Mitigation

Thermal mitigation is enabled in the NAD by limiting the uplink throughput, transmitter output power, 5G operation, and LTE category under extreme high temperature conditions.

T_{THERM} is the *maximum* of the temperatures read by the ADCs on the four PCB-mounted thermistors, which are located by the two largest power amplifiers and the SoC SA515M IC. These are all inside the NAD housing (underneath the shield).

T_{SENSE} is the die temperature of the Qualcomm SoC SA515M IC

The accuracy of the temperature sensors is about $\pm 3^{\circ}\text{C}$ at mitigation temperatures.

Table 12-1 shows the various thermal mitigation states and the sensor temperatures which trigger them:

Table 12-1 Thermal Mitigation States

Mitigation Level	Temp Sensor (T_{SENSE})	5G NR NAD Transceiver IC Mitigation	Temp Sensor (T_{THERM})	5G NR NAD PA Mitigation
0	$T_{SENSE} < 95^{\circ}\text{C}$	Full Operation without limitation	$T_{THERM} < 90^{\circ}\text{C}$	Full Operation without limitation
1	$95^{\circ}\text{C} < T_{SENSE} < 100^{\circ}\text{C}$	5G NR NSA: Disable 5G, fallback to 4G LTE 5G NR SA: Full operation without limitation	$90^{\circ}\text{C} < T_{THERM} < 95^{\circ}\text{C}$	LTE/NR: Uplink data rate on the PUSCH is limited to a percentage of the targeted rate starting at 90%, reducing in 10% steps as needed but maintaining at least 1 Mbps.
2	$100^{\circ}\text{C} < T_{SENSE} < 105^{\circ}\text{C}$	- The downlink reduces functionality from 4x4 MIMO to 2x2 MIMO. - Carrier Aggregation modes reduce to 1CC as needed to reduce the power dissipation.	$95^{\circ}\text{C} < T_{THERM} < 100^{\circ}\text{C}$	LTE: TX power is initially lowered by 5dB with a 50% duty cycle. A max reduction to 10dBm is allowed with no direct impact to the <i>download</i> rate. NR: TX power is initially lowered by 3dB with a maximum reduction to 15dBm.
3	$105^{\circ}\text{C} < T_{SENSE} < 120^{\circ}\text{C}$	Limited Service - Only eCalls (e911 and e112) and TSP related emergency calls are available until T_{SENSE} shutdown is reached or the temperature is reduced. - C-V2X implements flow control, and will be disabled above 115C (or >110C if eCall is active) - This has been disabled in the modem and needs to be managed by application SW.	$100^{\circ}\text{C} < T_{THERM}$	Limited Service - Only eCalls (e911 and e112) and TSP related emergency calls are available until T_{SENSE} shutdown is reached or the temperature is reduced. - This has been disabled in the modem and needs to be managed by application SW.
4	$120^{\circ}\text{C} < T_{SENSE}$	System Shutdown Thermal protective shutdown by the power management device. Hardware will wait 10 minutes before attempting to re-power up.		

Note 1: In the case of an eCall, all thermal mitigation features are deactivated. The eCall will continue up to T_{SENSE} of *at least* 115°C , but will end once T_{SENSE} reaches 120°C .

12.2 Thermal Management

The thermal mitigation SW is designed to protect the NAD components from being exposed to excessively high temperatures. However, the ability of the NAD to perform well in hot environments relies on the TCU design to achieve proper thermal performance.

The NAD should be placed on the main board **away from devices that dissipate large amounts of heat** such as audio amplifiers and DC power supplies, ethernet PHYs, HDBaseT PHYs etc.

A thermal heat sink that touches the main PCB under the NAD mounting area **needs to be used** in order to keep the internal temperature of the NAD at reasonable levels, see [Figure 12-1](#). The solder resist needs to be removed on the heat sink touch down area on the main PCB opposite the NAD. Thermal gap filler must be used between the heat sink and the main PCB to ensure good thermal contact.

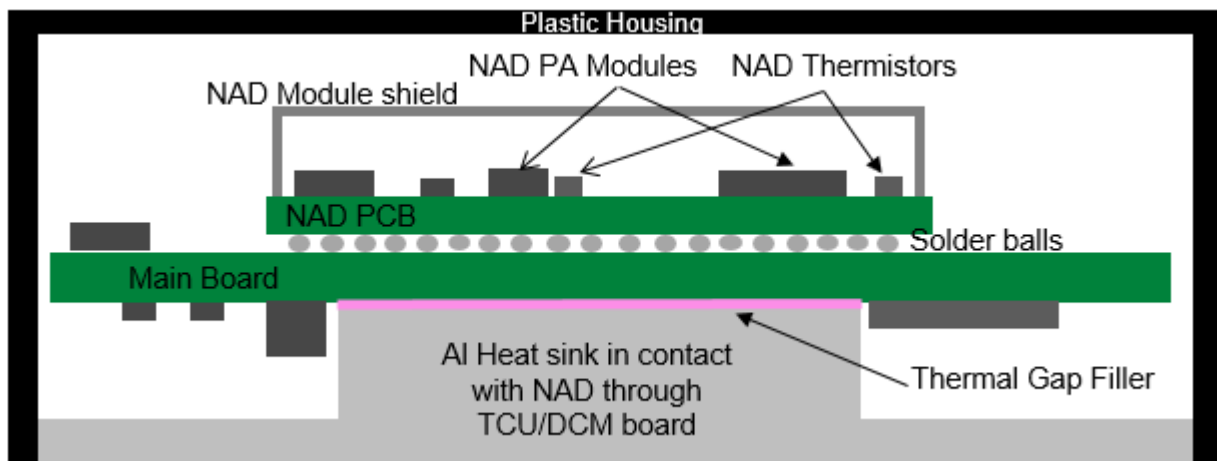


Figure 12-1: NAD Heatsink Method

A passive aluminum heat sink has proven to be a good compromise between cost and efficiency.

The darkened regions of [Figure 12-2](#) show where good thermal contact should be made on the PCB underneath the NAD.

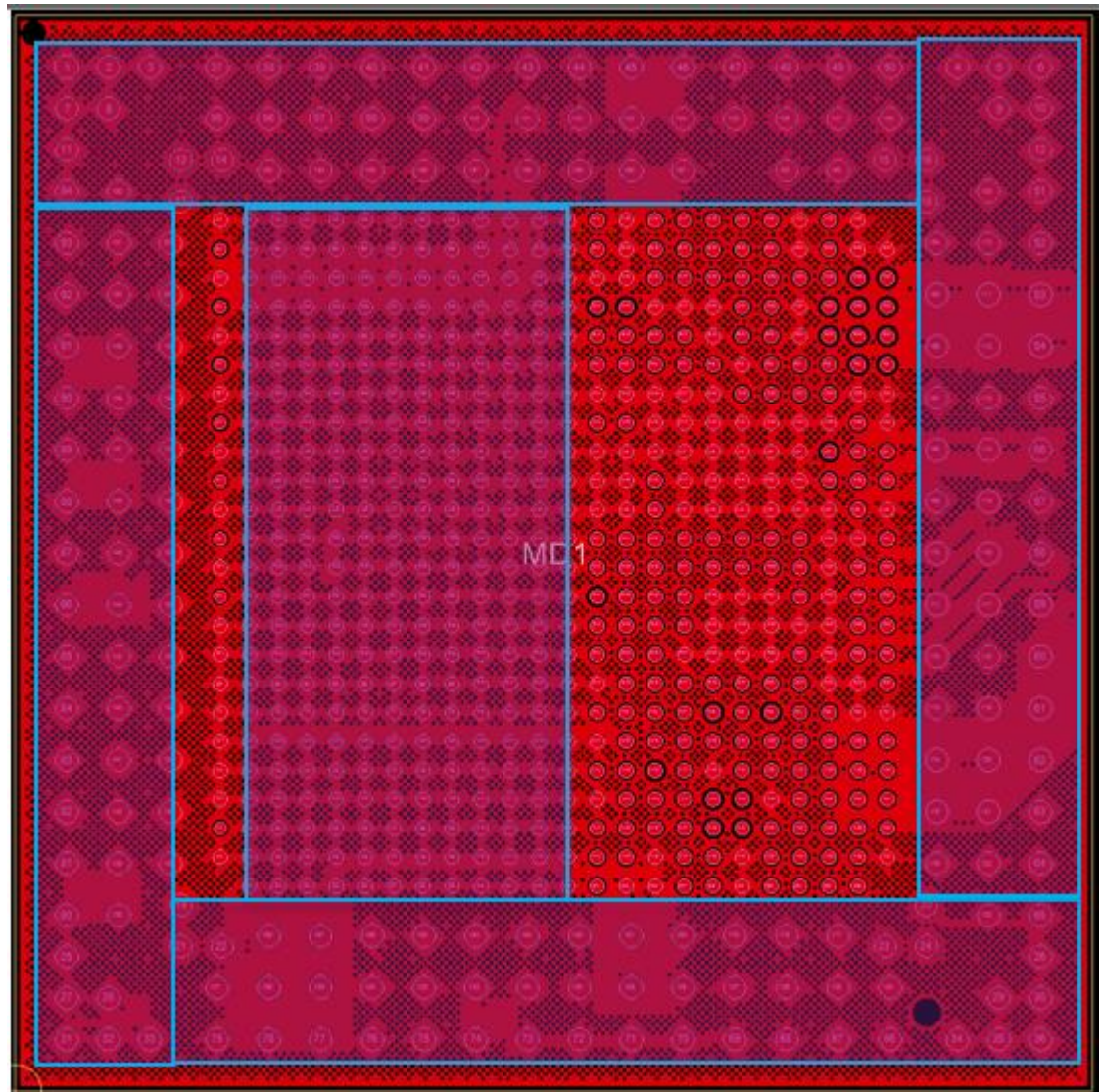


Figure 12-2: NAD Heatsink Pin Location

To maximize the thermal performance, the main board needs to have as many GND vias as possible under the NAD in the heat sink touch down areas. It is best to limit the routing under the NAD on ALL layers - and have as much GND as possible - in these areas as well, shown as darkened regions (see [Figure 12-2](#)) which correspond to NAD thermal GND pads.

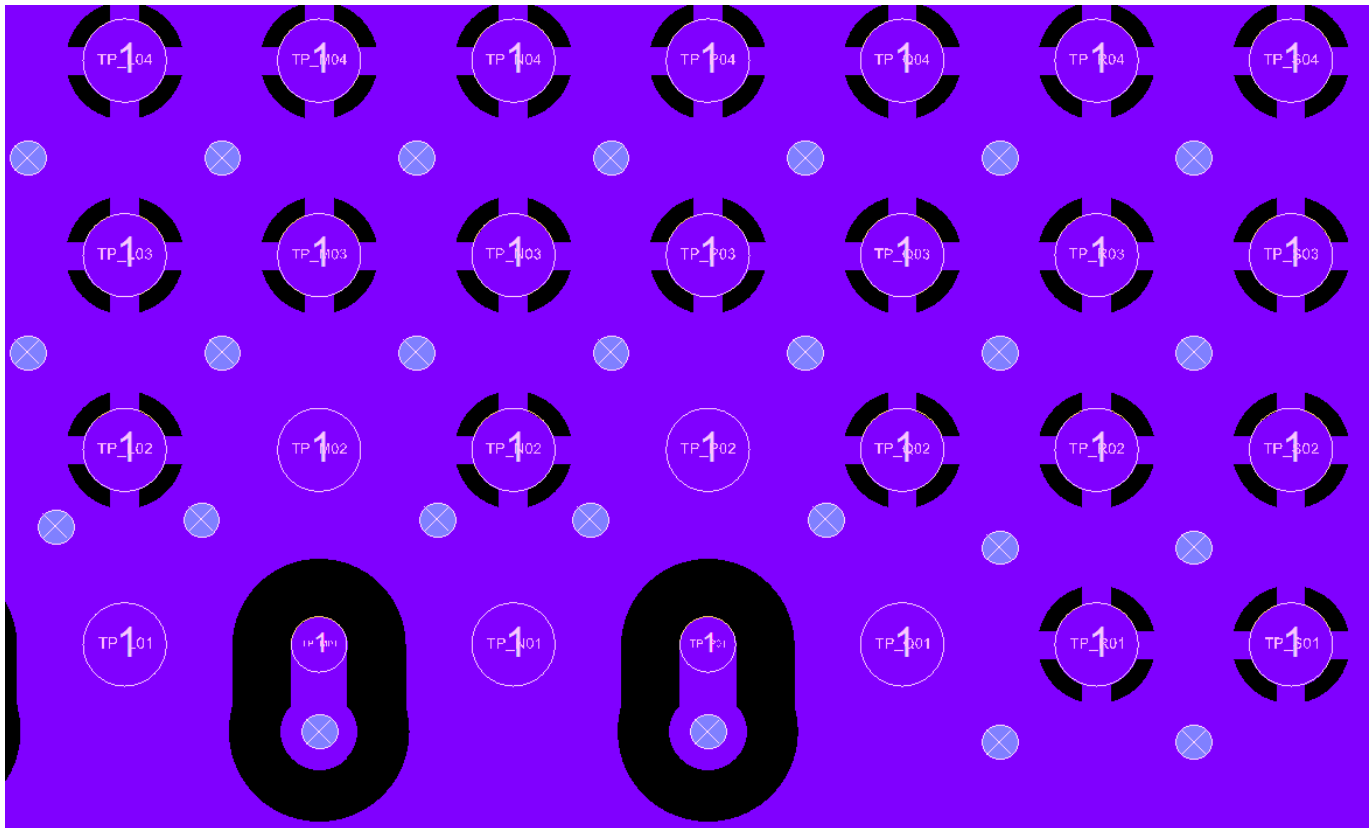


Figure 12-3: Main PCB NAD Heatsink Area GND Approach

More GND vias should be added as shown in [Figure 12-3](#).

12.3 Thermal Simulation Methods

Many factors will affect the maximum temperature at which the NAD can operate before temperature-related backoffs or shutdowns occur. To simulate the thermal behaviour, a suitable model must be created. Within Continental, the thermal characteristics of the NAD are modelled using commercial software by ANSYS, referred to as “Icepak”. The simulation profile used by the NAD development team is: “[SA515M_Thermal-ICEPACK MODEL v1.4 NAD Model](#)”.

13 Regulatory Compliance Notes

13.1 U.S.A. / Canada

To be updated in a later release.

14 Certification Information

To be updated in a later release.